

Compare and Contrast Real-World Examples

CSC105 – Prof. Earl

Created By: Jeffrey A. Shaffer

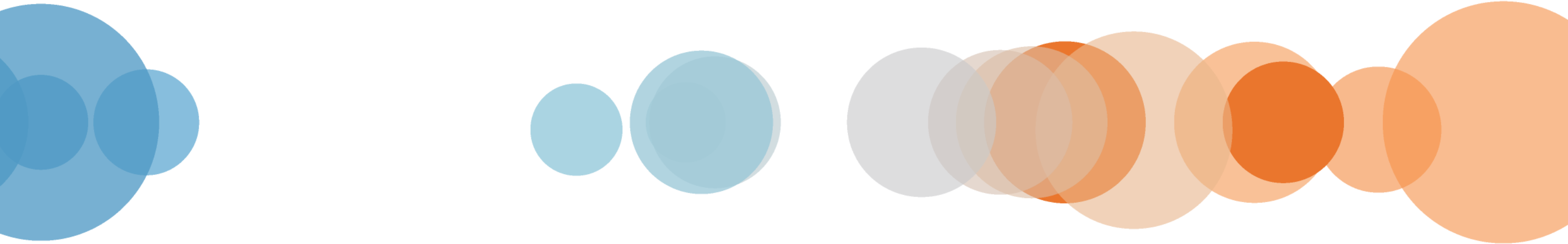
Goals

By completing the course modules, students will:

- Compare and contrast real-world examples
- Explore redesigns to improve the visualization of the data
- Understand various issues encountered with certain chart types

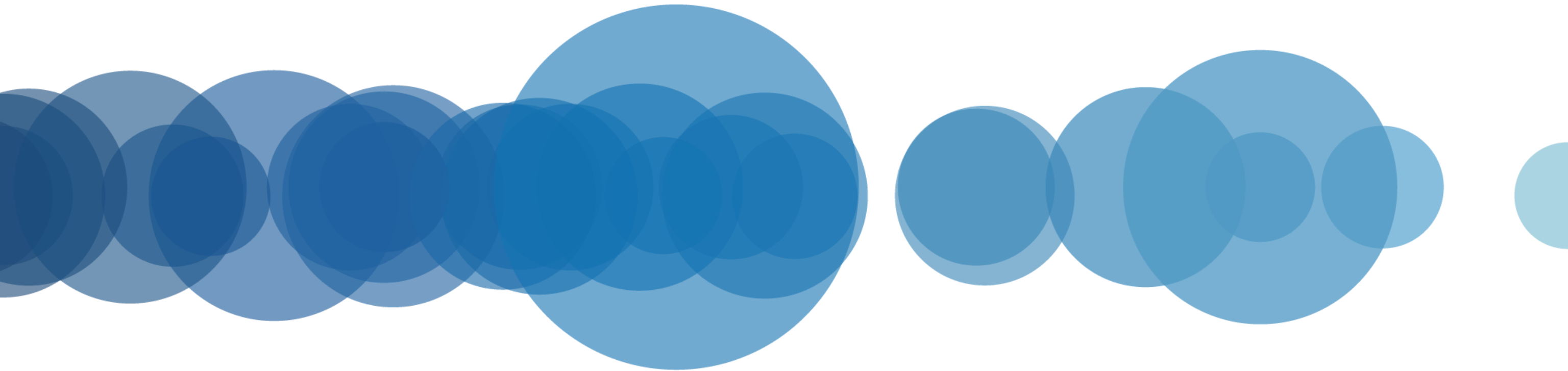


Compare and Contrast Real-World Examples



Real-World Example

Fox News



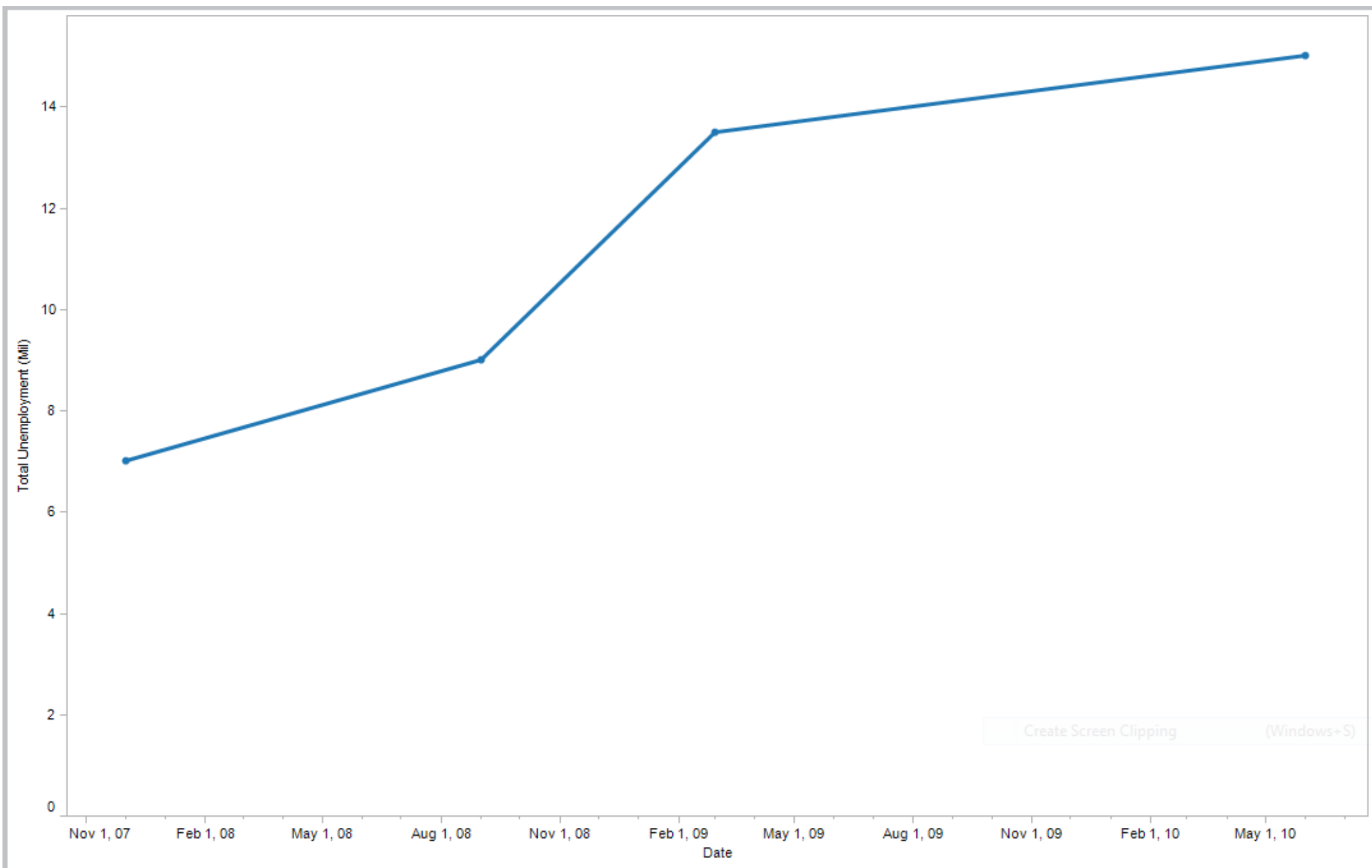
TOTAL UNEMPLOYED



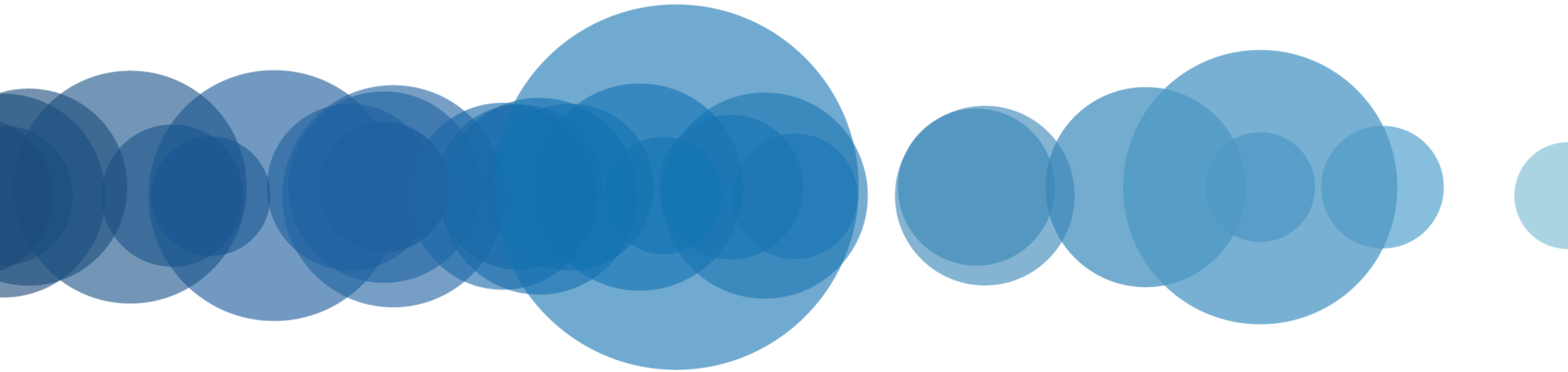
LIVE LIVE
FOX NEWS FOX NEWS
channel

SOURCE: BUREAU OF LABOR STATISTICS

THAT SURPASSES THE IXTOC I SPILL OFF MEXIC S&P FUT ▼ 3.00



Compare & Contrast



	Scenario #1	Scenario #2	Scenario #3
Sales	\$782,193	\$782,193	\$782,193
Royalty	173,965	173,965	173,965
Lease	1,089,496	1,089,496	1,089,496
Total Revenue	2,045,654	2,045,654	2,045,654
Management Fees	(590,550)	(590,550)	(514,285)
Direct Expenses:	(73,695)	(73,695)	(73,695)
SGA expenses	(3,750)	(3,750)	(3,750)
Interest Expenses	(105,650)	(105,650)	(105,650)
Total Expenses	(773,645)	(773,645)	(697,380)
Cash Available	1,272,009	1,272,009	1,348,274
Principal Payment	(2,273,950)	(1,125,718)	(1,125,717)
Net Cash	(\$1,001,941)	\$146,291	\$222,557

	Scenario #1	Scenario #2	Scenario #3
Sales	\$782,193	\$782,193	\$782,193
Royalty	173,965	173,965	173,965
Lease	1,089,496	1,089,496	1,089,496
Total Revenue	2,045,654	2,045,654	2,045,654
Management Fees	(590,550)	(590,550)	(514,285)
Direct Expenses:	(73,695)	(73,695)	(73,695)
SGA expenses	(3,750)	(3,750)	(3,750)
Interest Expenses	(105,650)	(105,650)	(105,650)
Total Expenses	(773,645)	(773,645)	(697,380)
Cash Available	1,272,009	1,272,009	1,348,274
Principal Payment	(2,273,950)	(1,125,718)	(1,125,717)
Net Cash	(\$1,001,941)	\$146,291	\$222,557

Flag of Mali

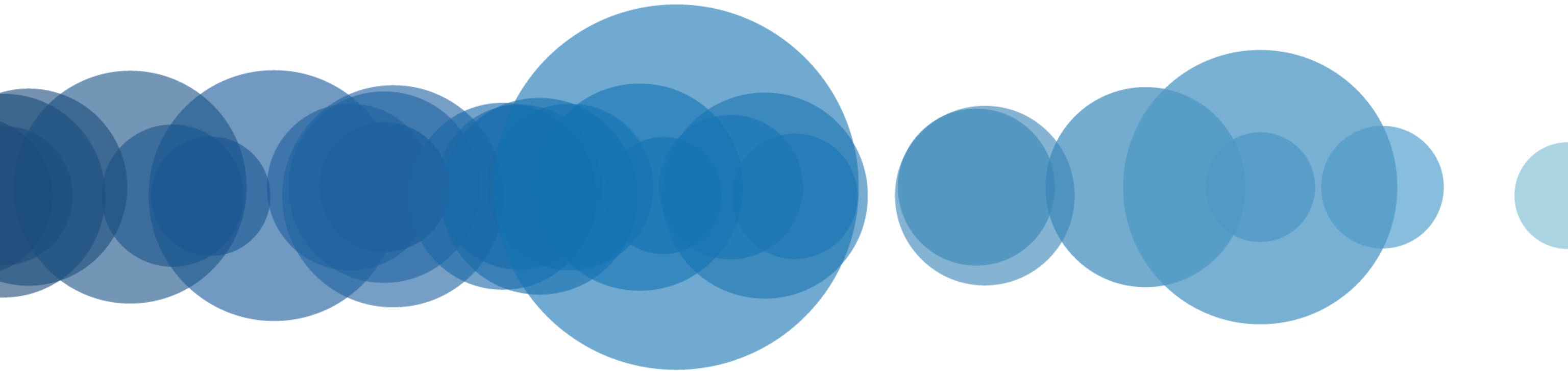


	Scenario #1	Scenario #2	Scenario #3
Sales	\$782,193	\$782,193	\$782,193
Royalty	173,965	173,965	173,965
Lease	1,089,496	1,089,496	1,089,496
Total Revenue	2,045,654	2,045,654	2,045,654
Management Fees	(590,550)	(590,550)	(514,285)
Direct Expenses:	(73,695)	(73,695)	(73,695)
SGA expenses	(3,750)	(3,750)	(3,750)
Interest Expenses	(105,650)	(105,650)	(105,650)
Total Expenses	(773,645)	(773,645)	(697,380)
Cash Available	1,272,009	1,272,009	1,348,274
Principal Payment	(2,273,950)	(1,125,718)	(1,125,717)
Net Cash	(\$1,001,941)	\$146,291	\$222,557

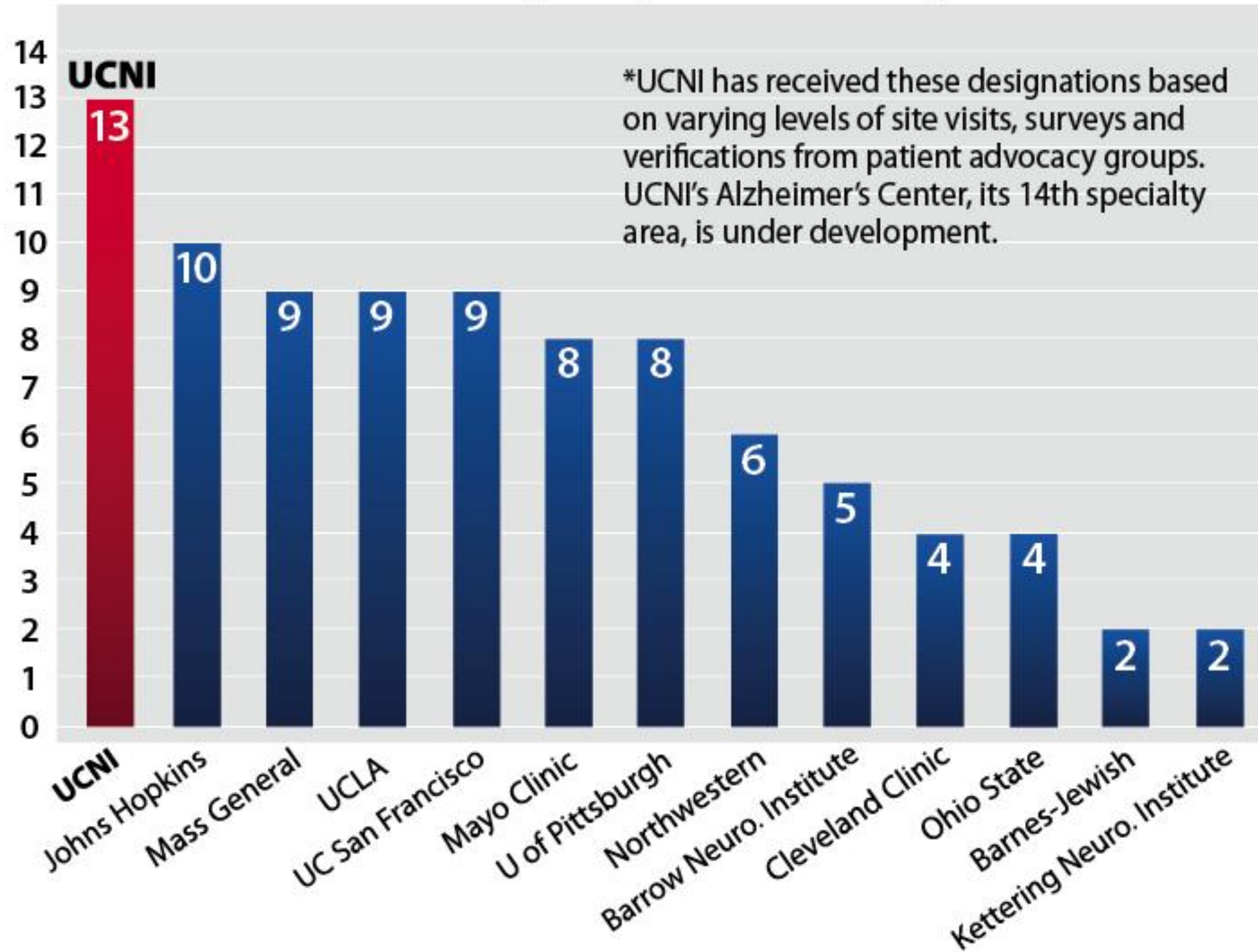
	Scenario #1	Scenario #2	Scenario #3
Sales	\$782,193	\$782,193	\$782,193
Royalty	173,965	173,965	173,965
Lease	1,089,496	1,089,496	1,089,496
Total Revenue	2,045,654	2,045,654	2,045,654
Management Fees	(590,550)	(590,550)	(514,285)
Direct Expenses:	(73,695)	(73,695)	(73,695)
SGA expenses	(3,750)	(3,750)	(3,750)
Interest Expenses	(105,650)	(105,650)	(105,650)
Total Expenses	(773,645)	(773,645)	(697,380)
Cash Available	1,272,009	1,272,009	1,348,274
Principal Payment	(2,273,950)	(1,125,718)	(1,125,717)
Net Cash	(\$1,001,941)	\$146,291	\$222,557

Real-World Example

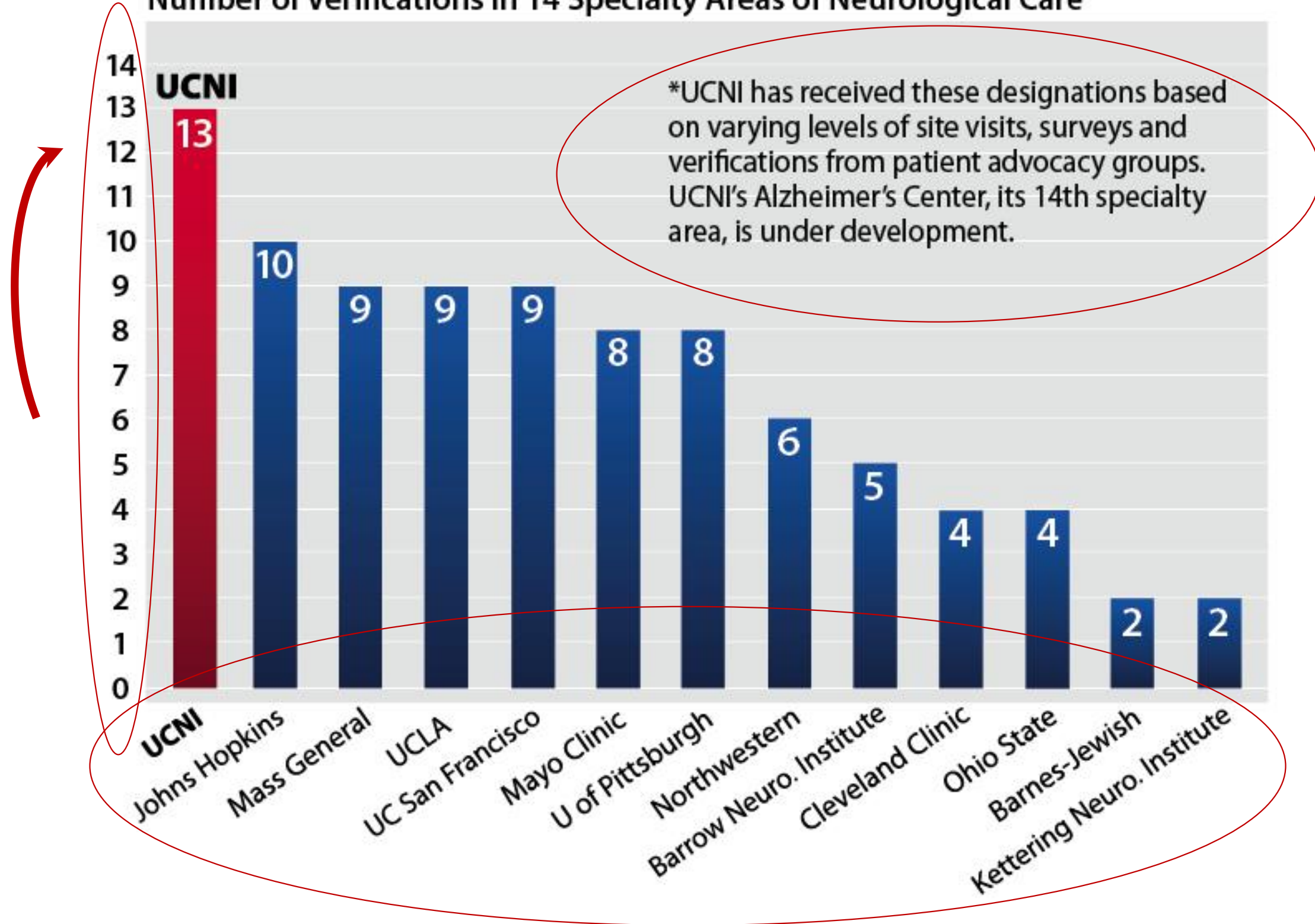
Highlight Table



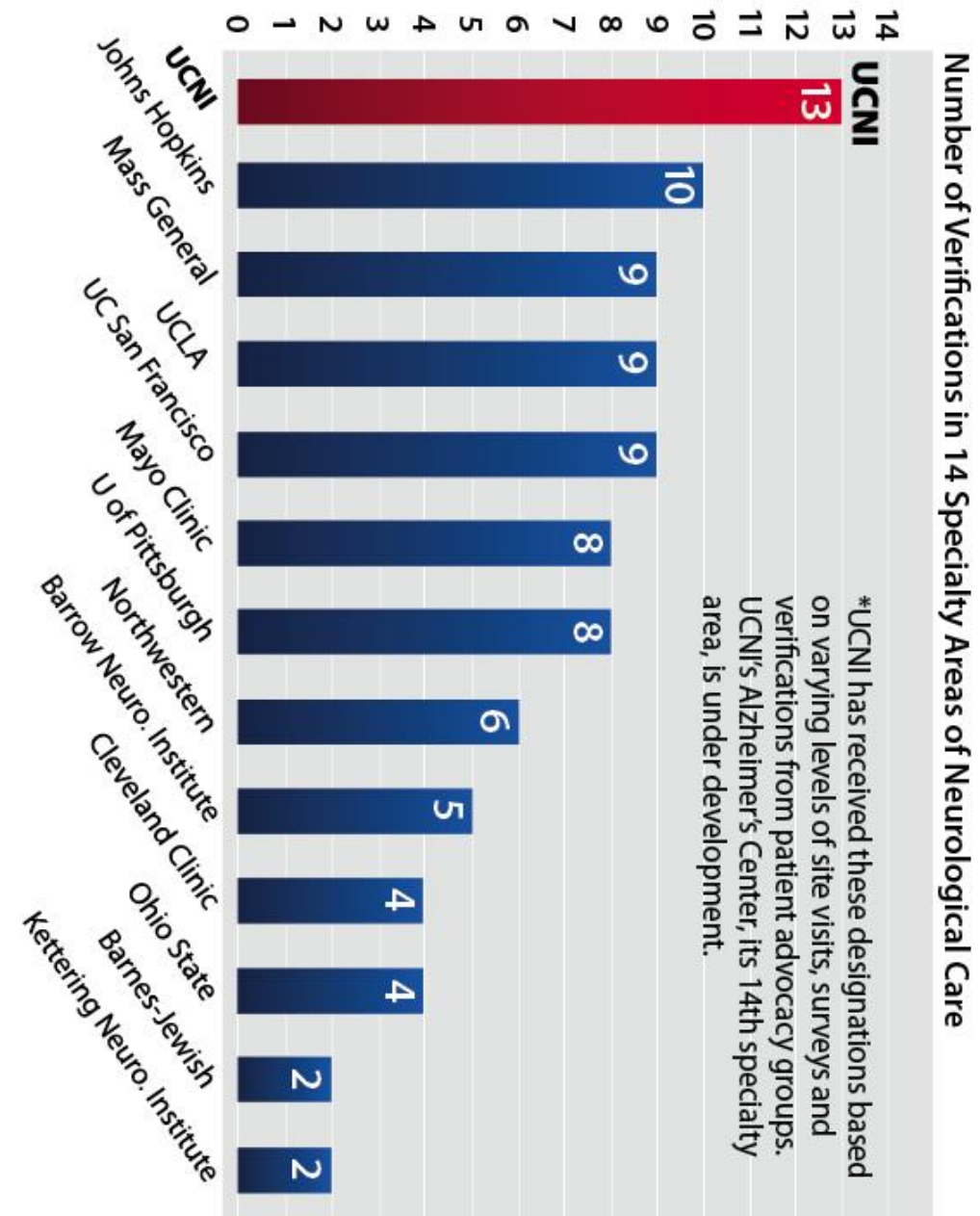
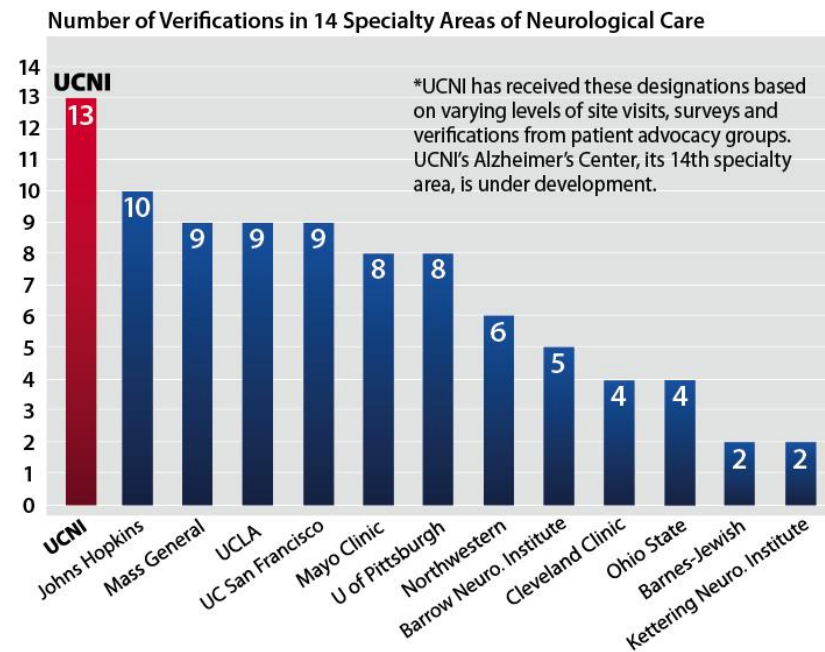
Number of Verifications in 14 Specialty Areas of Neurological Care



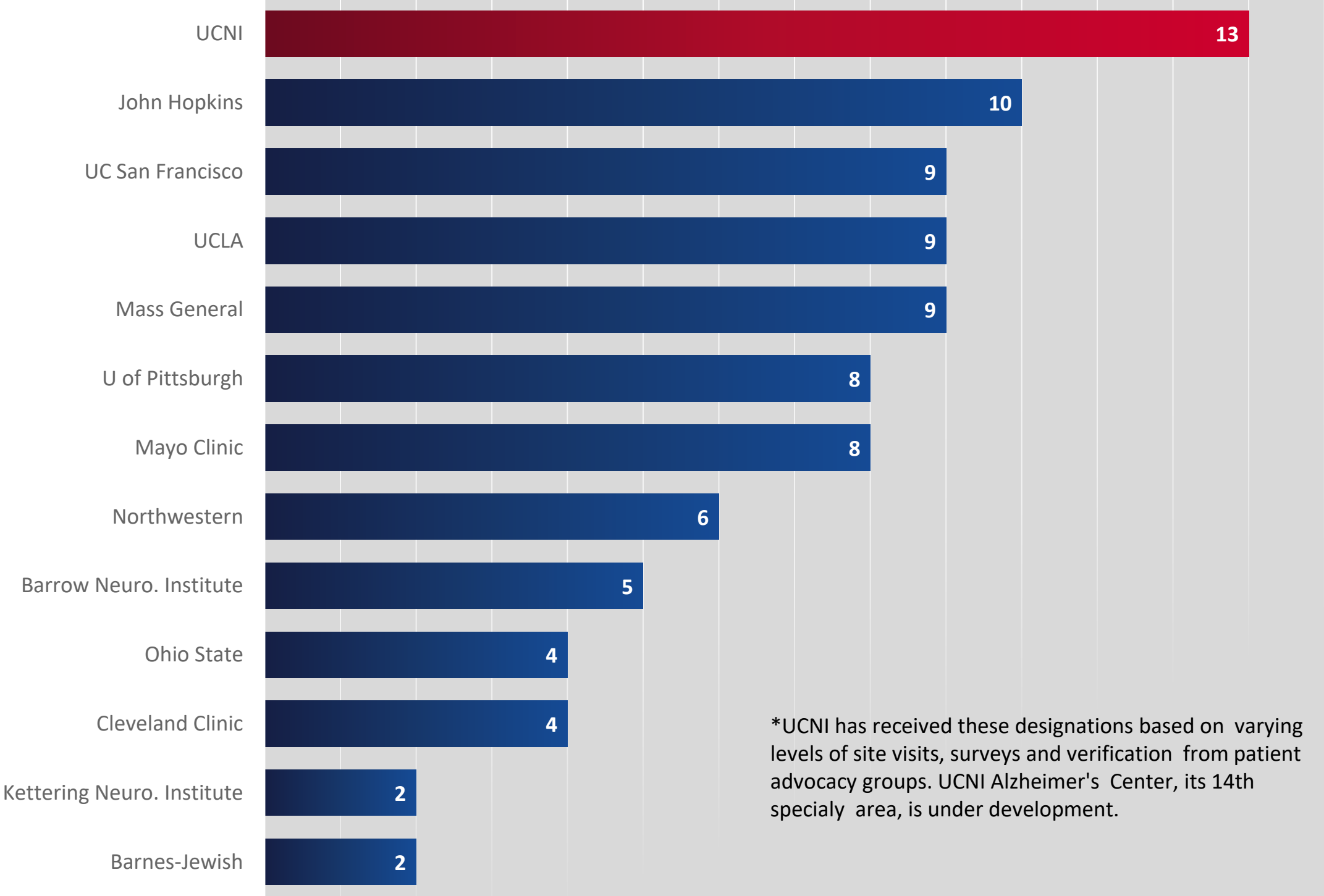
Number of Verifications in 14 Specialty Areas of Neurological Care



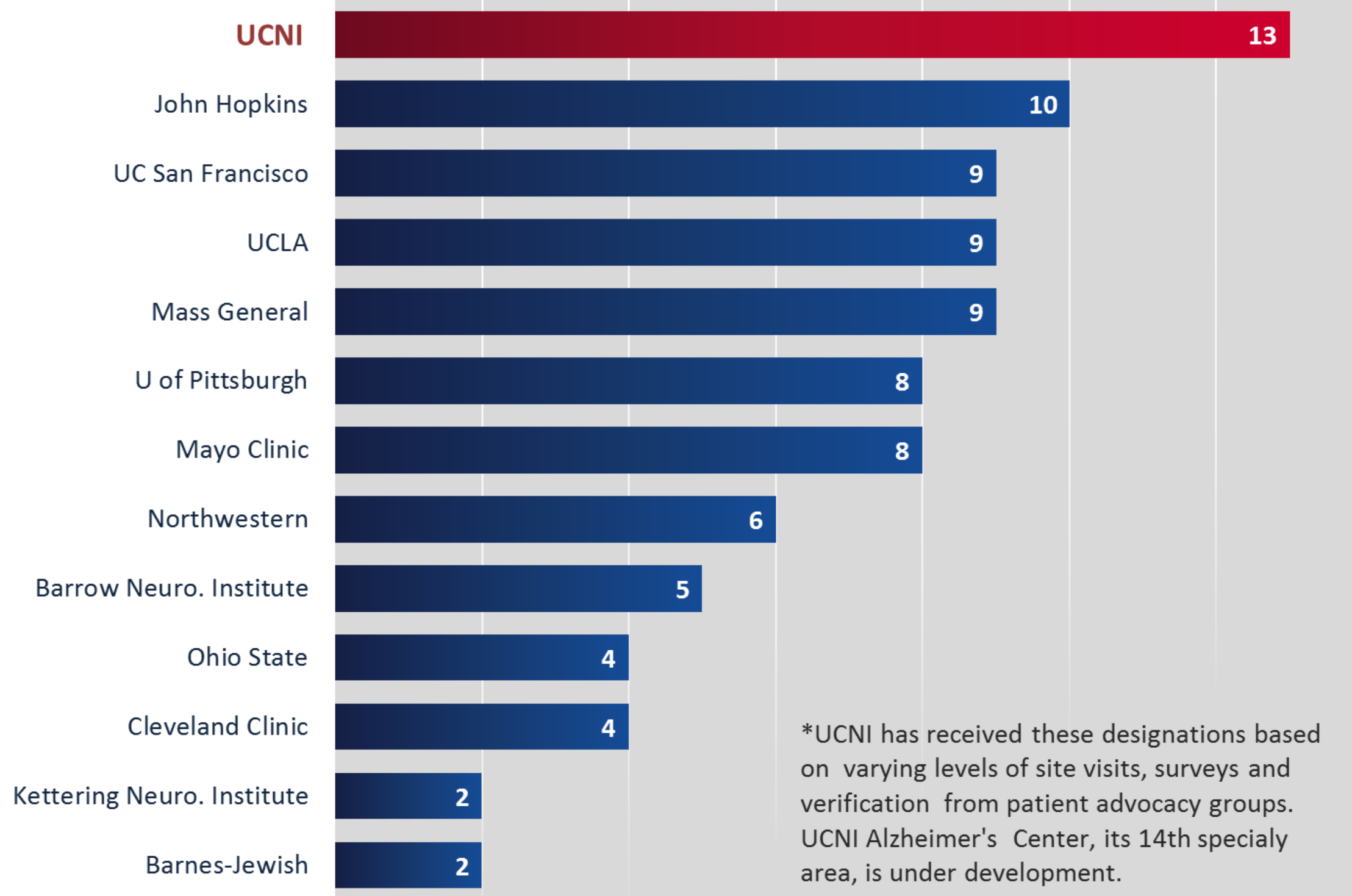
1. Make horizontal
2. Horizontal text
3. Remove Axis labels
4. Fix gridlines on top of paragraph
5. Used ColorPic



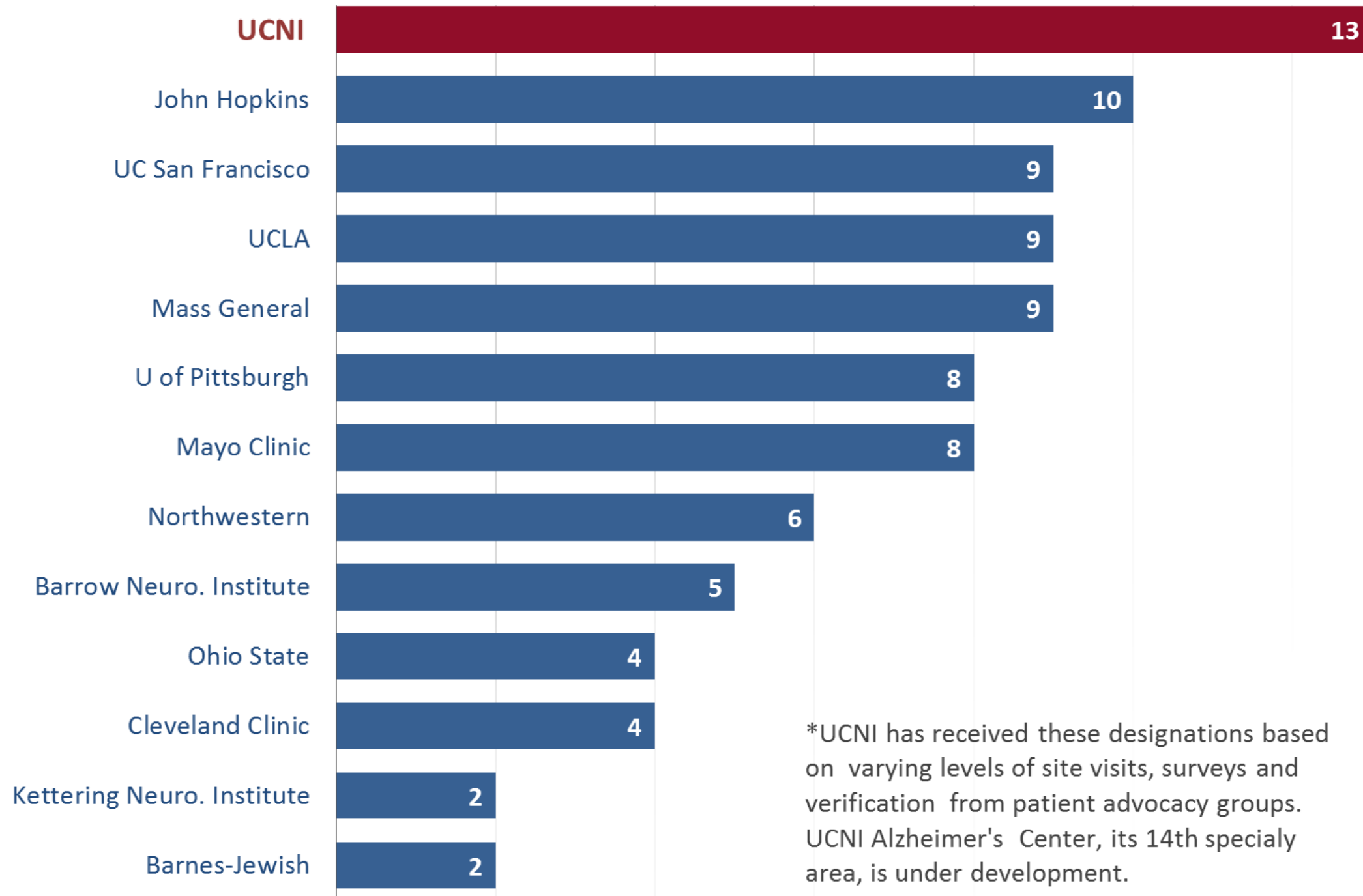
Number of Verifications in 14 Specialty Areas of Neurological Care



Number of Verifications in 14 Specialty Areas of Neurological Care

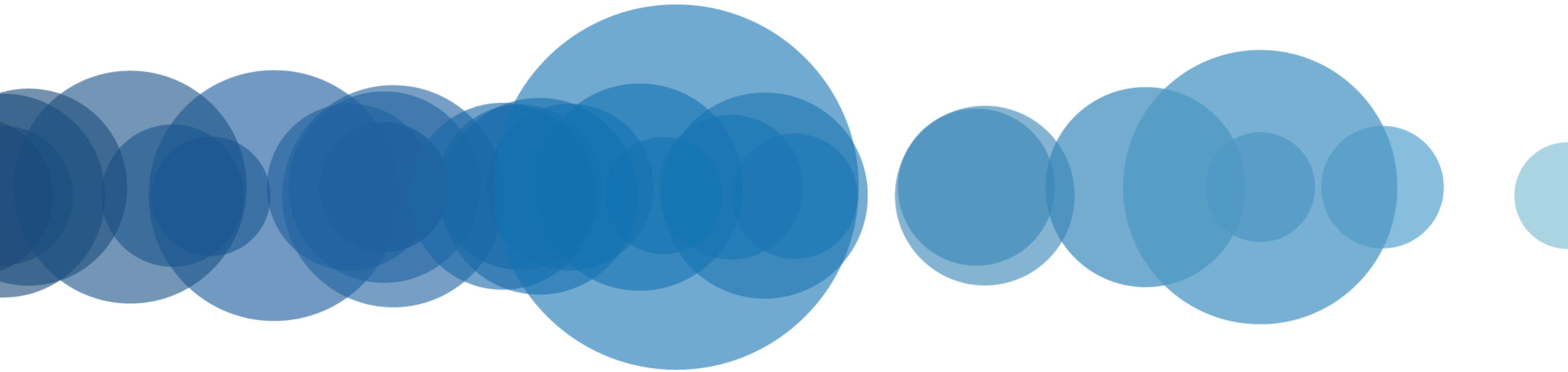


Number of Verifications in 14 Specialty Areas of Neurological Care

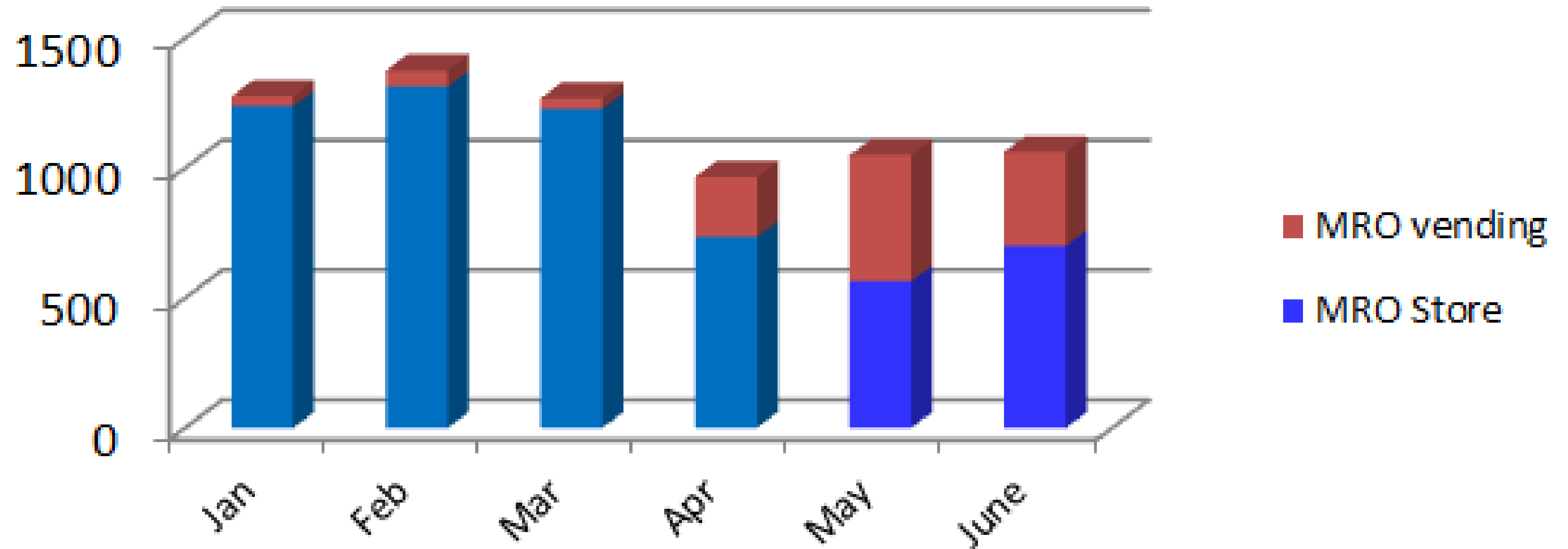


Real-World Example

Sales Comparison

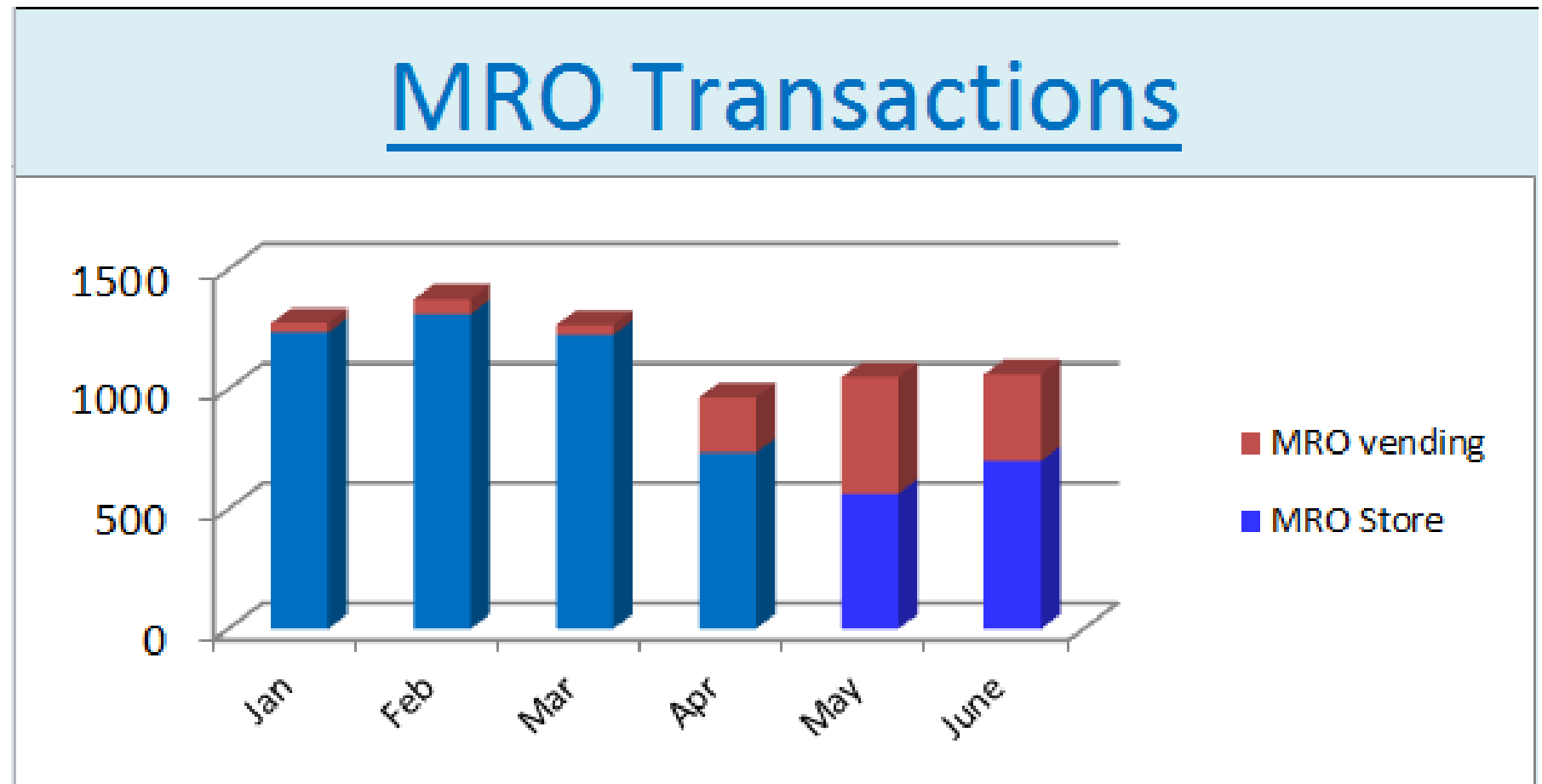


MRO Transactions

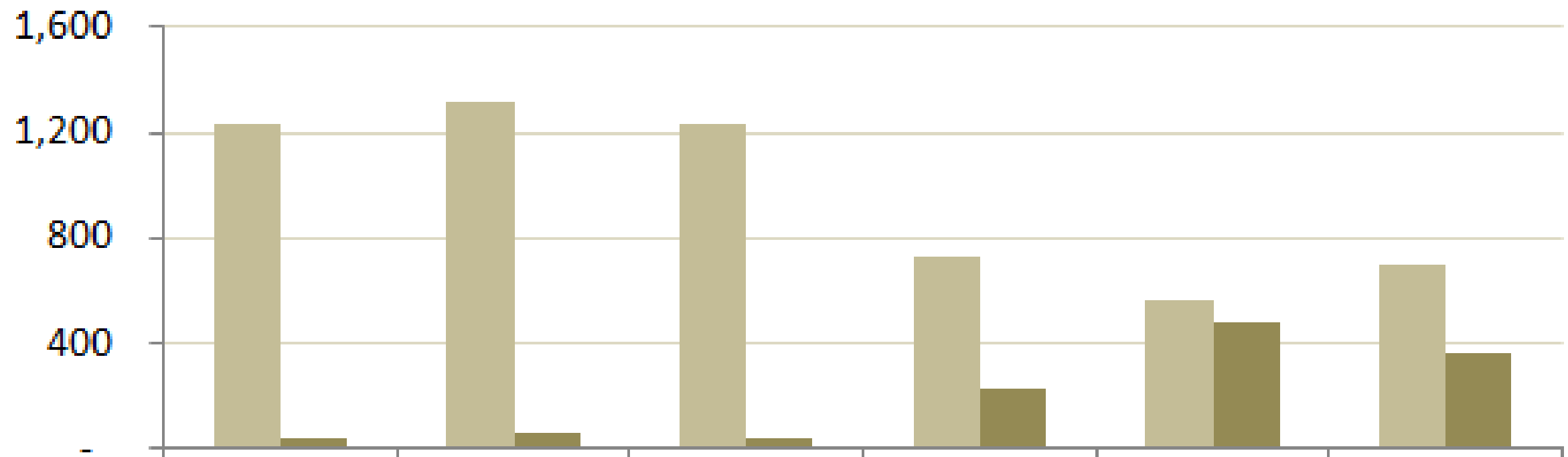


Design Issues

- Title Format
- Three Dimensional
- Axis Labels on Angle
- Dark Gridlines
- No Data
- Can't easily compare
- Color Choice
- Legend Space



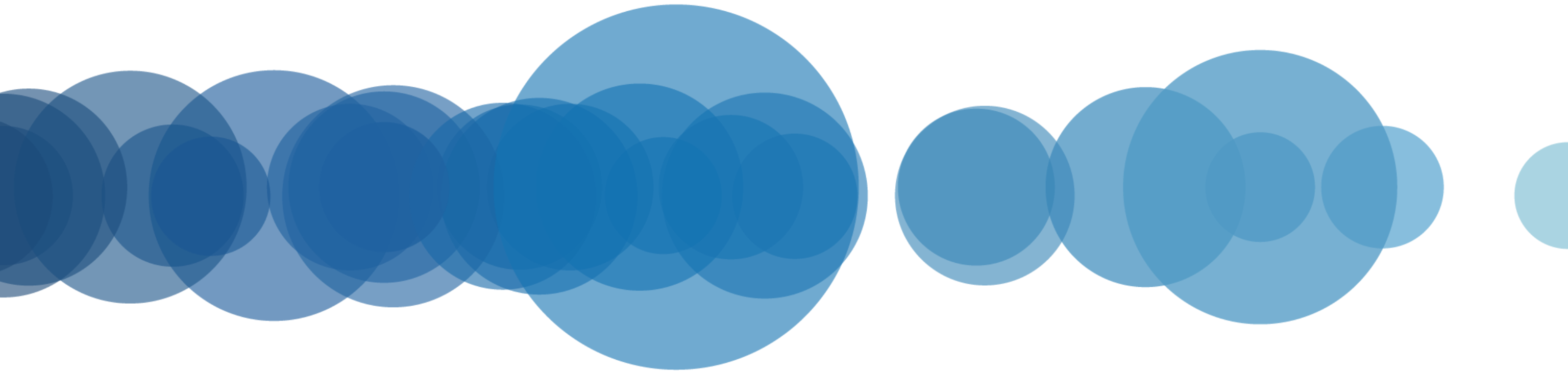
MRO Transactions



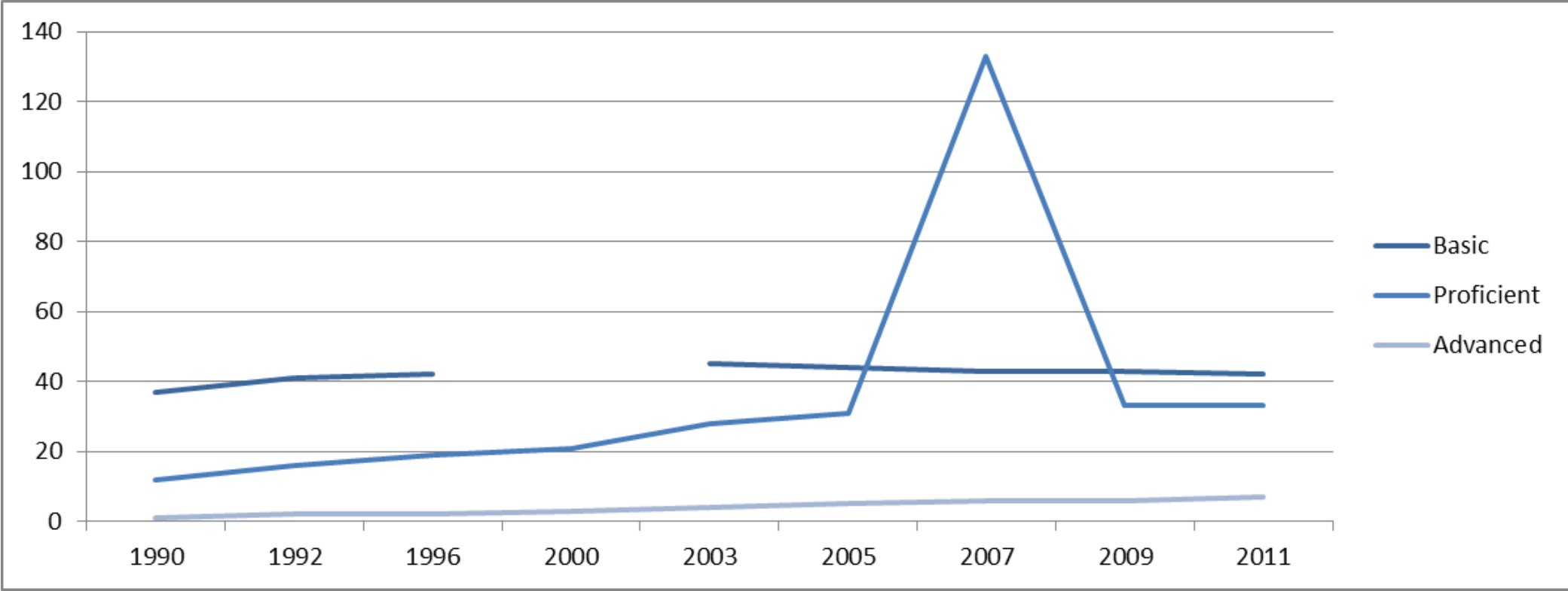
	Jan	Feb	Mar	Apr	May	June
MRO Store	1,237	1,313	1,227	734	564	700
MRO vending	39	63	41	231	485	362

Real-World Example

Technology Proficiency

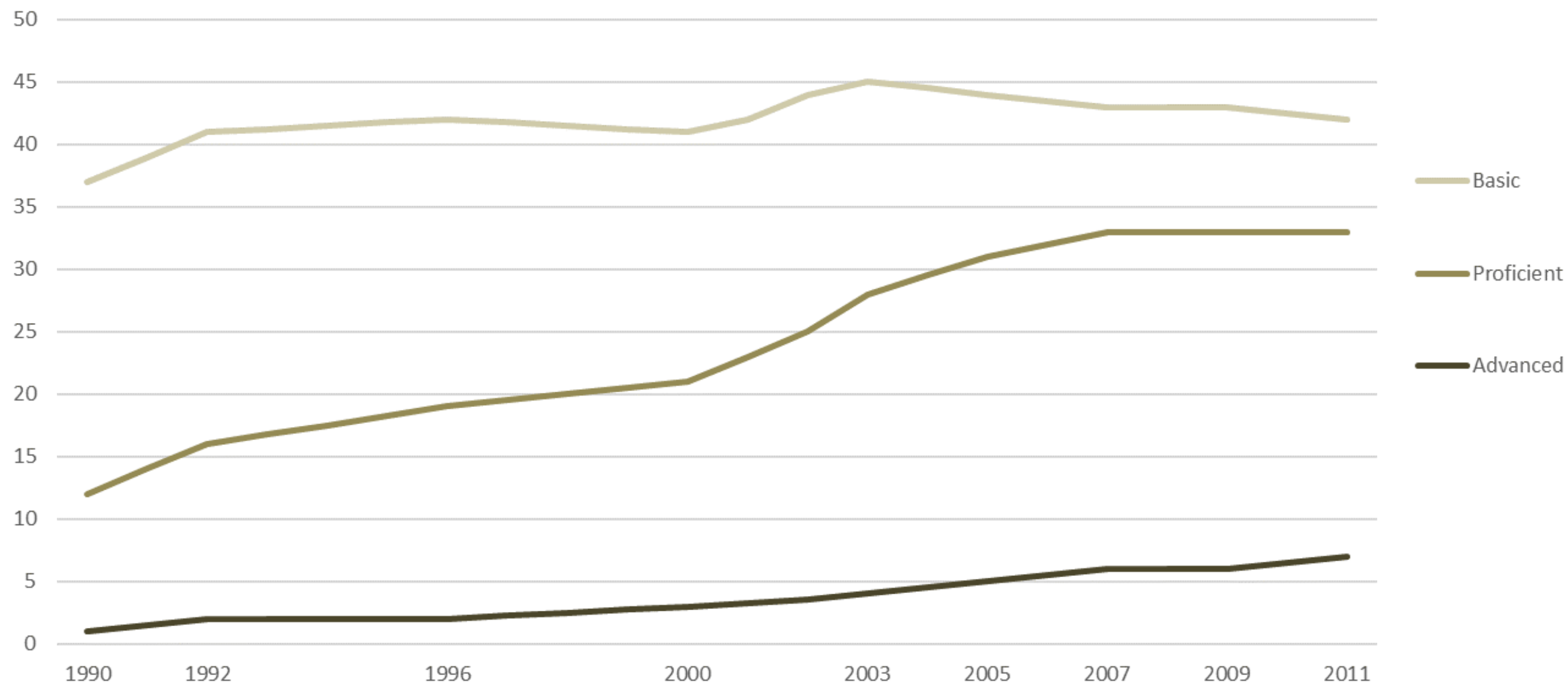


Technology Proficiency Over Time

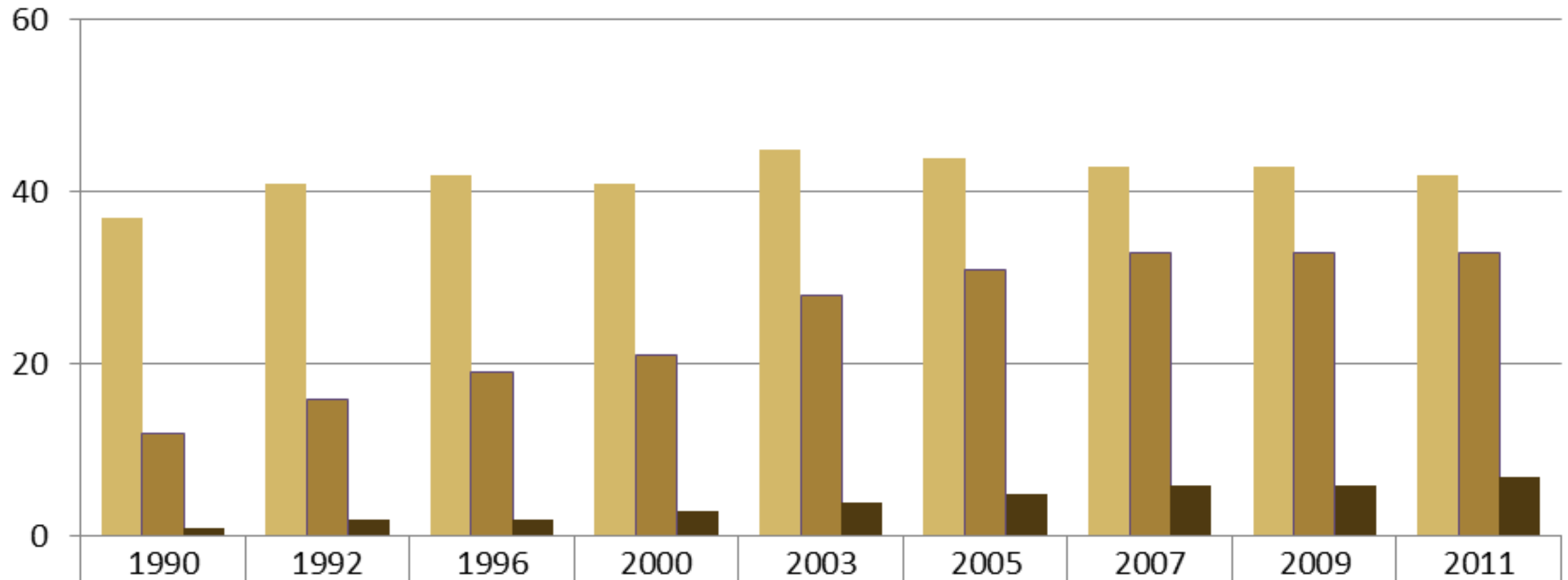


	Basic	Proficient	Advanced
1990	37	12	1
1992	41	16	2
1996	42	19	2
2000		21	3
2003	45	28	4
2005	44	31	5
2007	43	133	6
2009	43	33	6
2011	42	33	7

Technology Proficiency Over Time

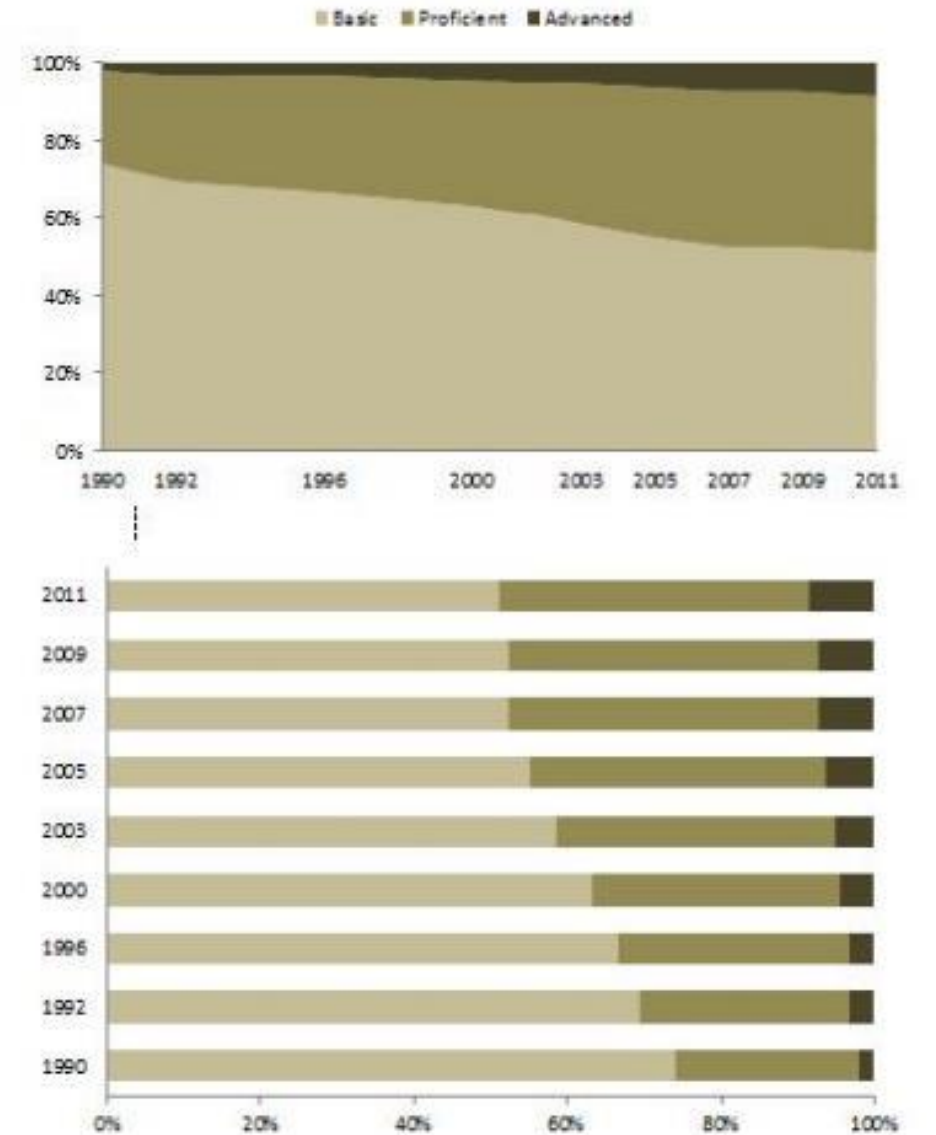
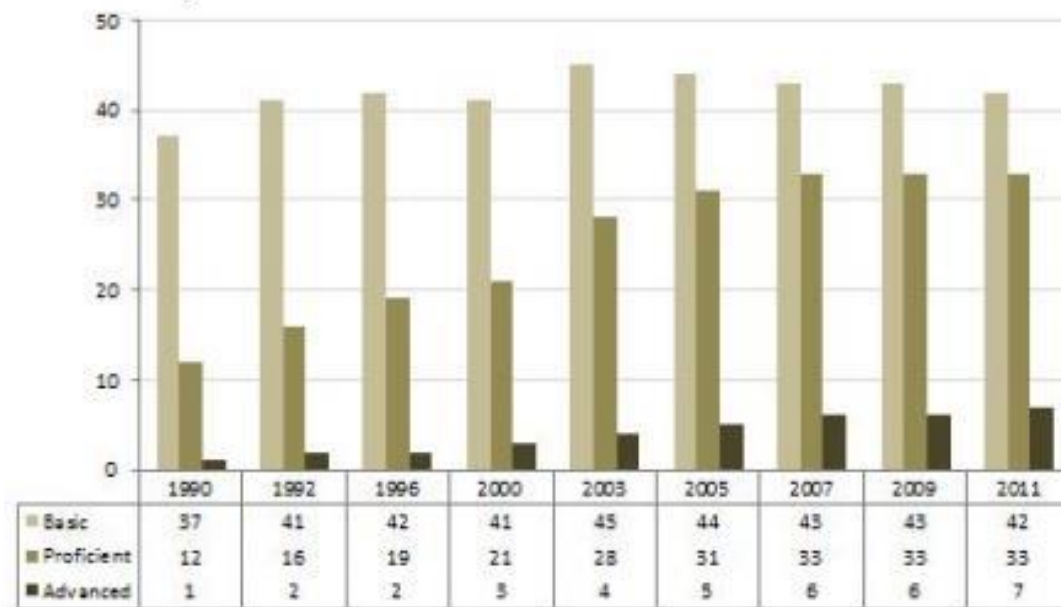
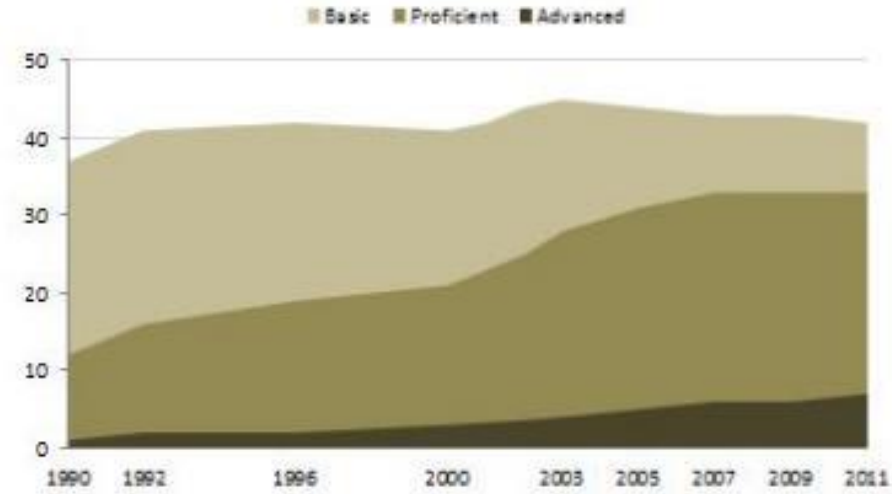
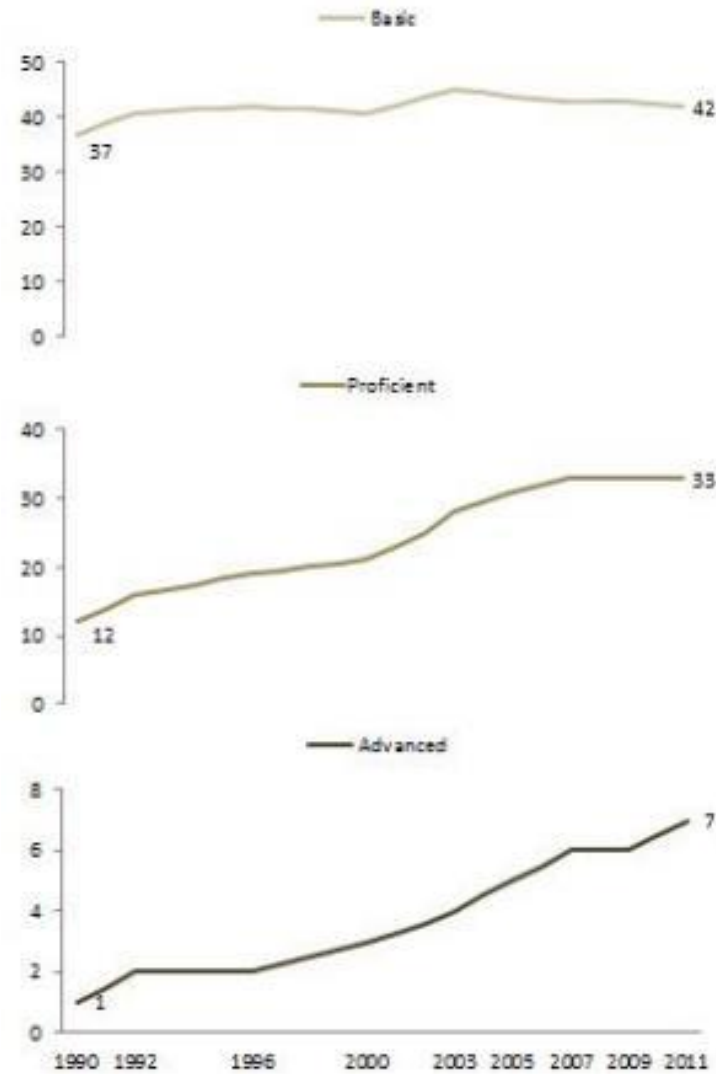


Technology Proficiency Over Time



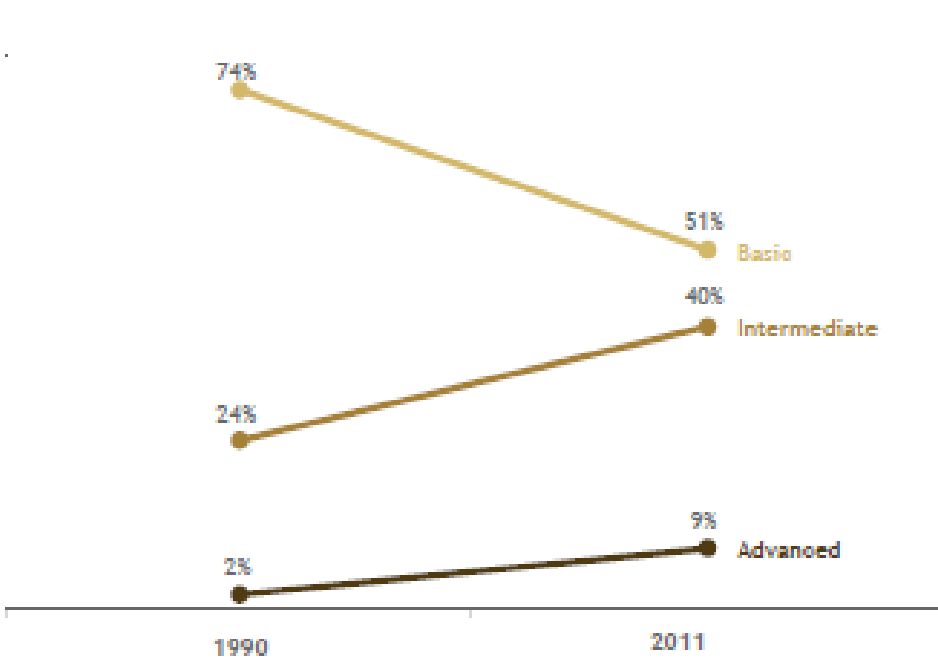
Basic	37	41	42	41	45	44	43	43	42
Intermediate	12	16	19	21	28	31	33	33	33
Advanced	1	2	2	3	4	5	6	6	7

Technology Proficiency Over Time

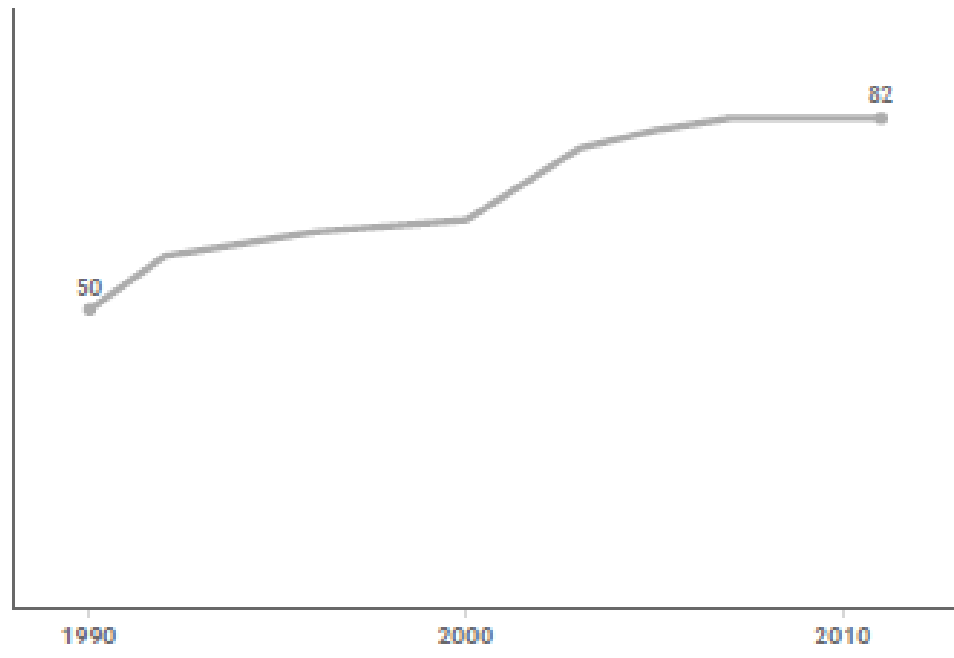


Overall Company Proficiency: Technology

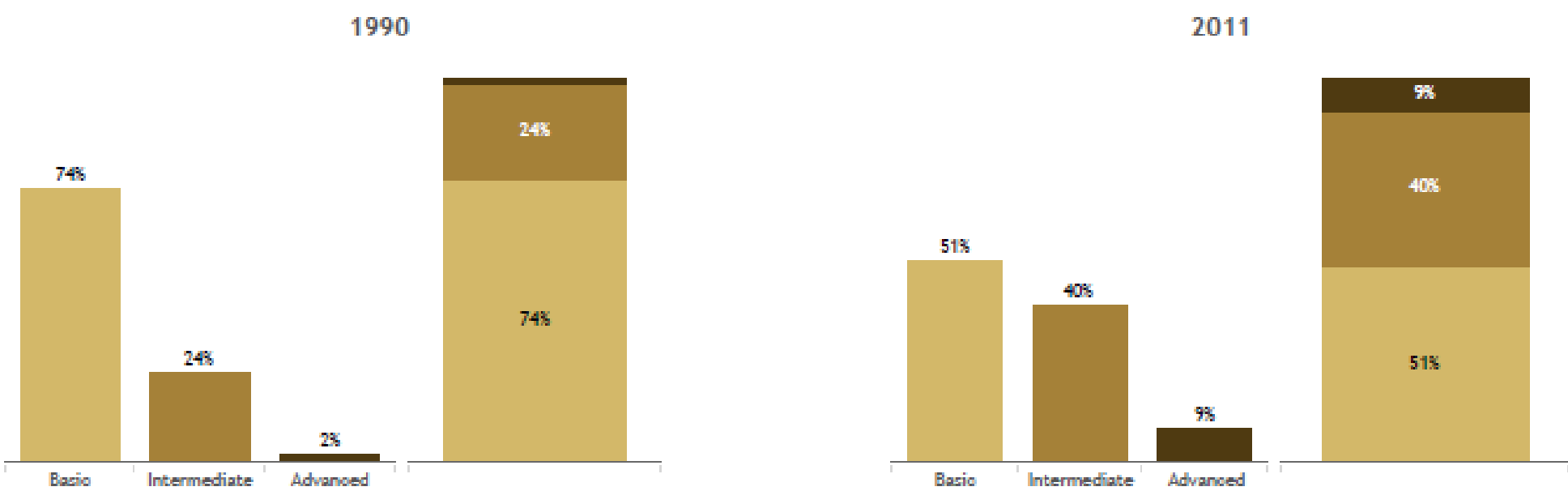
% of Employees by Proficiency Level



Total # of Employees



% of Employees by Proficiency Level



Overall Company Proficiency: Technology

From 1990

3 out of 4
employees have
Basic Proficiency



Only 24%
of employees are
Intermediate



Only 1
employee in the
entire company with
Advanced Proficiency



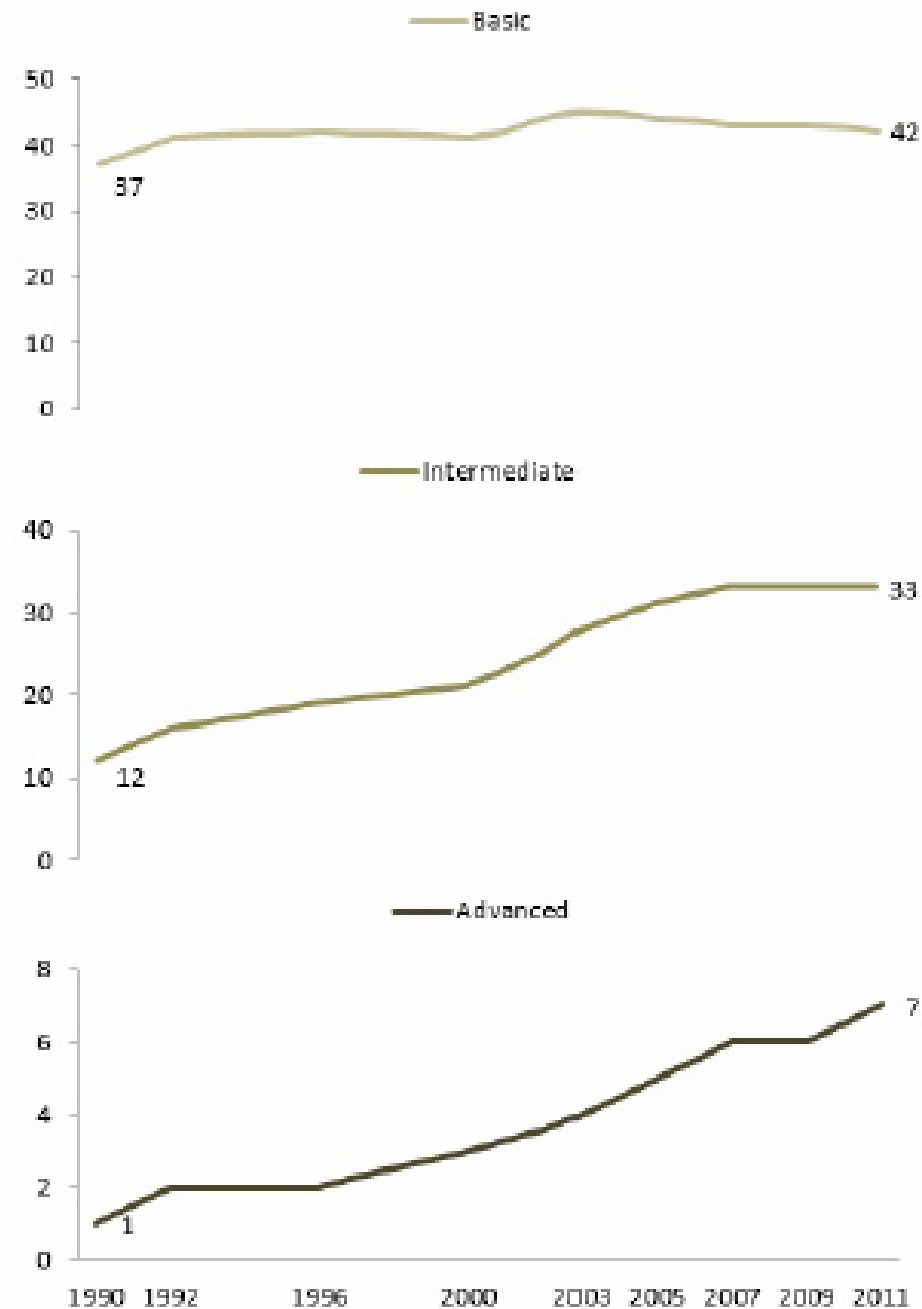
to 2011

The company has grown 64%
adding 32 employees
but only increasing 5 with
Basic Proficiency

49%
of employees are
Intermediate or
Advanced

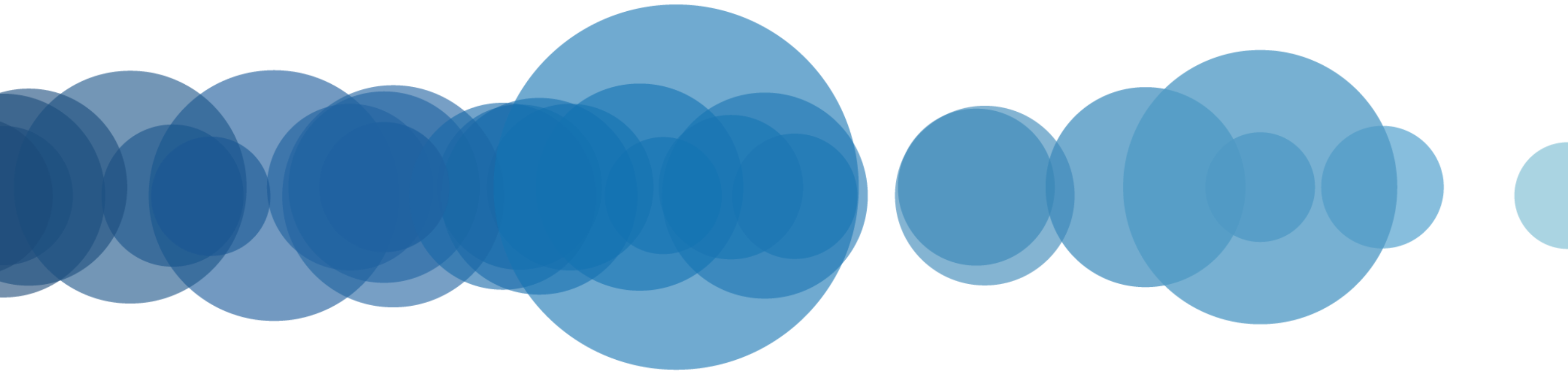


1 out of 10
employees are
Advanced Proficiency

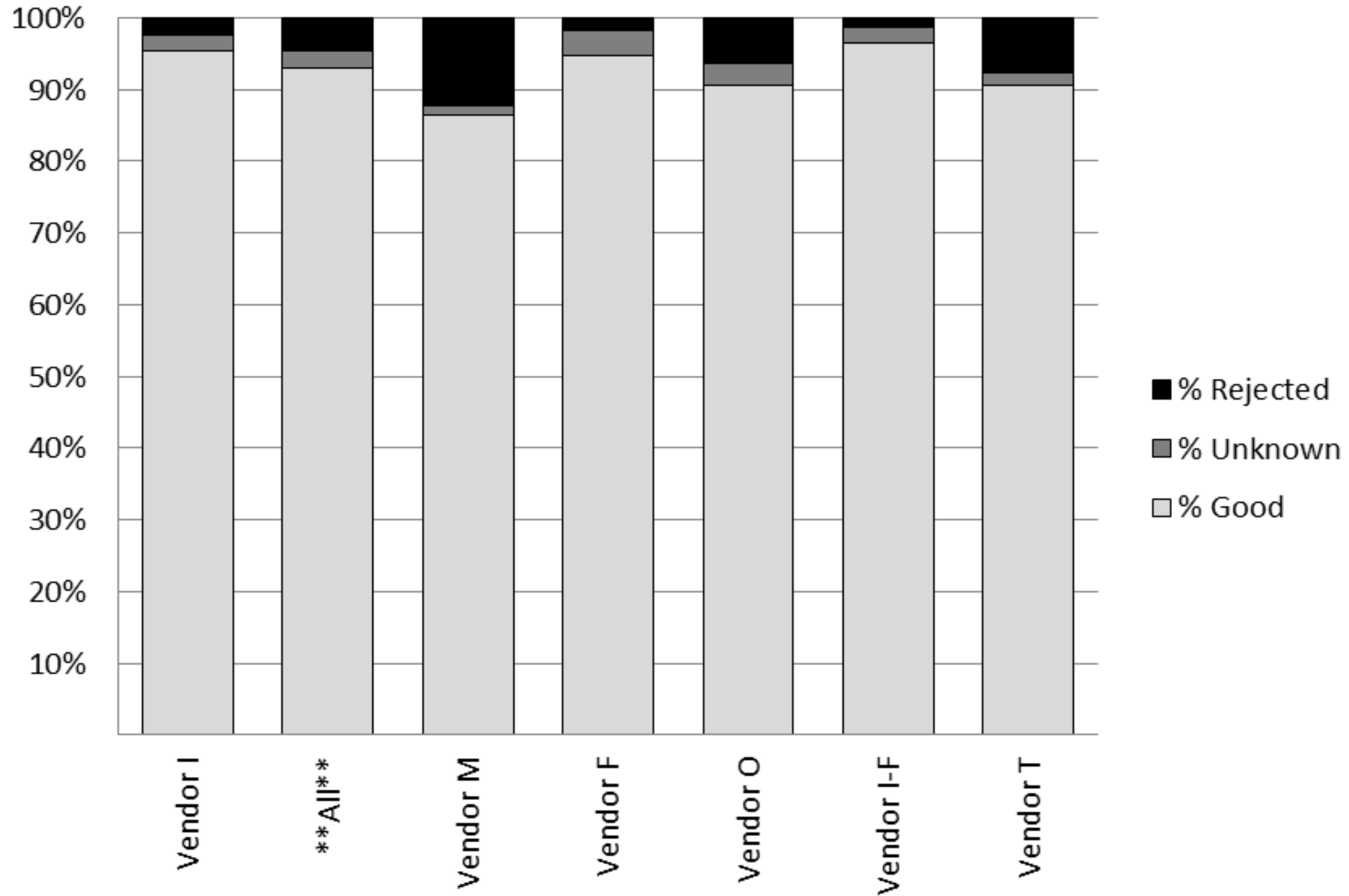


Real-World Example

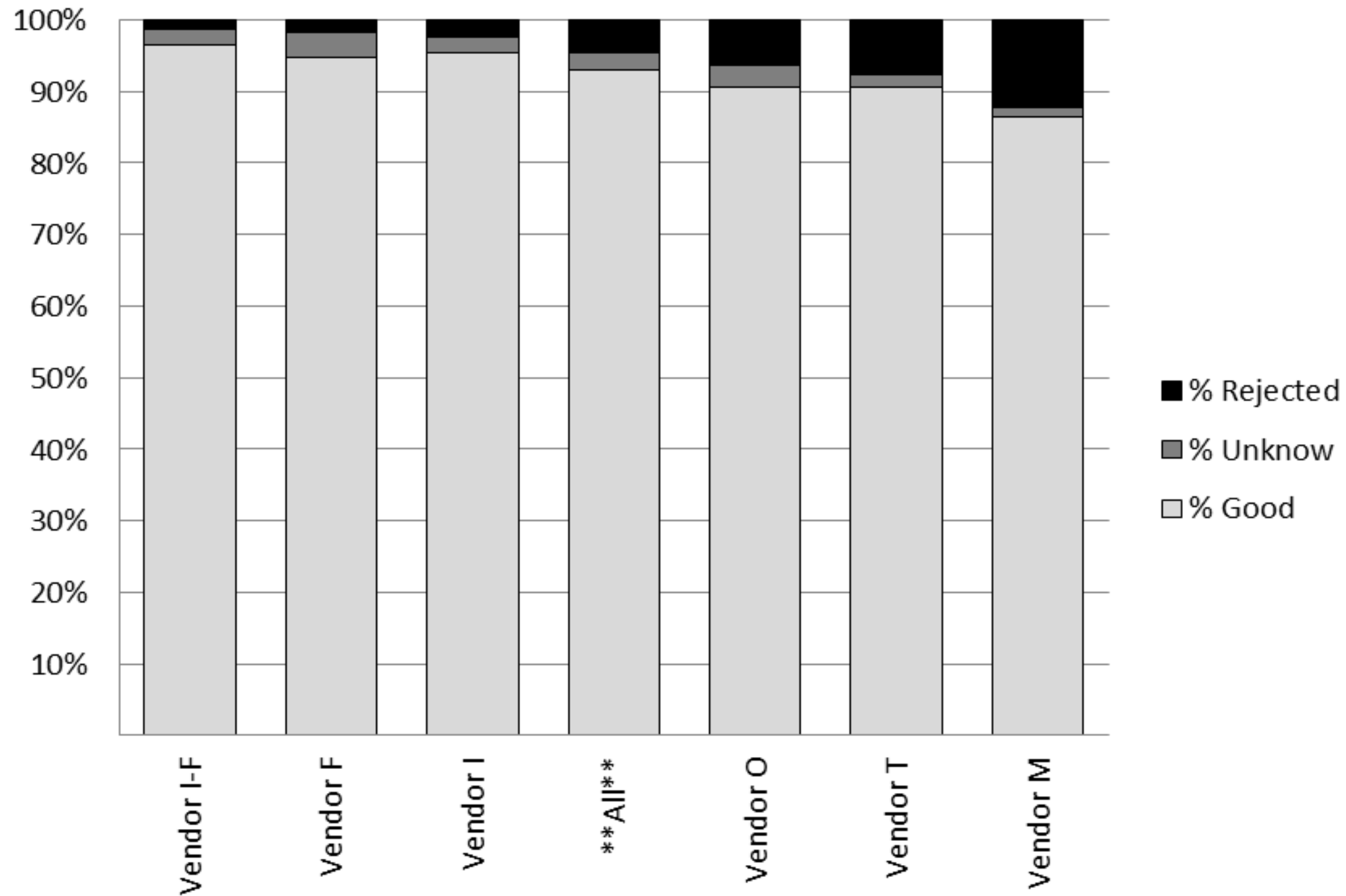
Comparing Vendors



Outcome Breakdown

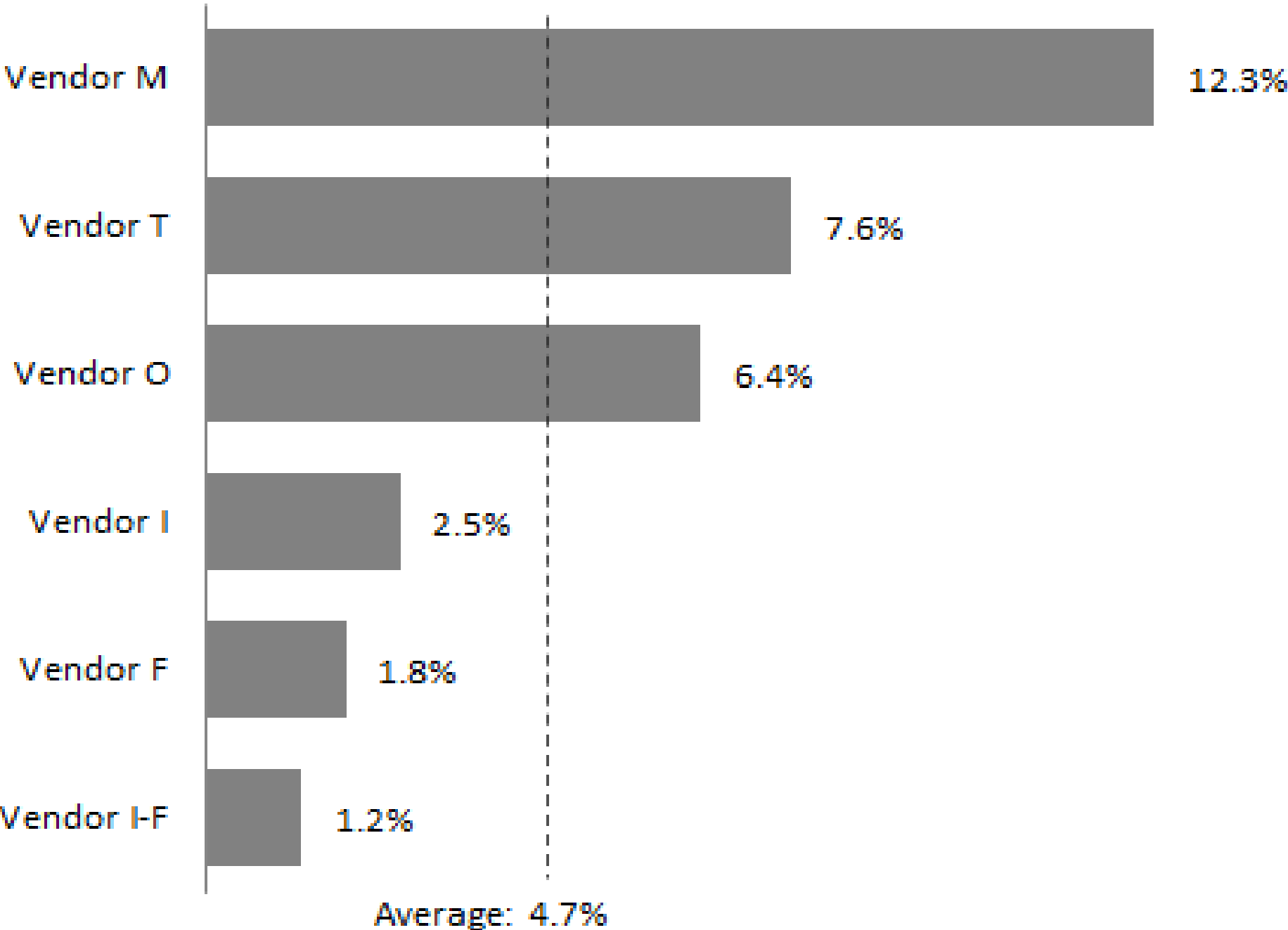


Outcome Breakdown



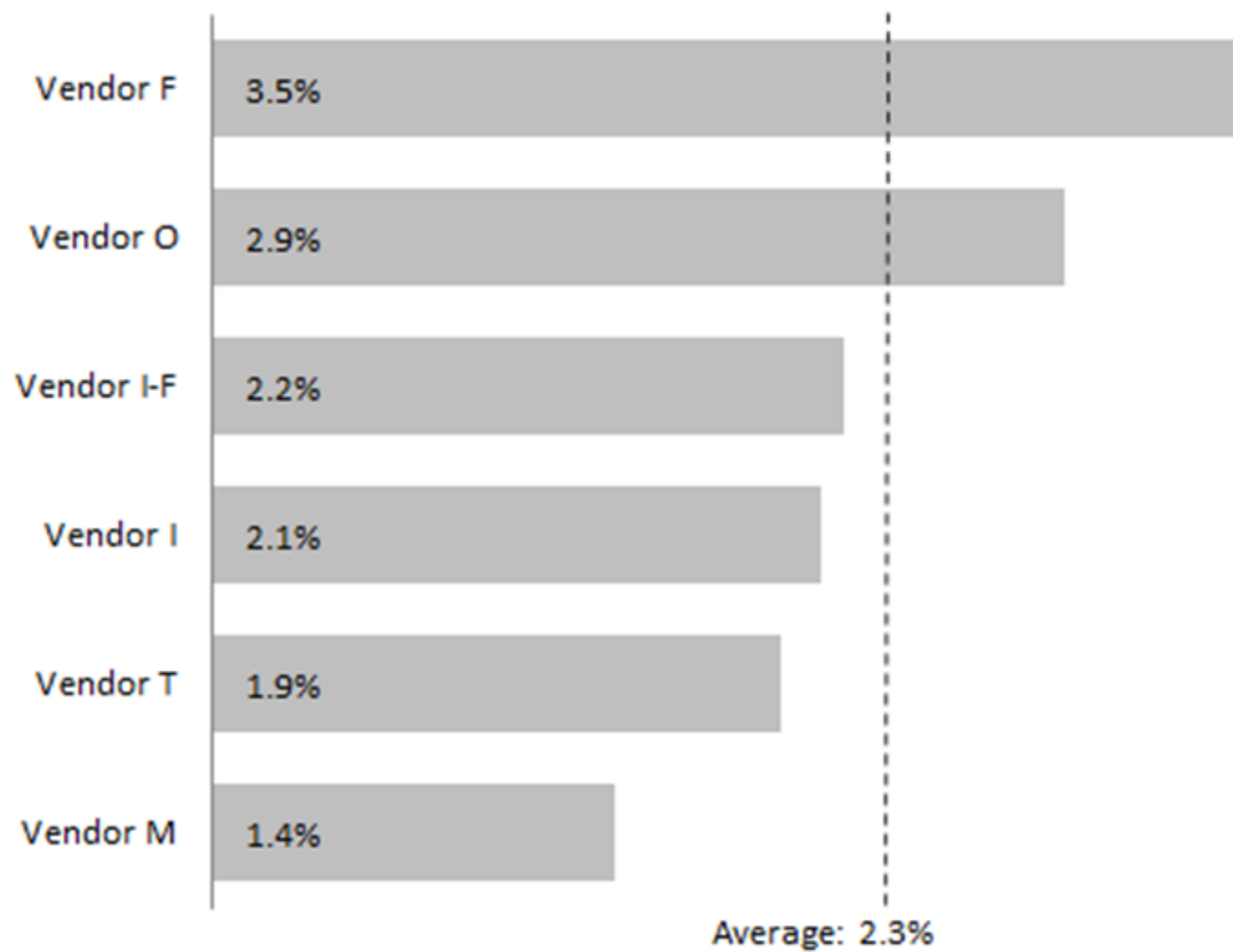
Outcome Breakdown

■ % Rejected



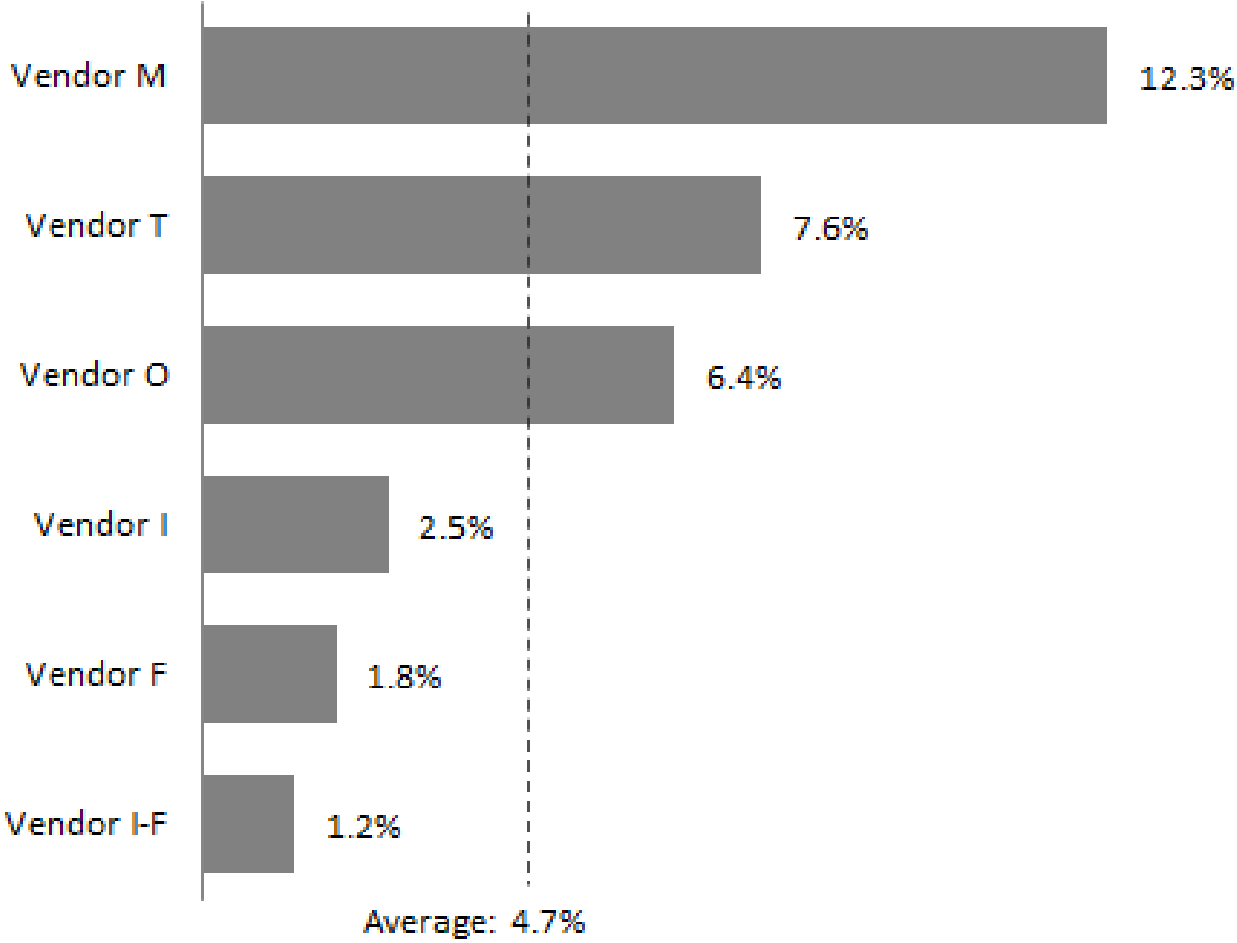
Outcome Breakdown

■ % Unknown



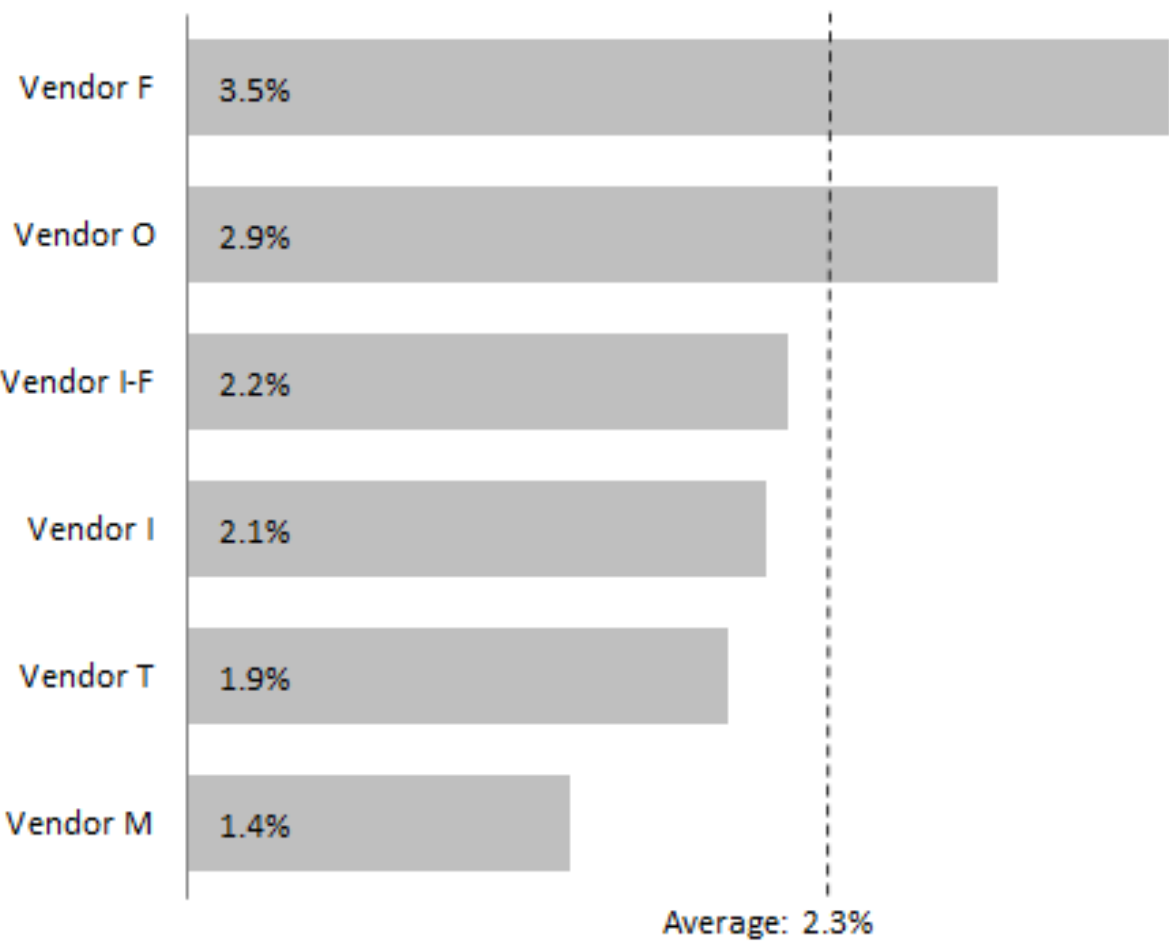
Outcome Breakdown

■ % Rejected



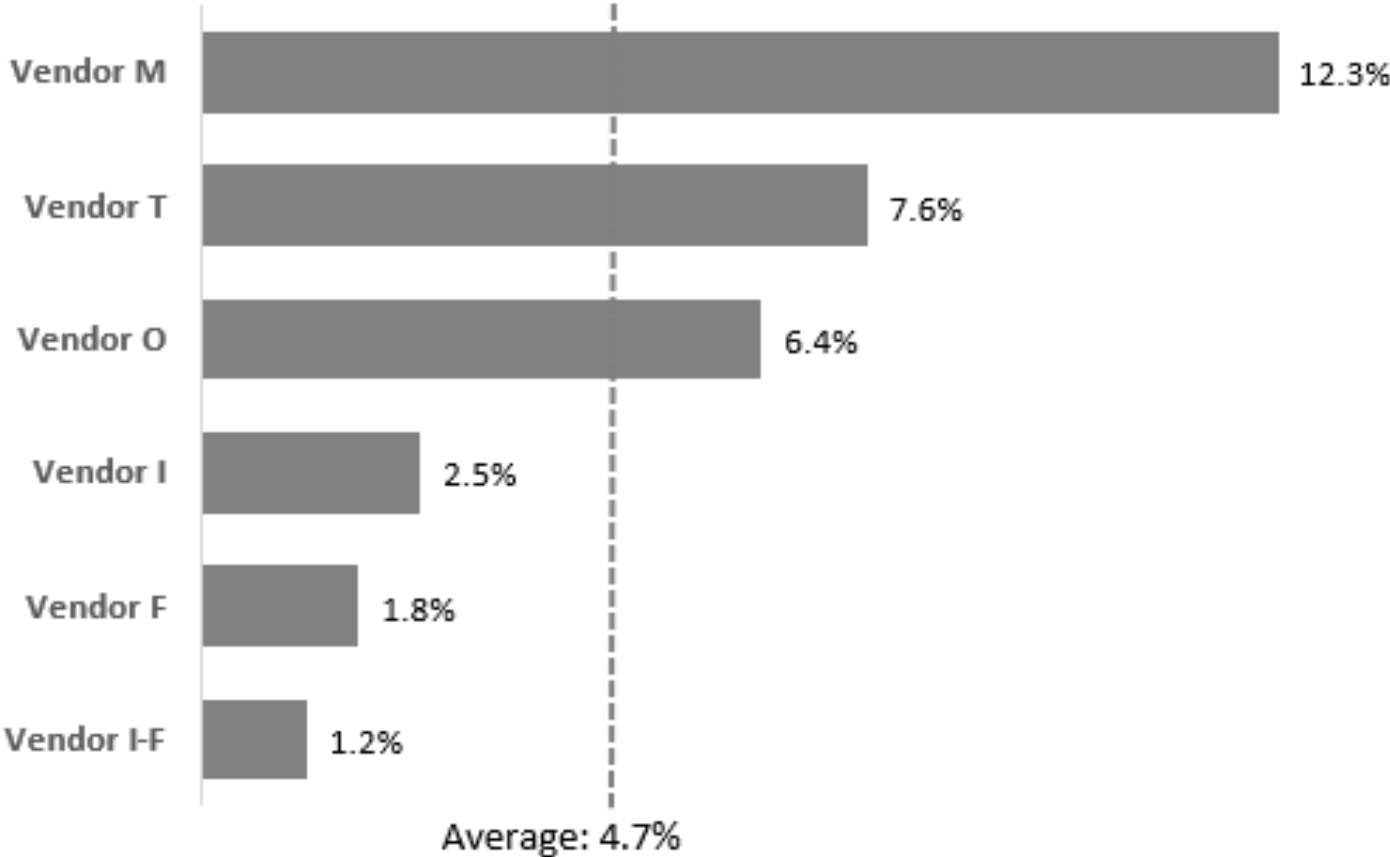
Outcome Breakdown

■ % Unknown



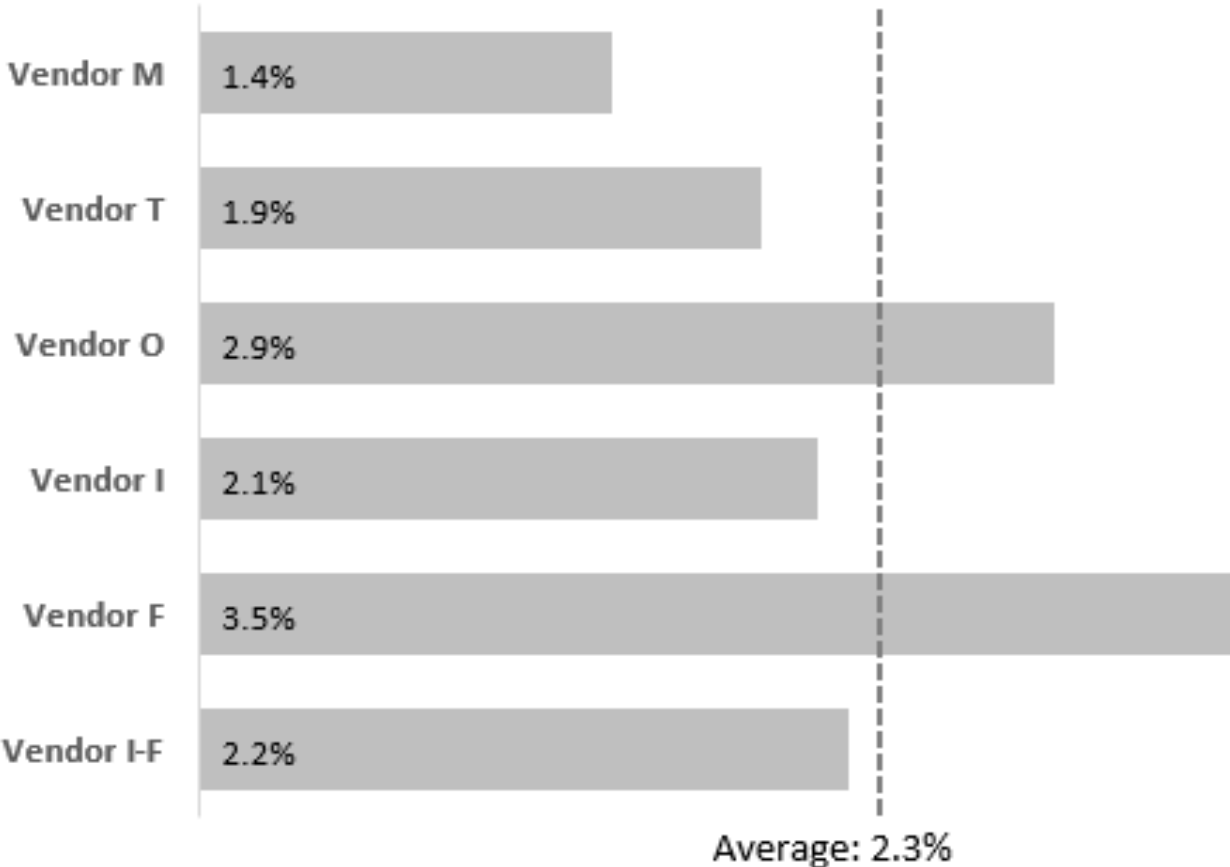
Outcome Breakdown

■ Rejected



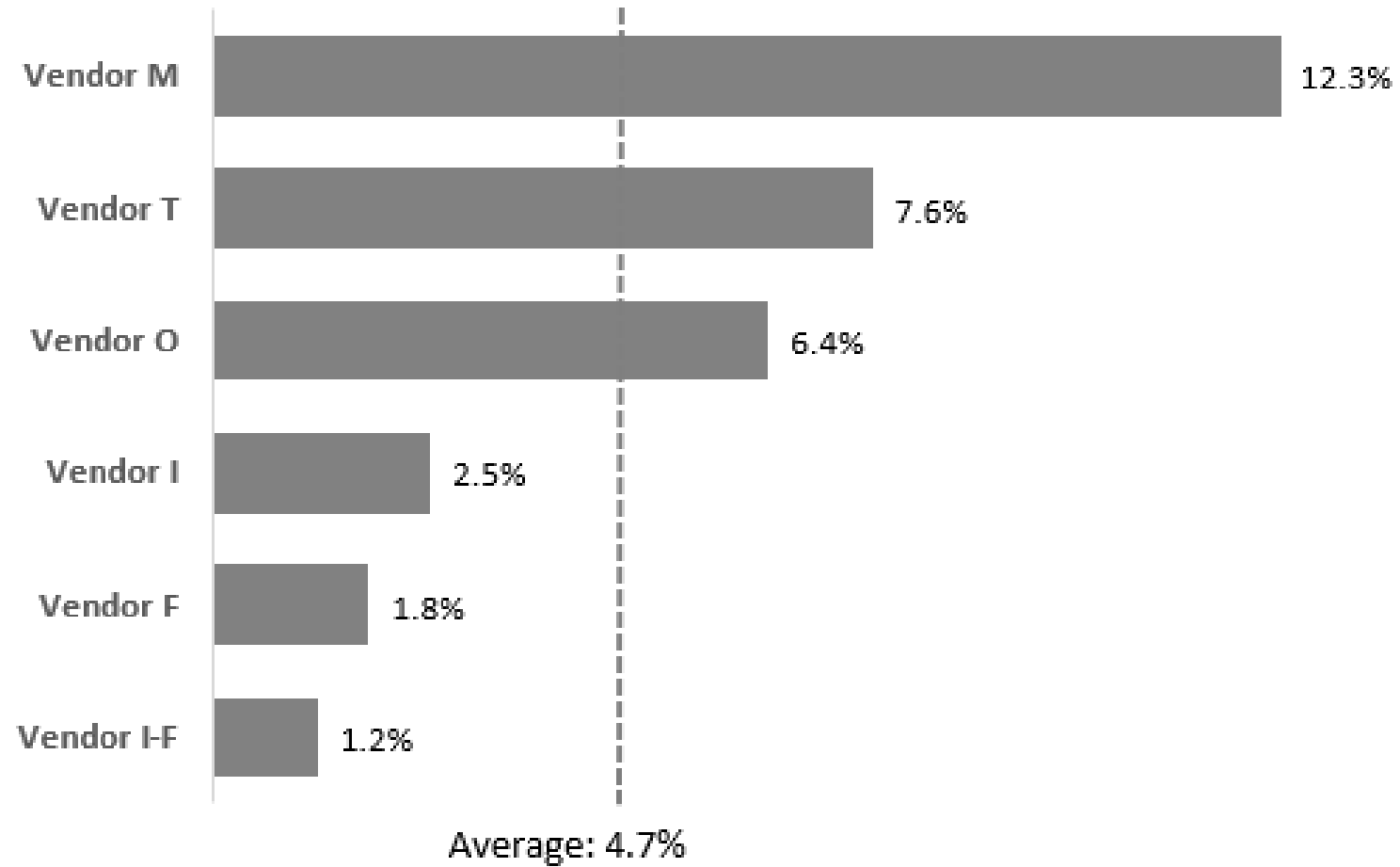
Outcome Breakdown

■ Unknown



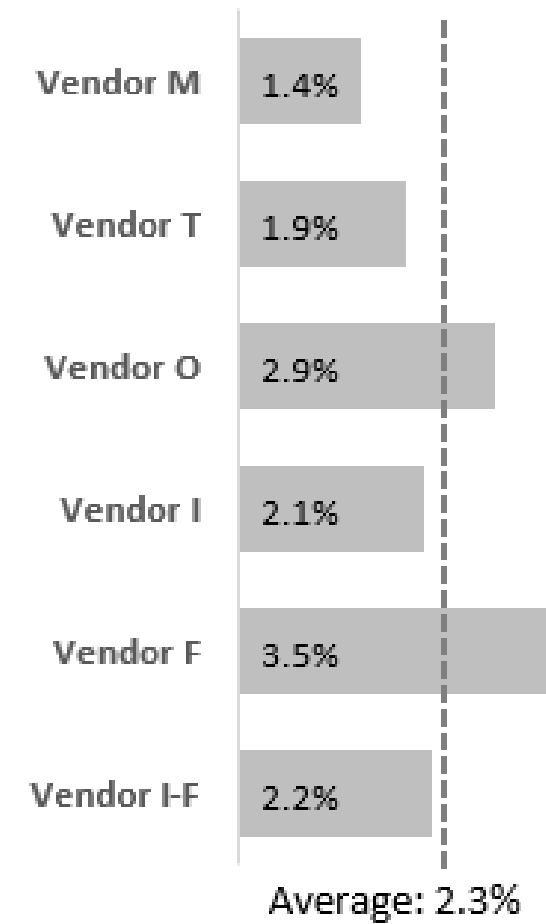
Outcome Breakdown

■ Rejected

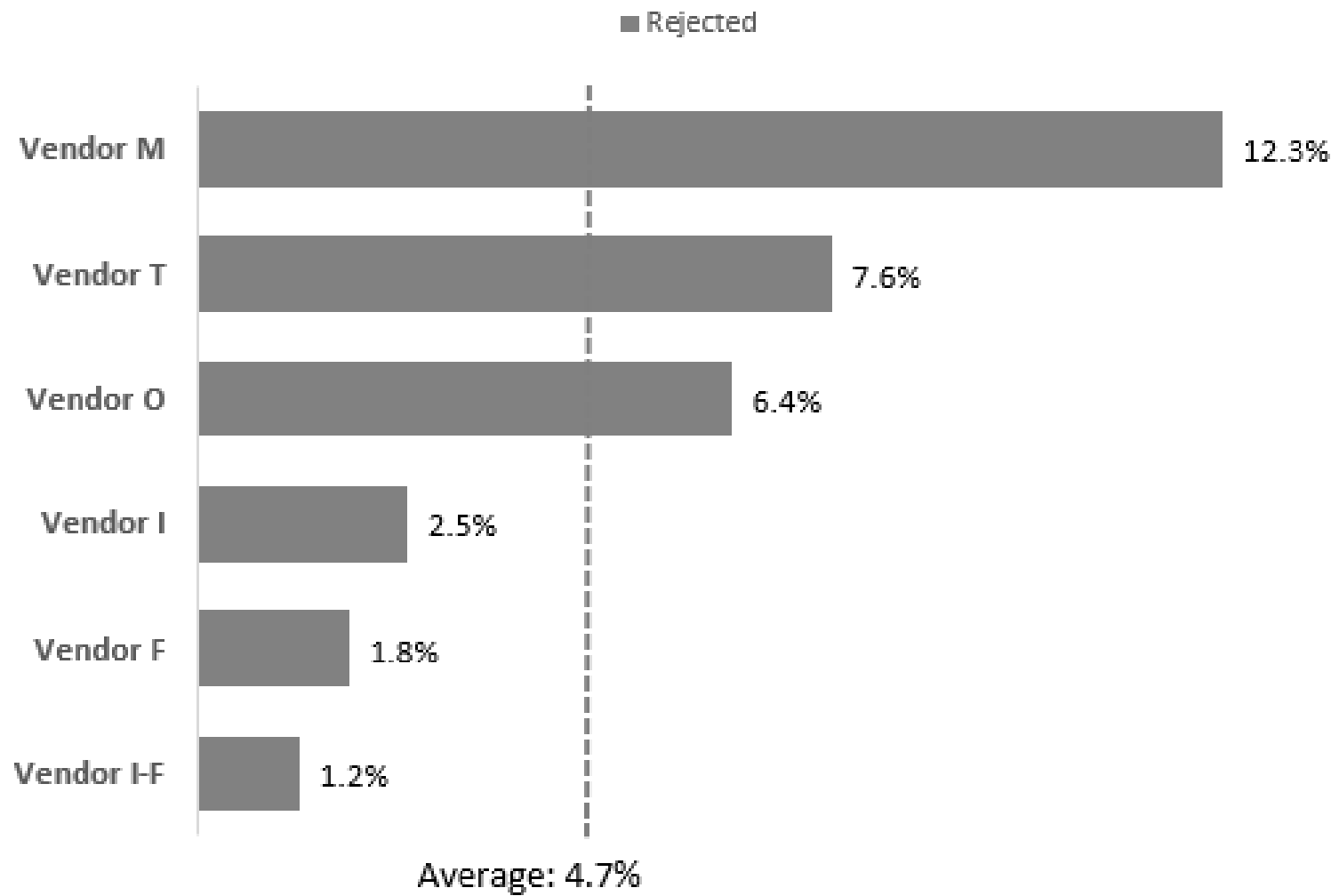


Outcome Breakdown

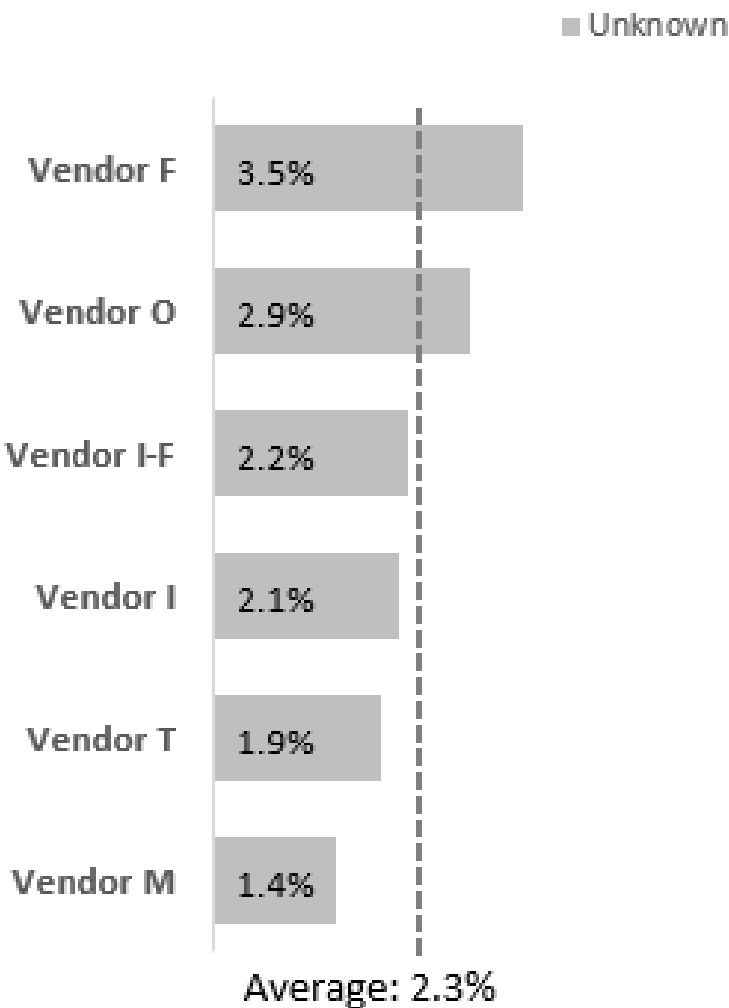
■ Unknown



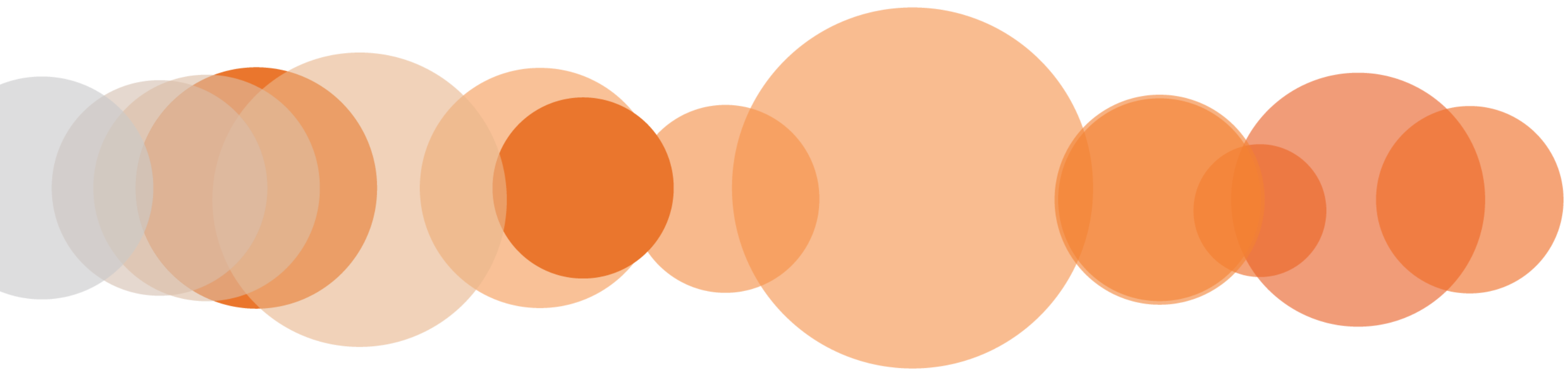
Outcome Breakdown

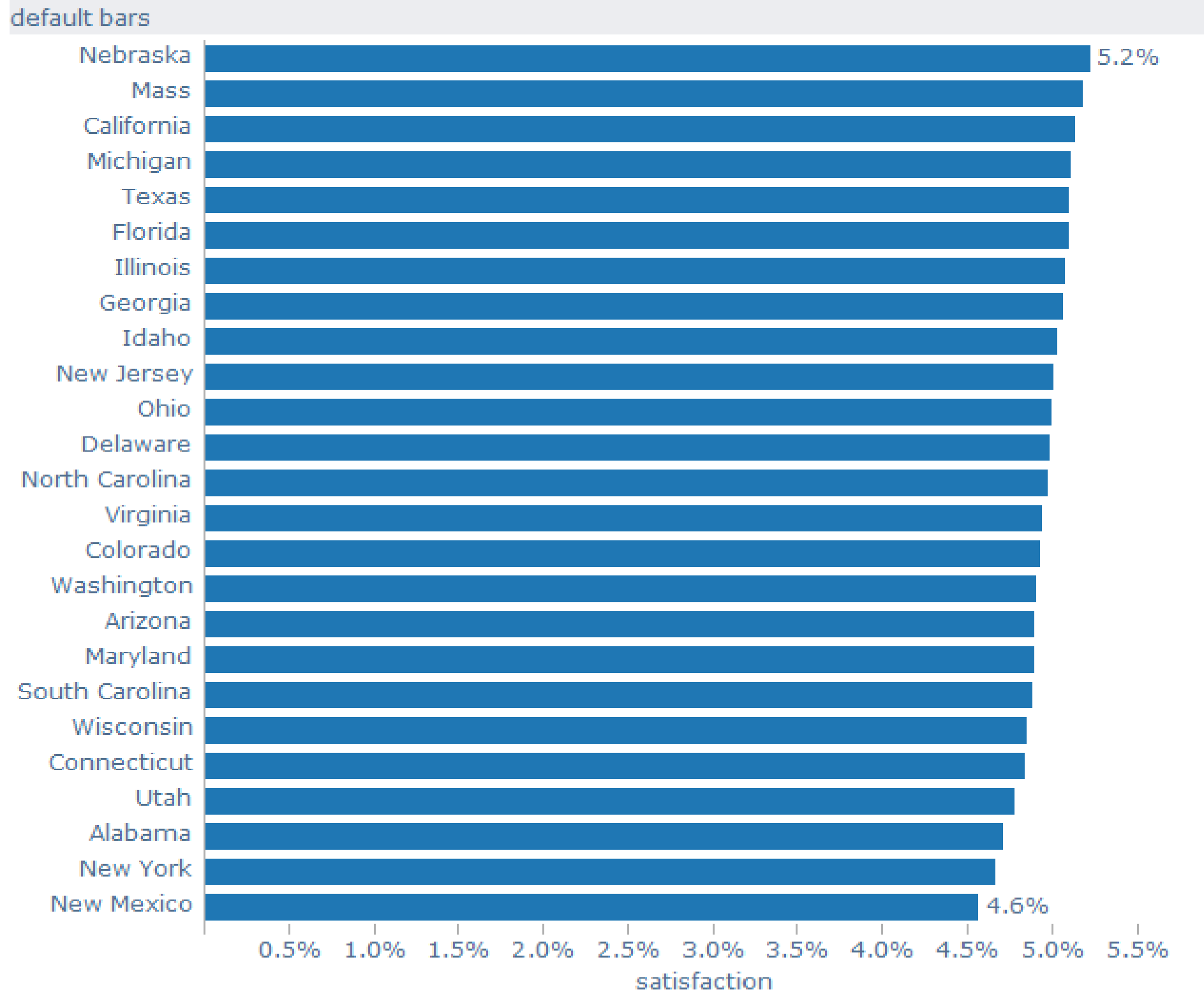


Outcome Breakdown

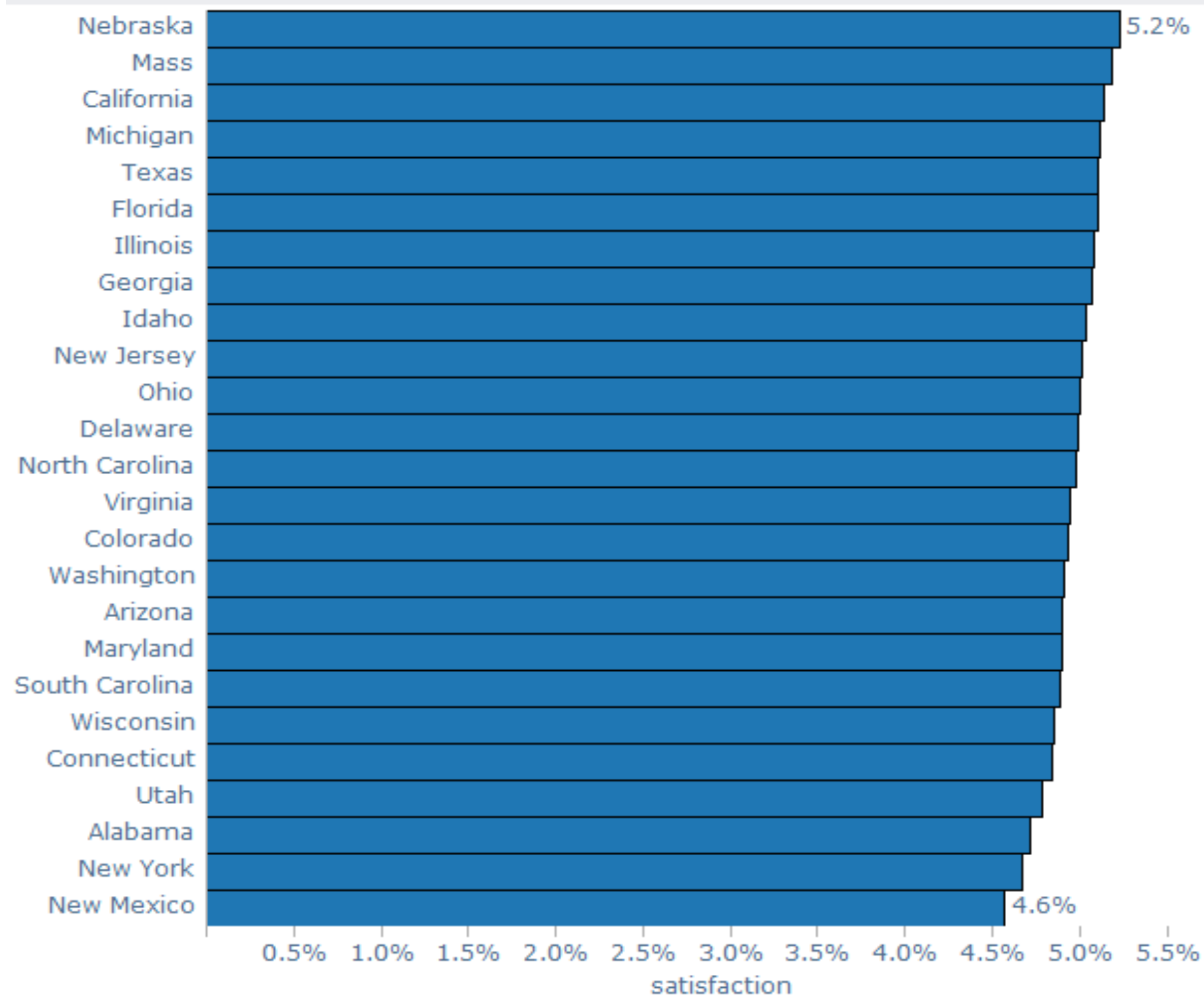


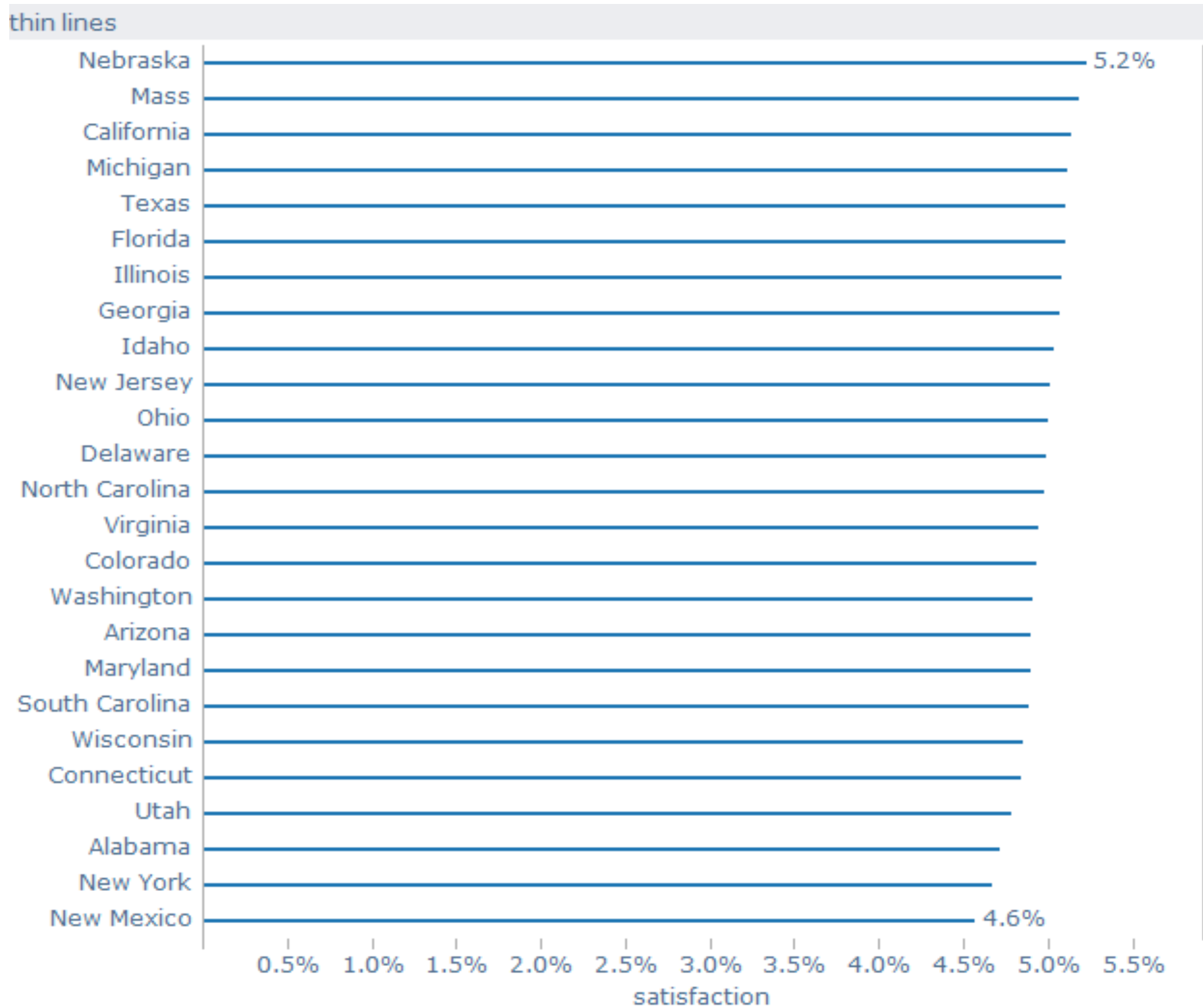
Bar Chart Issue

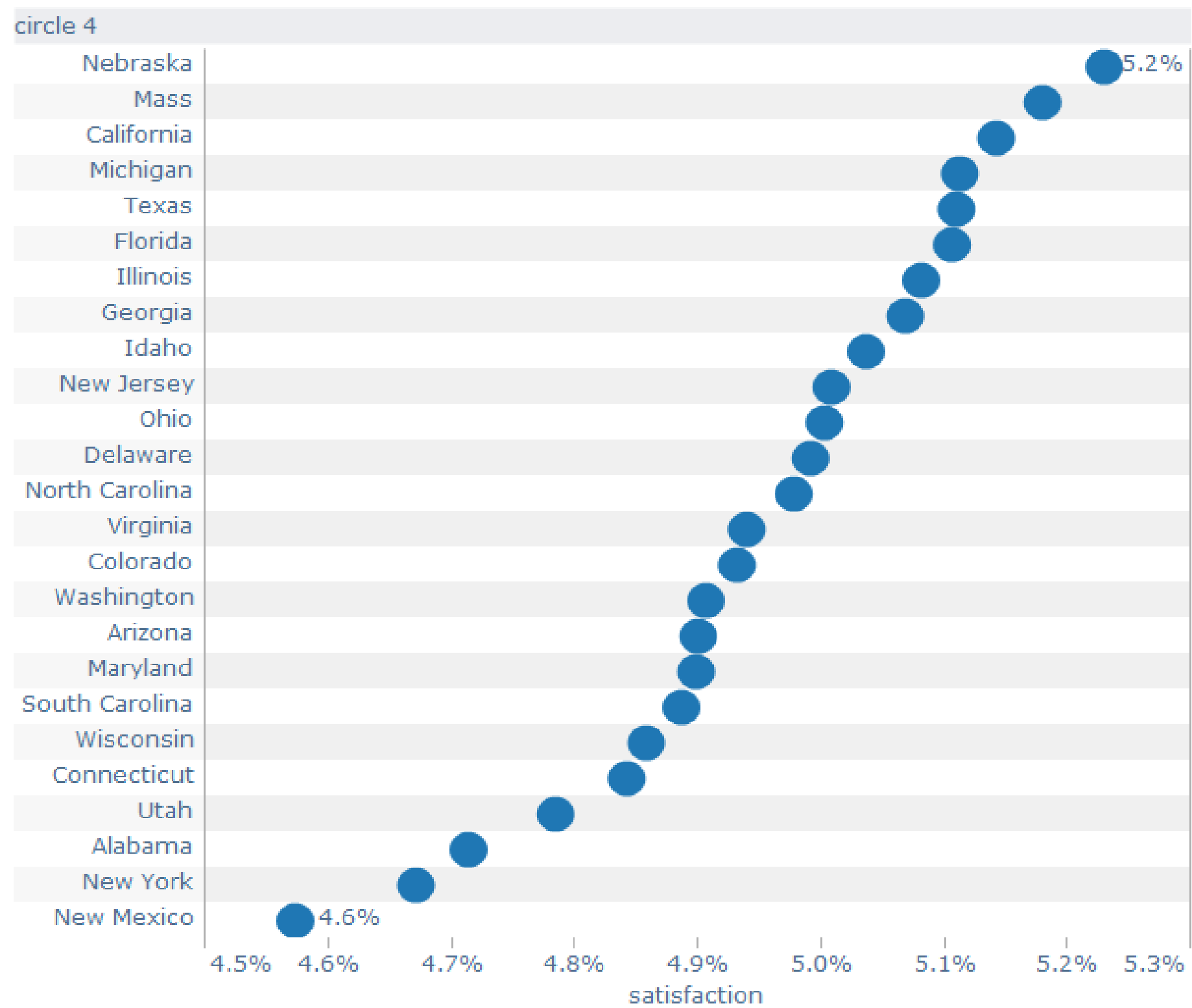


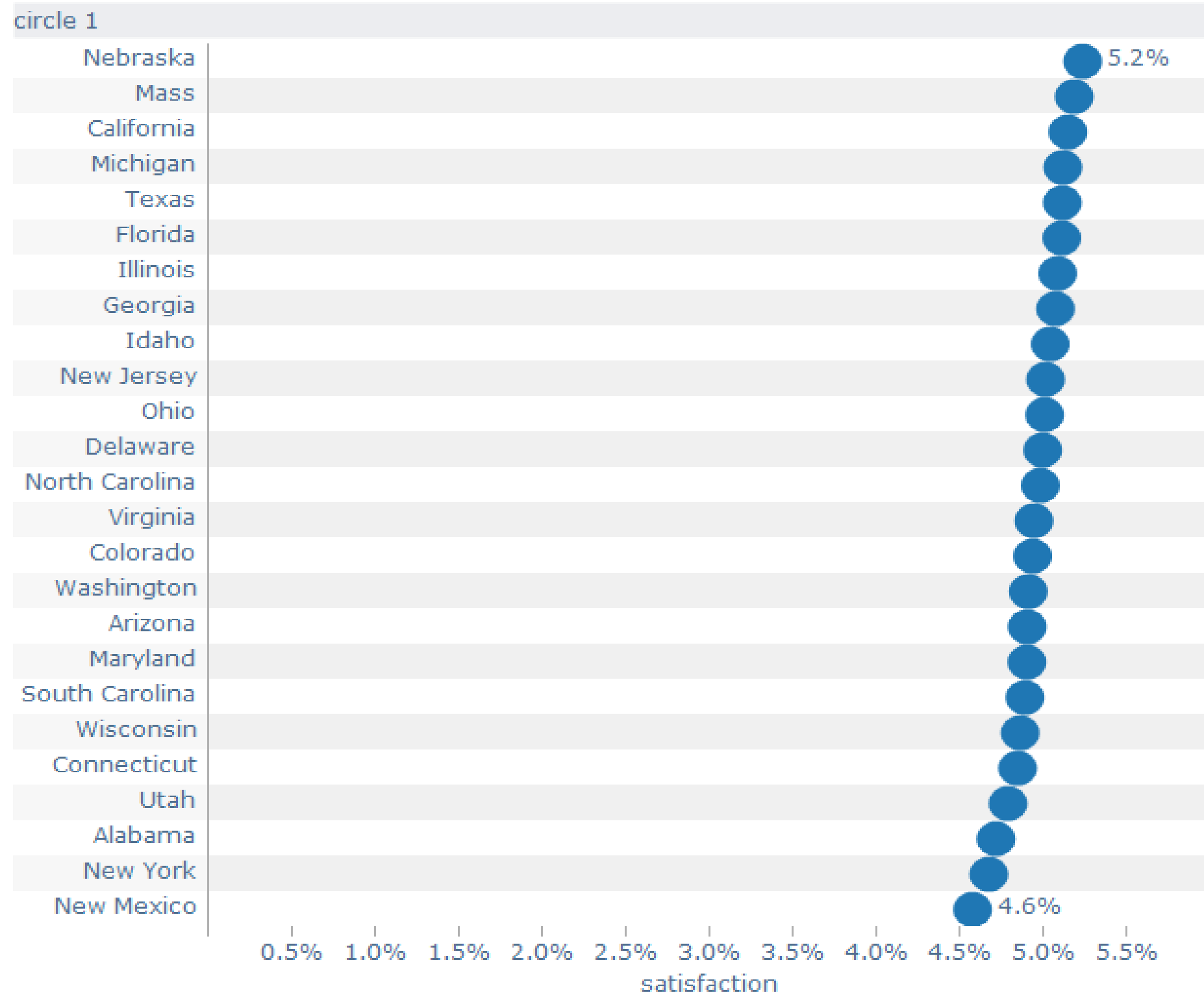


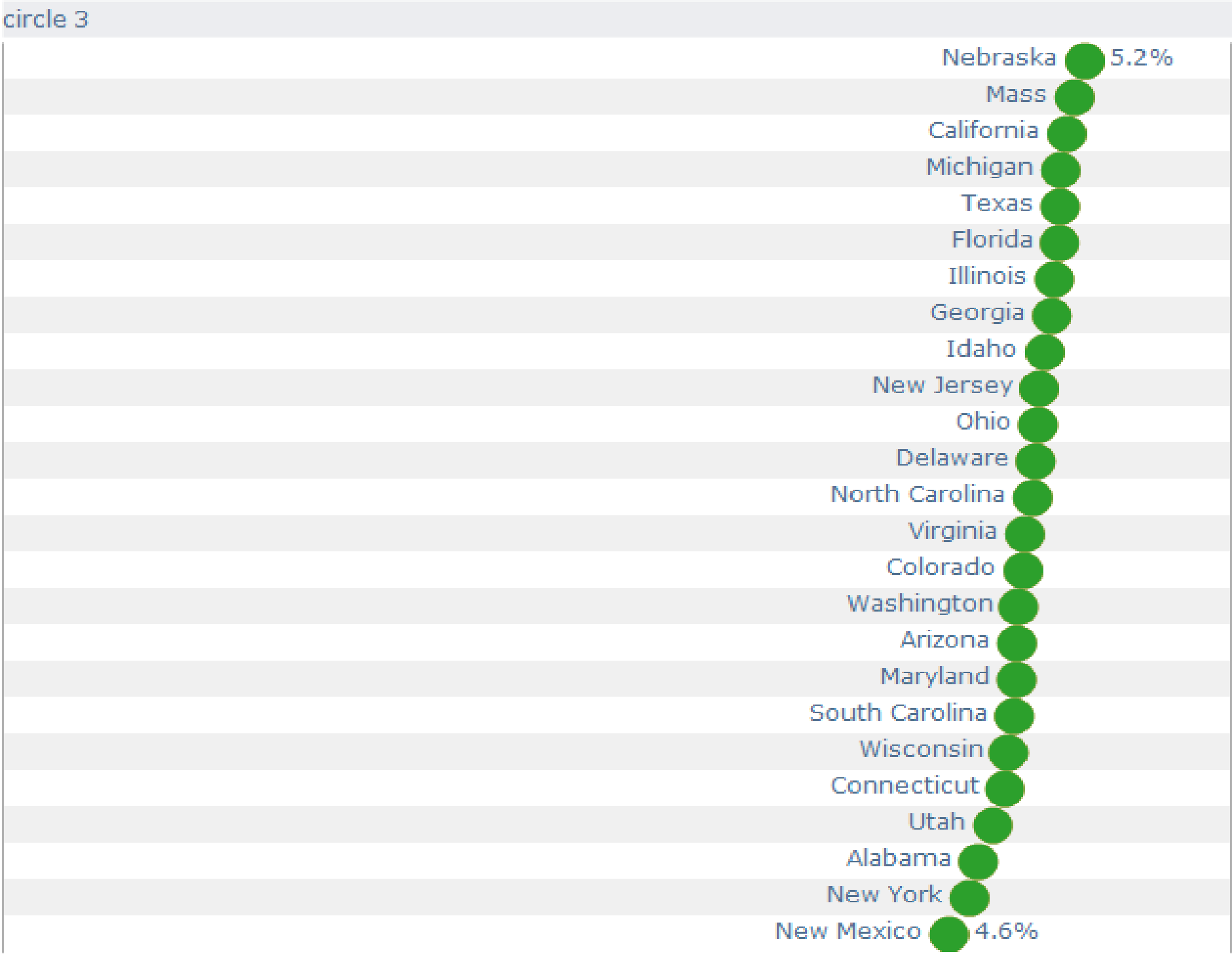
fat bars and borders

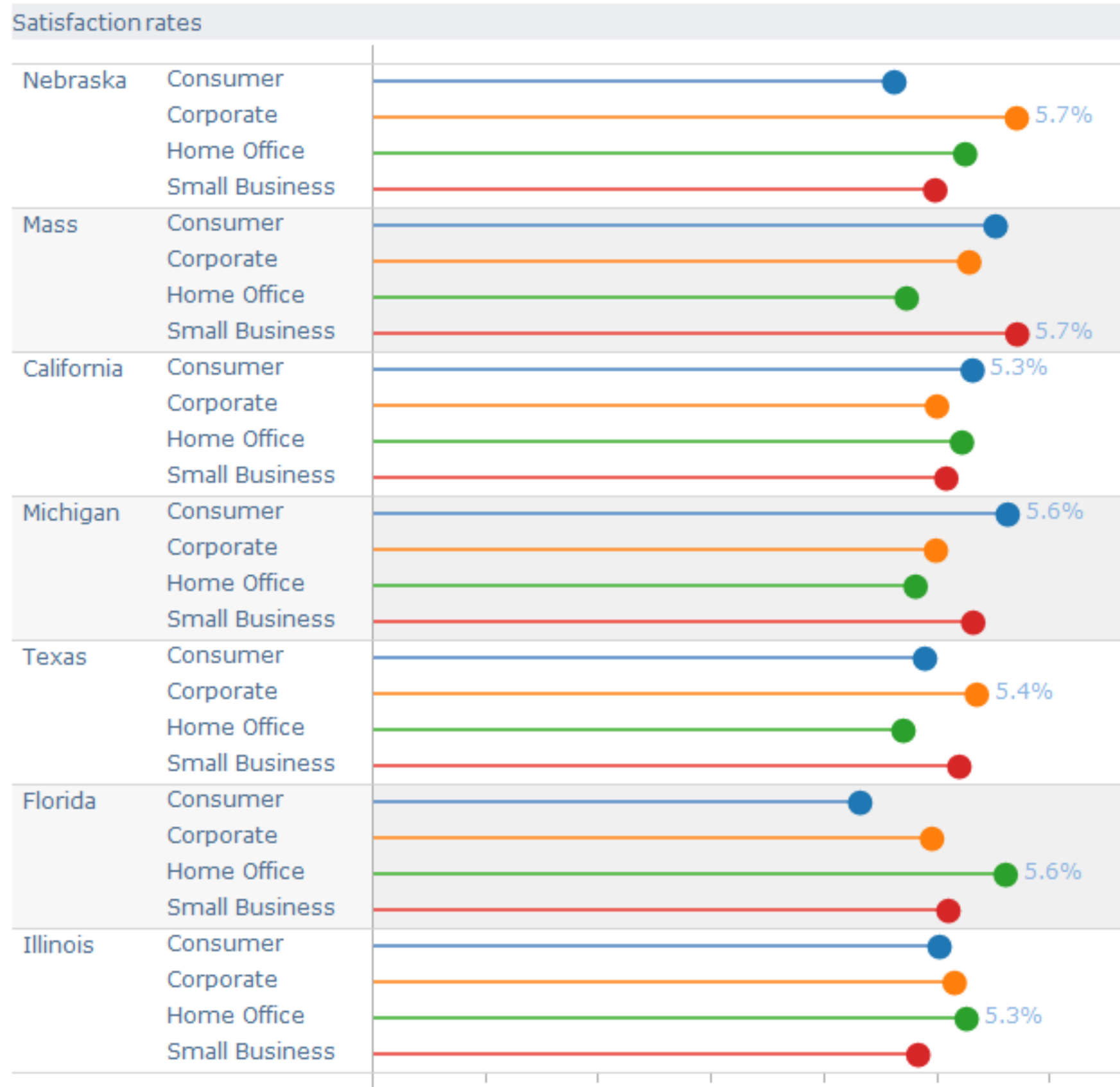










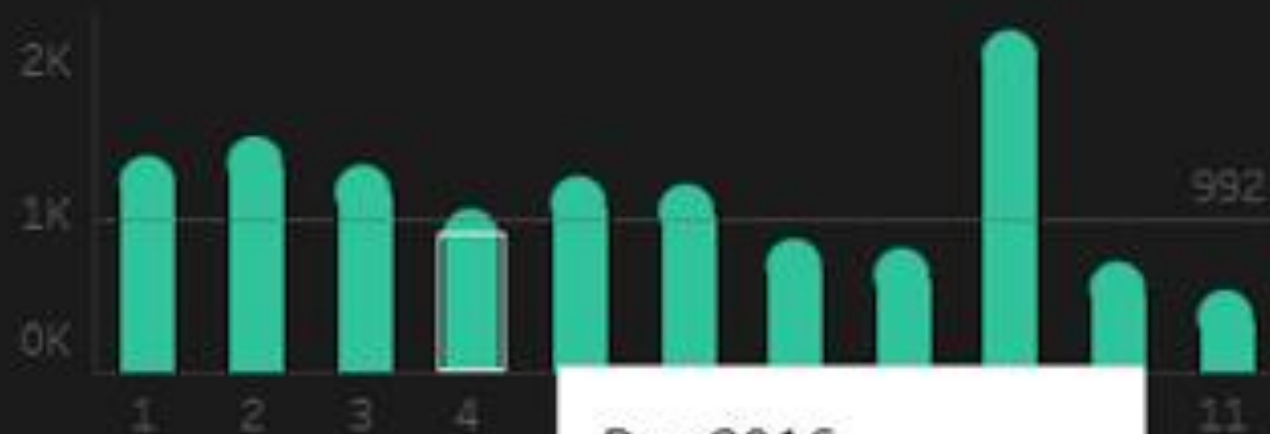




#DearDuo View Review



10,911 Viz Views

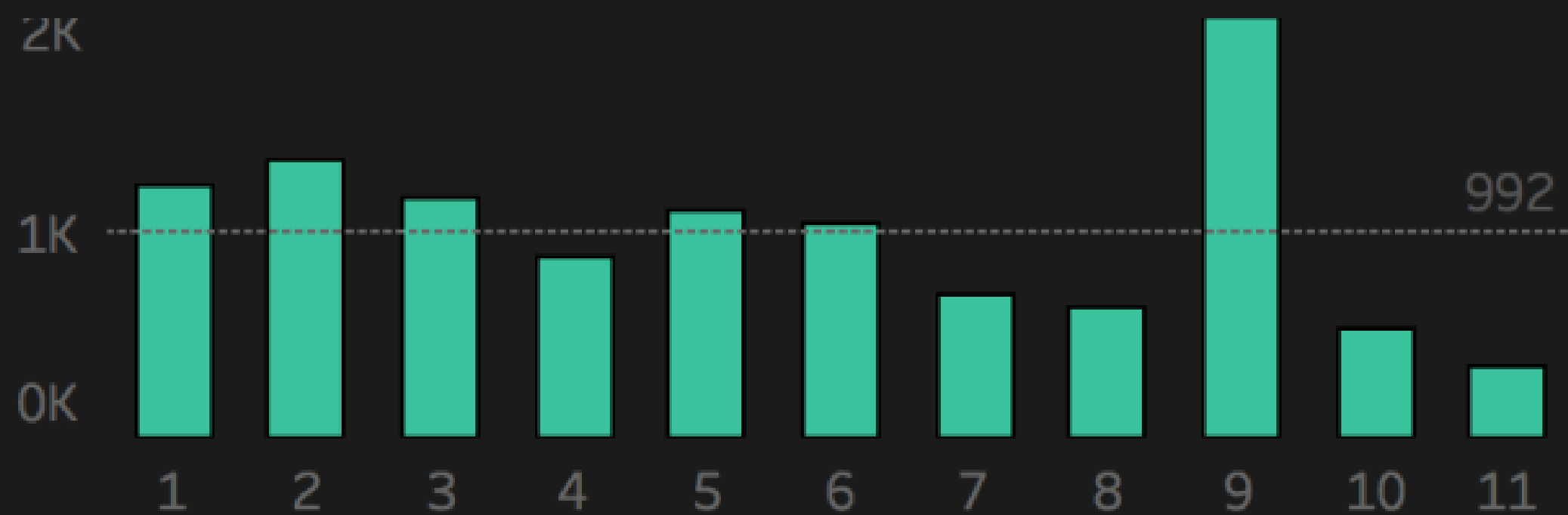
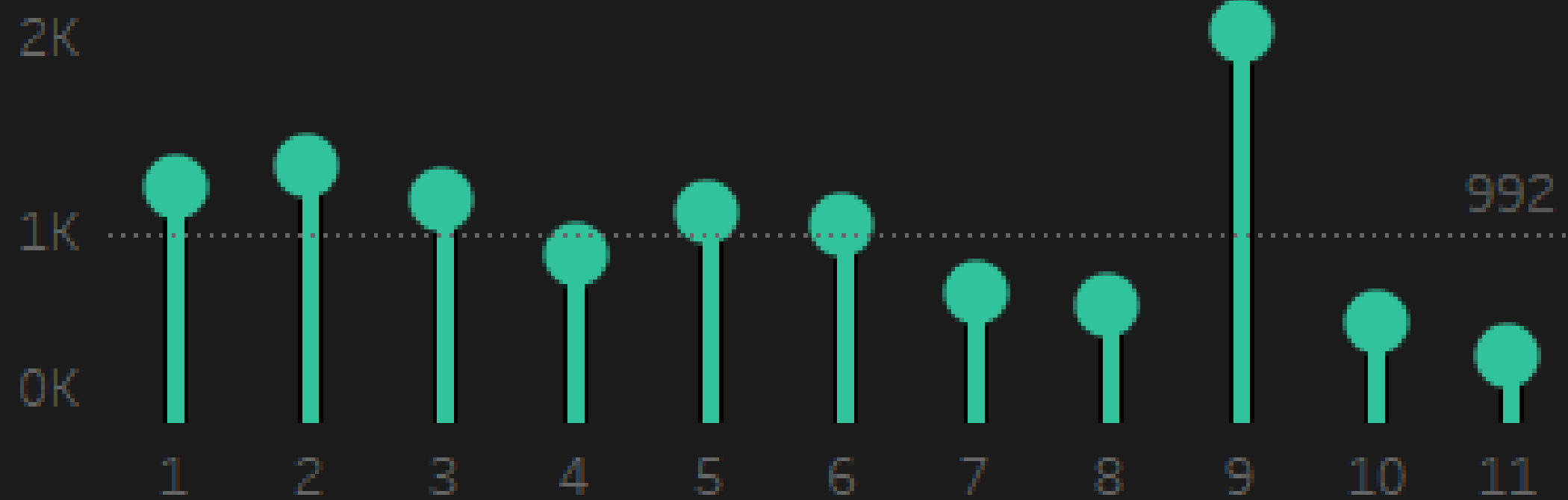


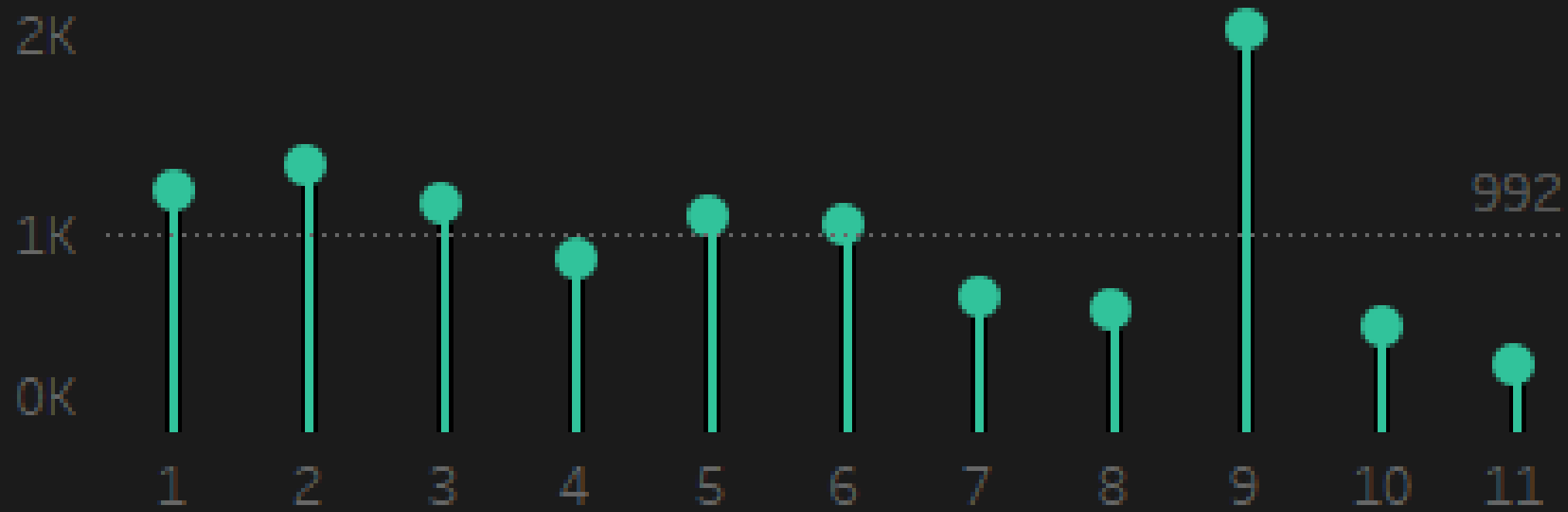
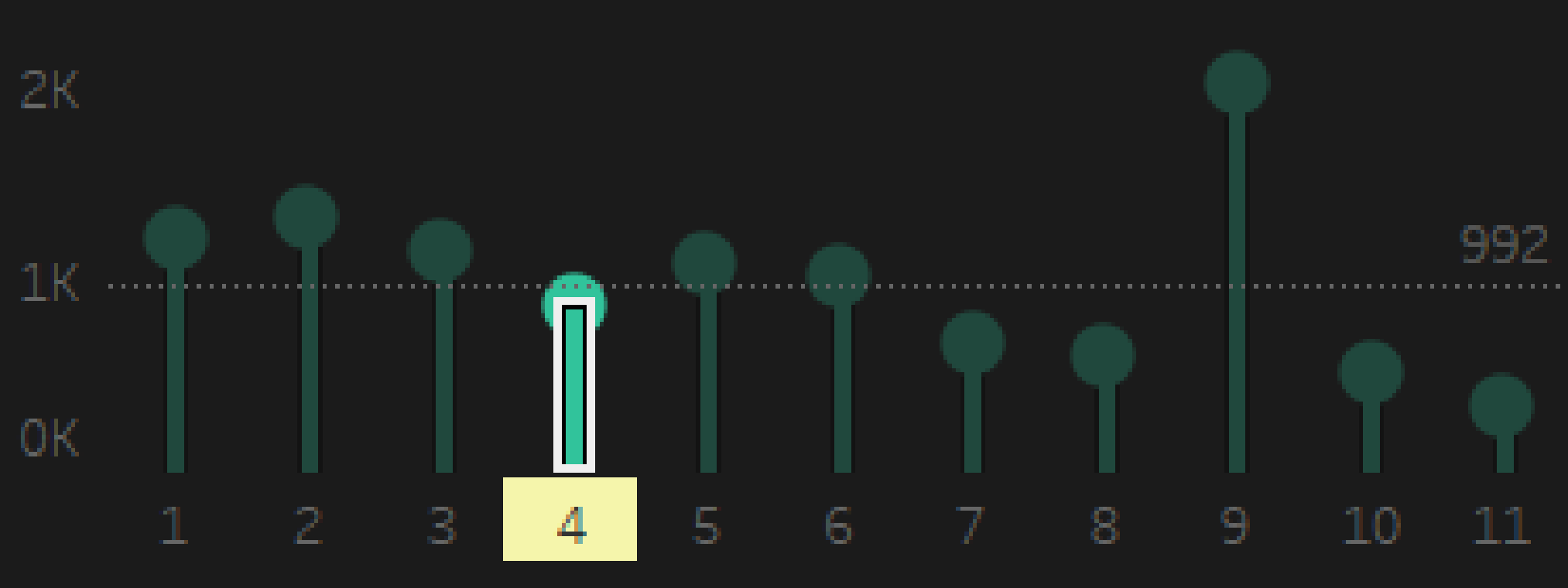
The viz views were pretty consistent over the last 11 months. We expect the more recent vizzes to have more views. We haven't been around as long and as a result, we haven't

Dec 2016

Visualized Music

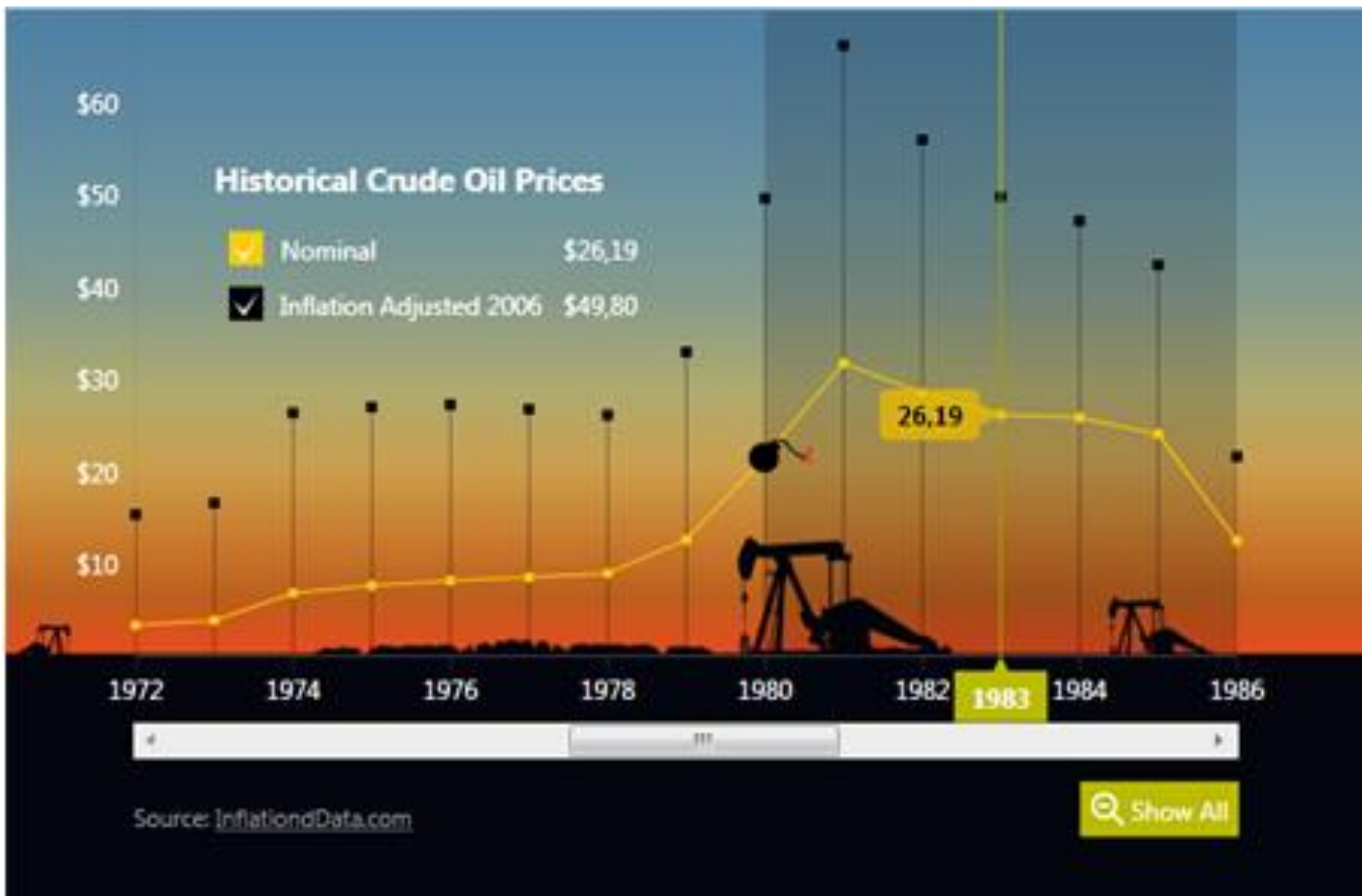
874 Viz Views





Historical Crude Oil Prices

 Nominal \$26,19
 Inflation Adjusted 2006 \$49,80

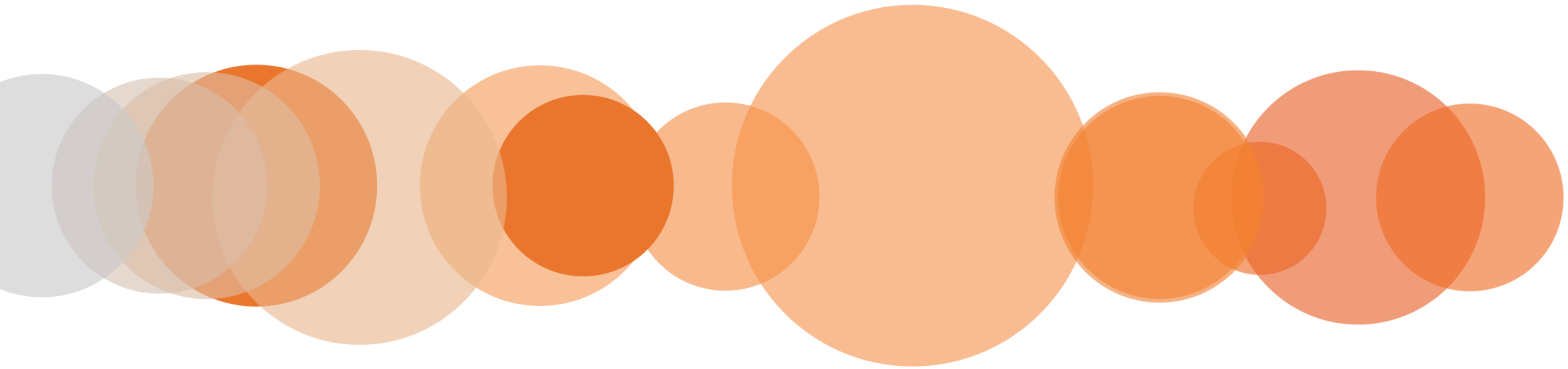


Source: inflationdata.com

 Show All

A Problem with Scale

An Example from The Big Book of Dashboard



Power Plant Monitor

Trend

TOTAL CAPACITY



10,858 MW

SPINNING RESERVE



758 MW

WATER PRODUCTION



394 MIGD

TOTAL LOAD



6,646 MW

STANDBY RESERVE



1,847 MW

WATER EXPORT



356 MIGD

Local Metrics (Real-Time)

Target

SYSTEM
FREQUENCY

54.43 HZ

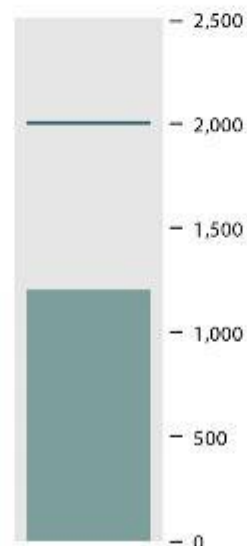
AMBIENT
TEMP

37.0°C

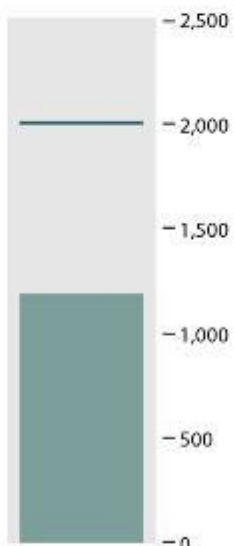
RELATIVE
HUMIDITY

39.8%

NG TOTAL FLOW



GD HEAT RATE



REAL-TIME DATA FEED

Disabled

Details (All Locations)

LOCATION	CURRENT OUTPUT (kW)	EXPECTED OUTPUT (kW)	ACTUAL VS. EXPECTED	14-DAY TREND	LAST INSPECTION
1833 Appleby Line	350,370	420,453	83 %		3 days
1833 Highway 4	3,374	3,917	86 %		19 days
48 Davis Dr	5,058	5,602	90 %		2 days
144 Howard Cavasos	655,698	714,991	92 %		7 days
2 Jack Hanoka Dr	5,342	5,521	97 %		13 days
22 Daybreak Dr	3,053	3,148	97 %		7 days
1 Whitestone Rd	4,579	4,711	97 %		2 days
1552 Flintrock Rd	5,342	5,488	97 %		18 days
1335 County Line	5,087	5,210	98 %		4 days
1 Jack Hanoka Dr	3,307	3,338	99 %		12 days
1525 West Line	946,311	954,645	99 %		4 days
1335 Omoo Rd	847,260	854,289	99 %		3 days
18819 Guelph St	1,060,307	1,068,858	99 %		15 days
1 Adam West Rd	652,373	650,426	100 %		4 days
11820 Dover Tr	900,606	884,292	102 %		6 days
15 Rural Rd	1,200,261	1,174,570	102 %		4 days

Details (All Locations)

LOCATION	CURRENT OUTPUT (kW)	EXPECTED OUTPUT (kW)	ACTUAL VS. EXPECTED
1833 Appleby Line	350,370	420,453	 83 %
1833 Highway 4	3,374	3,917	 86 %
48 Davis Dr	5,058	5,602	 90 %
144 Howard Cavasos	655,698	714,991	 92 %
2 Jack Hanoka Dr	5,342	5,521	 97 %
22 Daybreak Dr	3,053	3,148	 97 %
1 Whitestone Rd	4,579	4,711	 97 %
1552 Flintrock Rd	5,342	5,488	 97 %

Source: *The Big Book of Dashboards* (Figure 9.9)

Details (All Locations)

LOCATION	CURRENT OUTPUT (kW)	EXPECTED OUTPUT (kW)		
1833 Appleby Line	350,370	420,453		83 %
1833 Highway 4	3,374	3,917		86 %
48 Davis Dr	5,058	5,602		90 %
144 Howard Cavasos	655,698	714,991		92 %
2 Jack Hanoka Dr	5,342	5,521		97 %
22 Daybreak Dr	3,053	3,148		97 %
1 Whitestone Rd	4,579	4,711		97 %
1552 Flintrock Rd	5,342	5,488		97 %

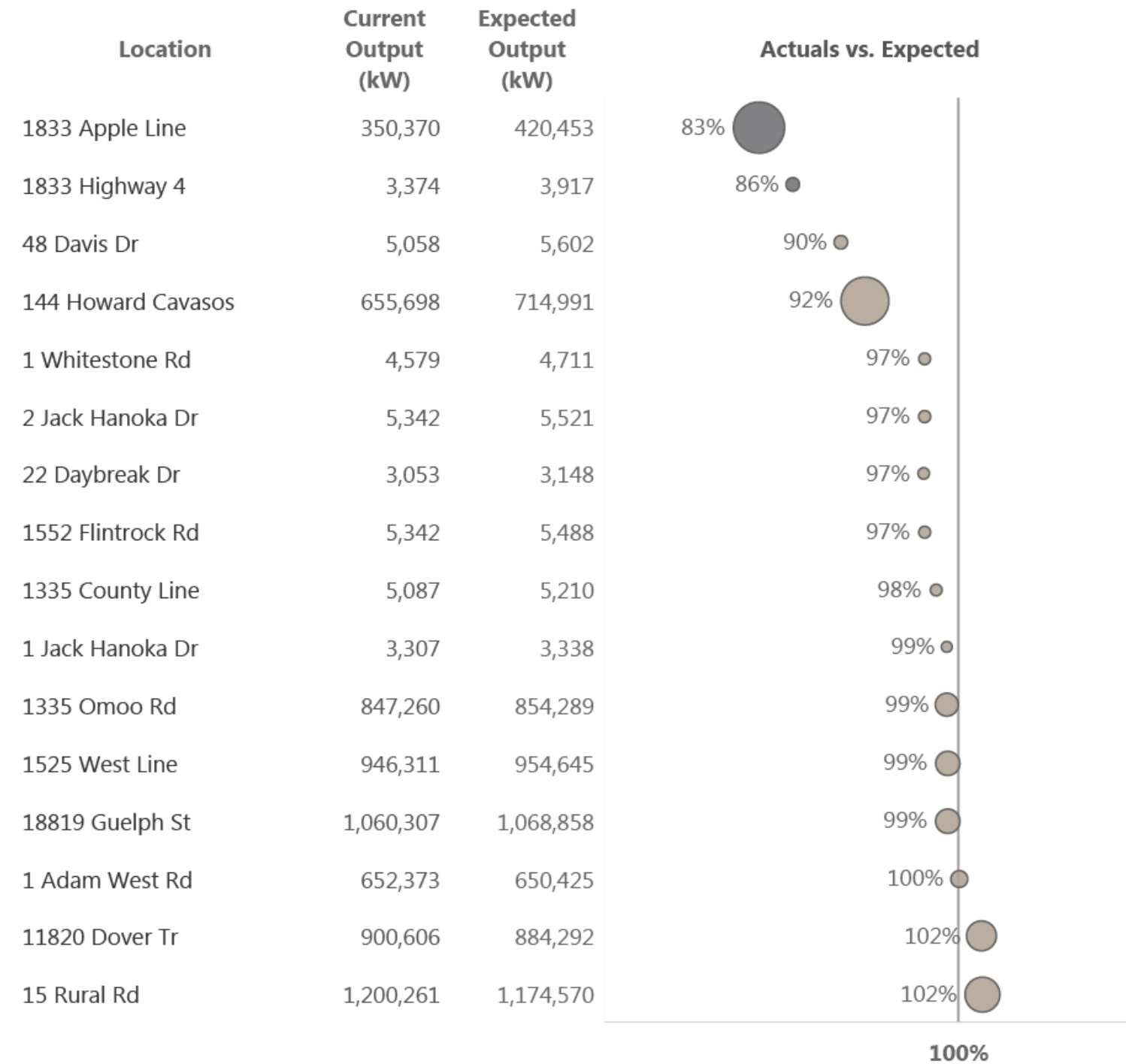
Source: *The Big Book of Dashboards* (Figure 9.10)

Details (All Locations)

LOCATION	CURRENT OUTPUT (kW)	EXPECTED OUTPUT (kW)	ACTUAL VS. EXPECTED	
1833 Appleby Line	350,370	420,453		83 %
1833 Highway 4	3,374	3,917		86 %
48 Davis Dr	5,058	5,602		90 %
144 Howard Cavasos	655,698	714,991		92 %
2 Jack Hanoka Dr	5,342	5,521		97 %
22 Daybreak Dr	3,053	3,148		97 %
1 Whitestone Rd	4,579	4,711		97 %
1552 Flintrock Rd	5,342	5,488		97 %

Source: *The Big Book of Dashboards* (Figure 9.11)

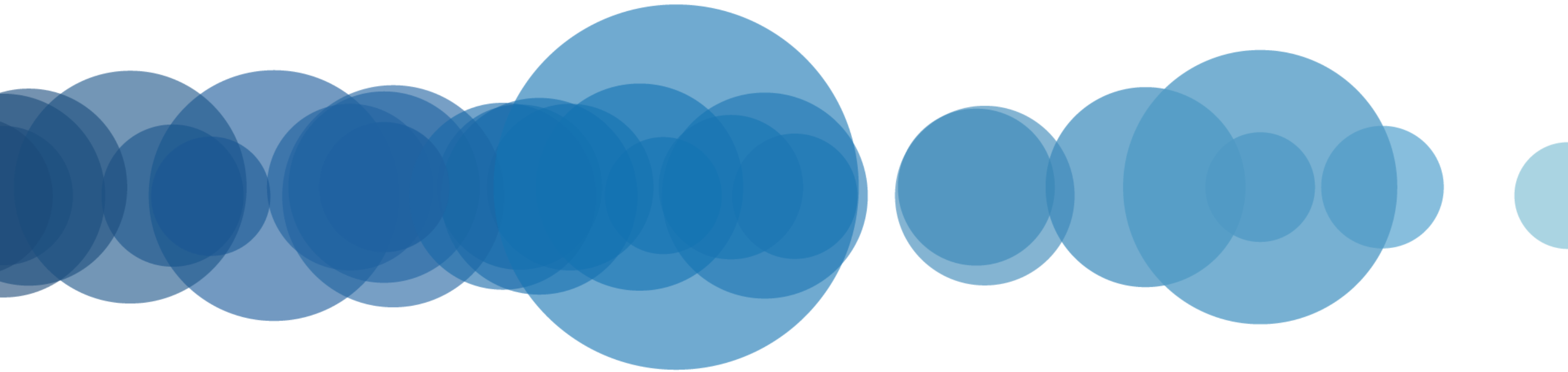
Details (All Locations)



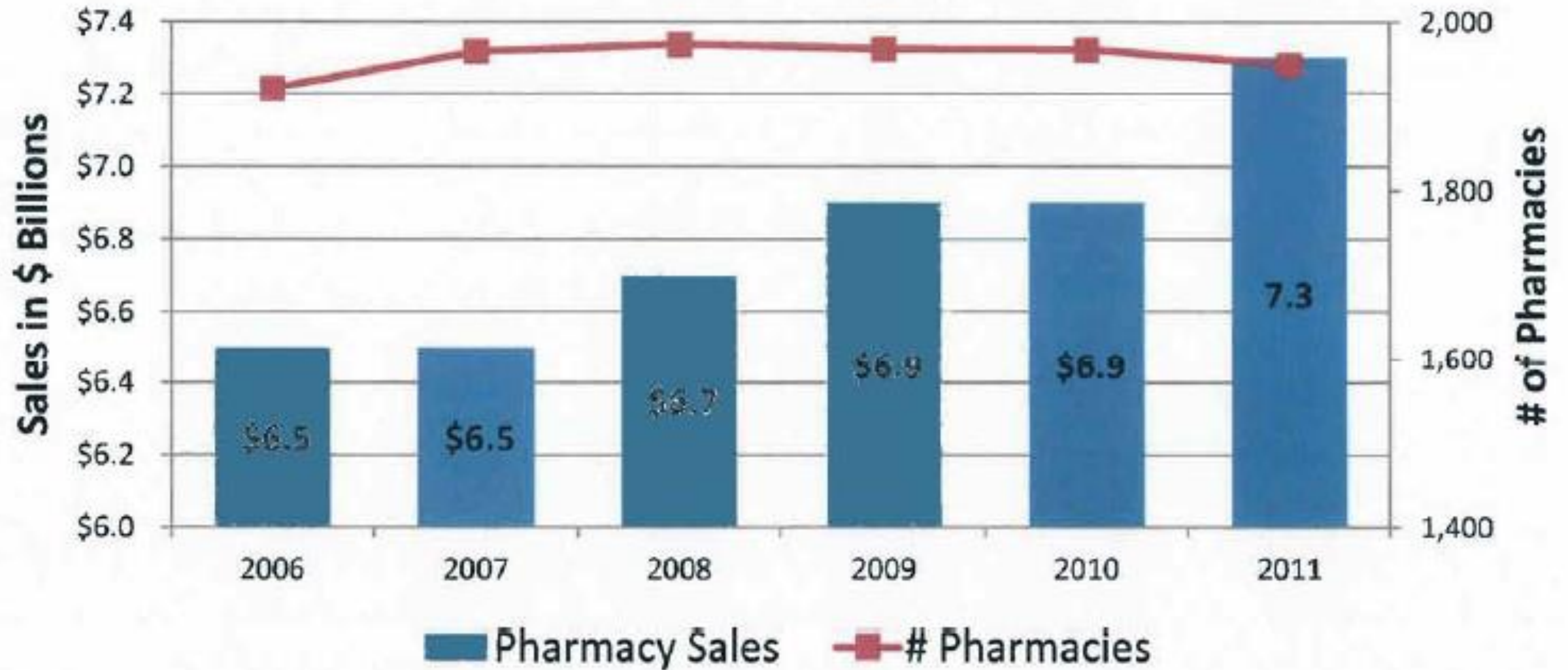
Source: *The Big Book of Dashboards* (Figure 9.12)

Real-World Example

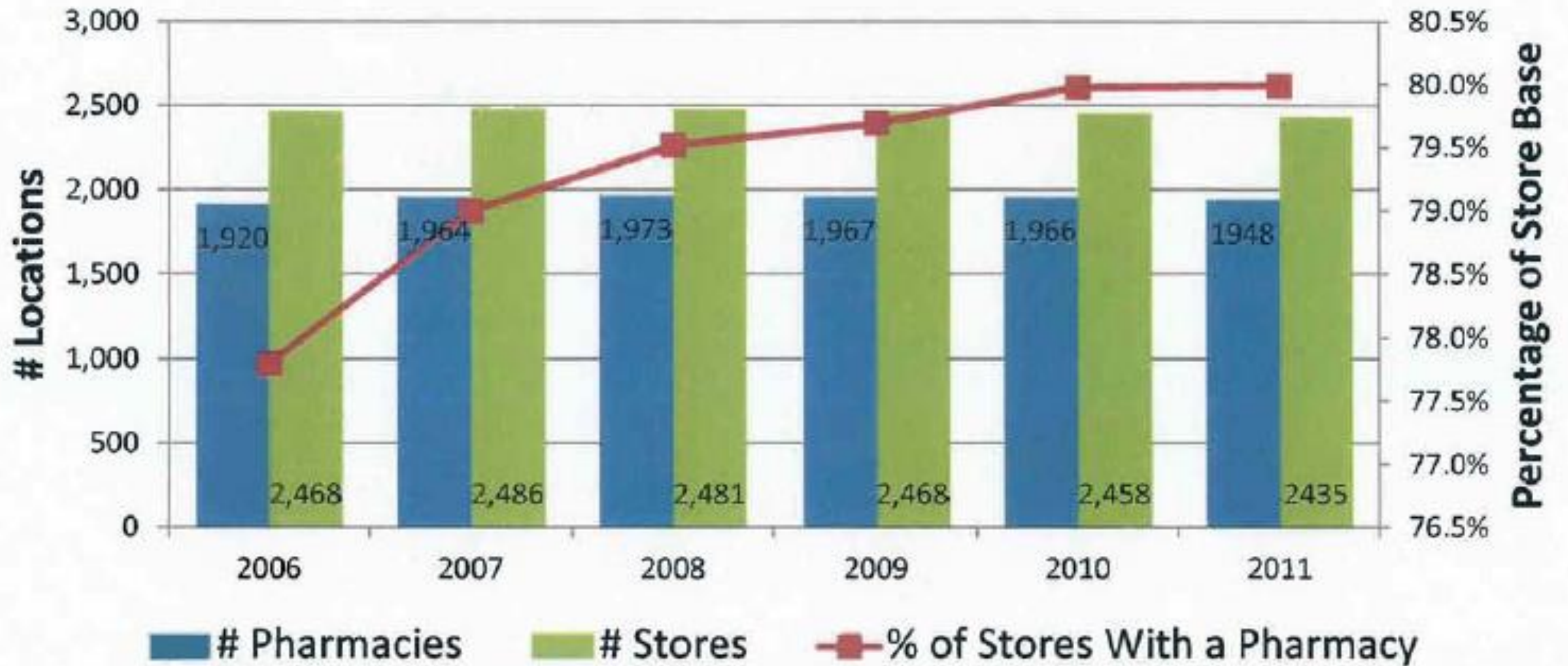
Kroger Investor Fact Book



Kroger Pharmacy Growth



Kroger Pharmacy Growth



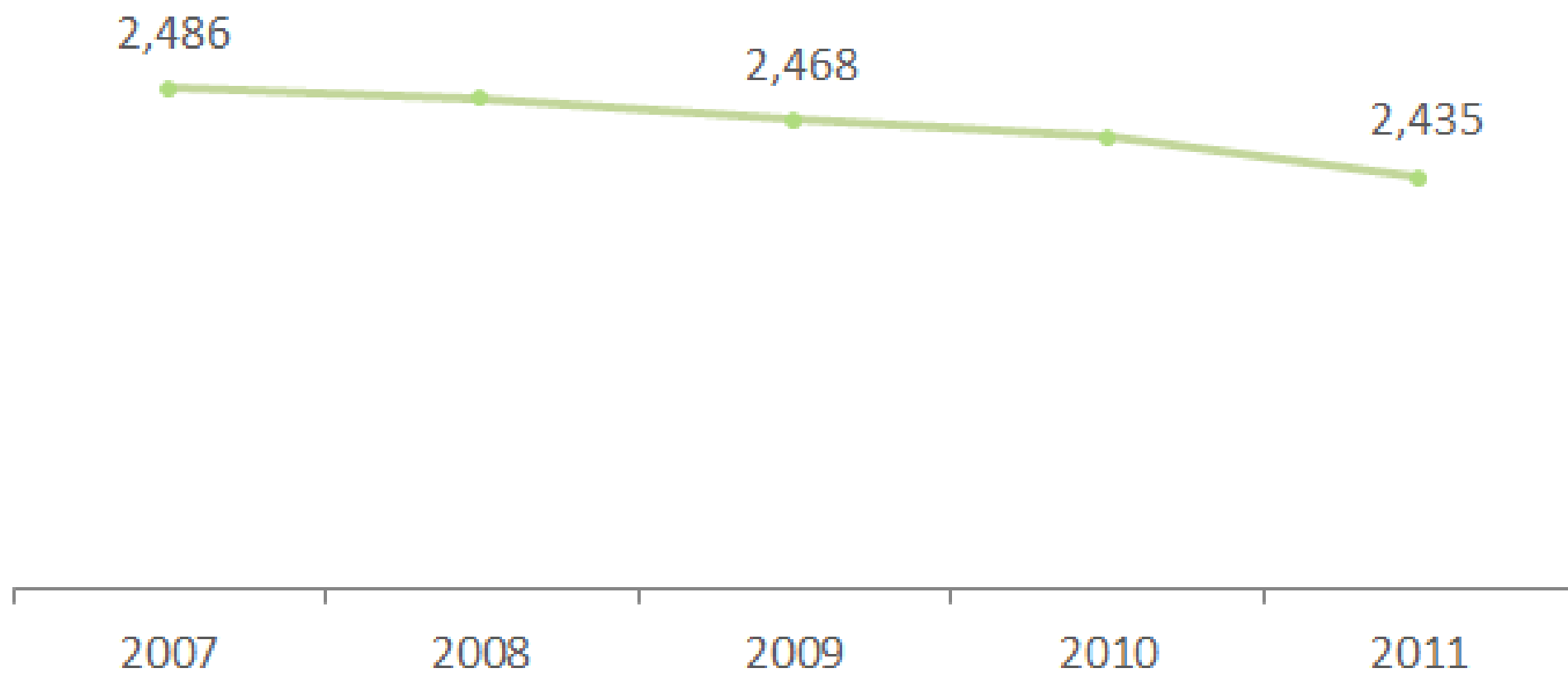
Kroger Pharmacy Growth?

What the data actually shows:

	<u>2006</u>	<u>2007</u>	<u>2008</u>	<u>2009</u>	<u>2010</u>	<u>2011</u>
	77.8%	79.0%	79.5%	79.7%	80.0%	80.0%
Pharmacies	1,920	1,964	1,973	1,967	1,966	1,948
Stores	2,468	2,486	2,481	2,468	2,458	2,435

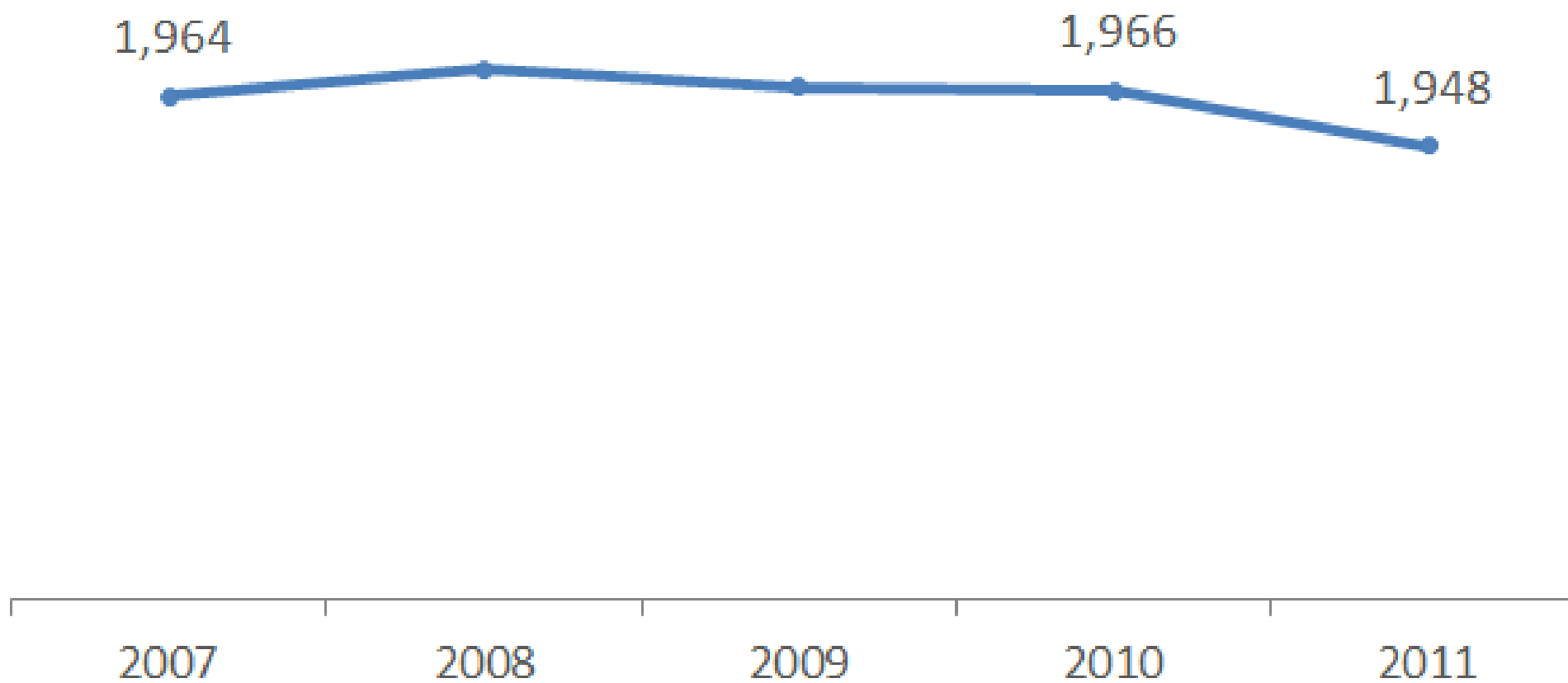
The Real Story

Total Number of Kroger Stores

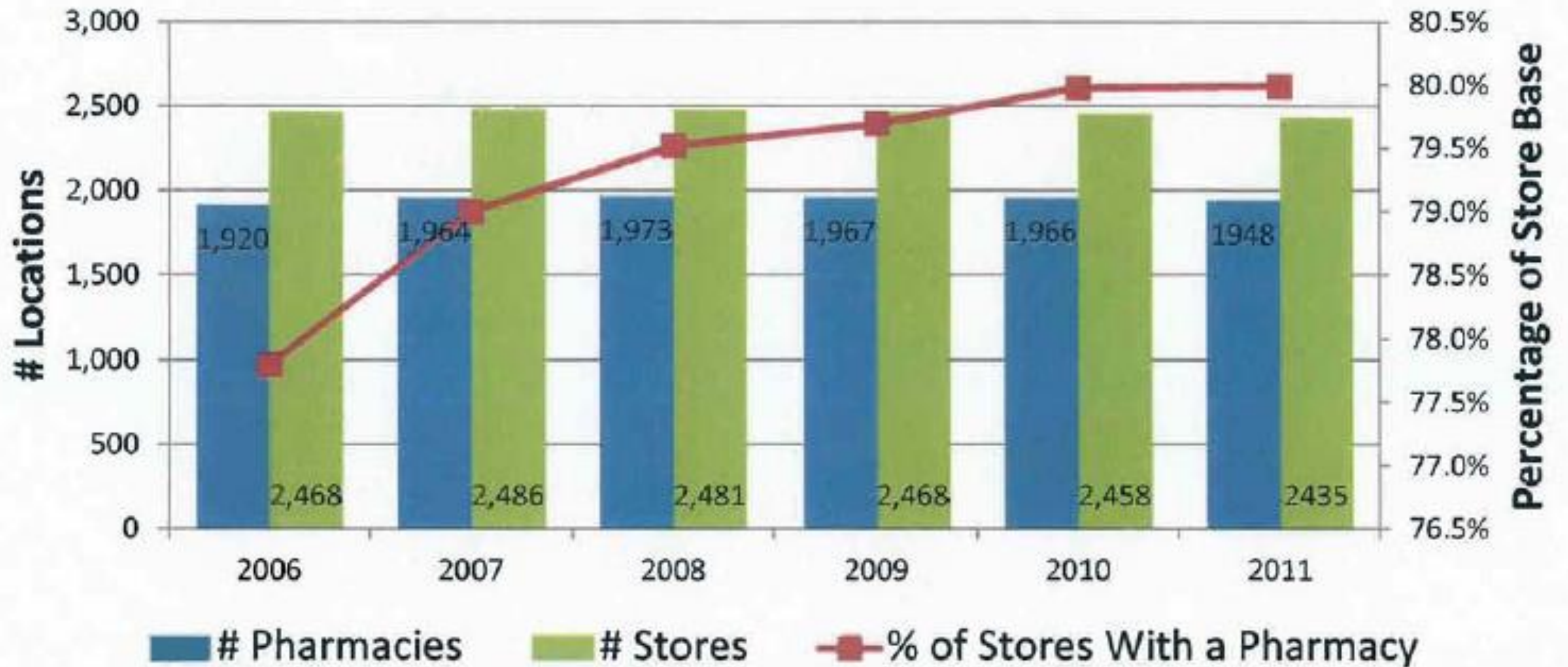


The Real Story

Total Number of Kroger Pharmacies



Kroger Pharmacy Growth



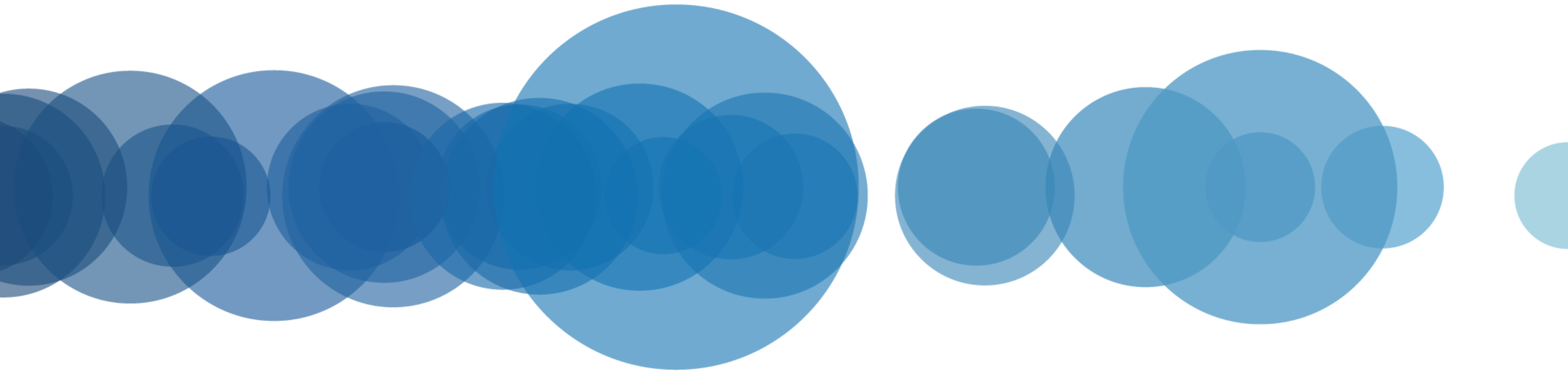
What is the real message?

1. From 2007 to 2010 Kroger closed stores that don't have pharmacies thus increasing the % of stores with a pharmacy.
2. Kroger has increased Pharmacy Sales from \$6.5B to \$7.3B with fewer stores and fewer pharmacies

They missed a very compelling Story
12% increase in Sales with Lower Costs!

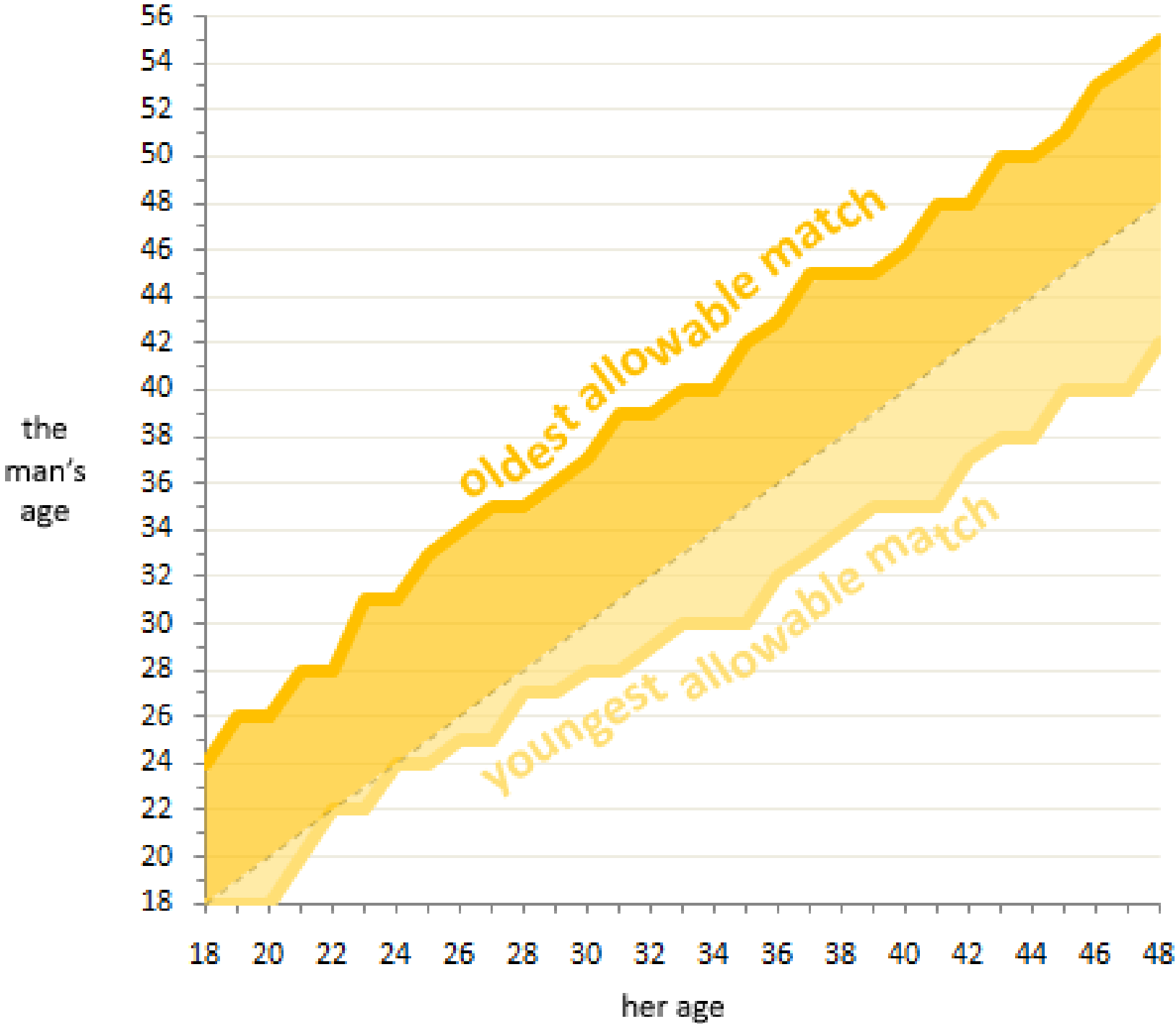
Just for Fun

OkCupid



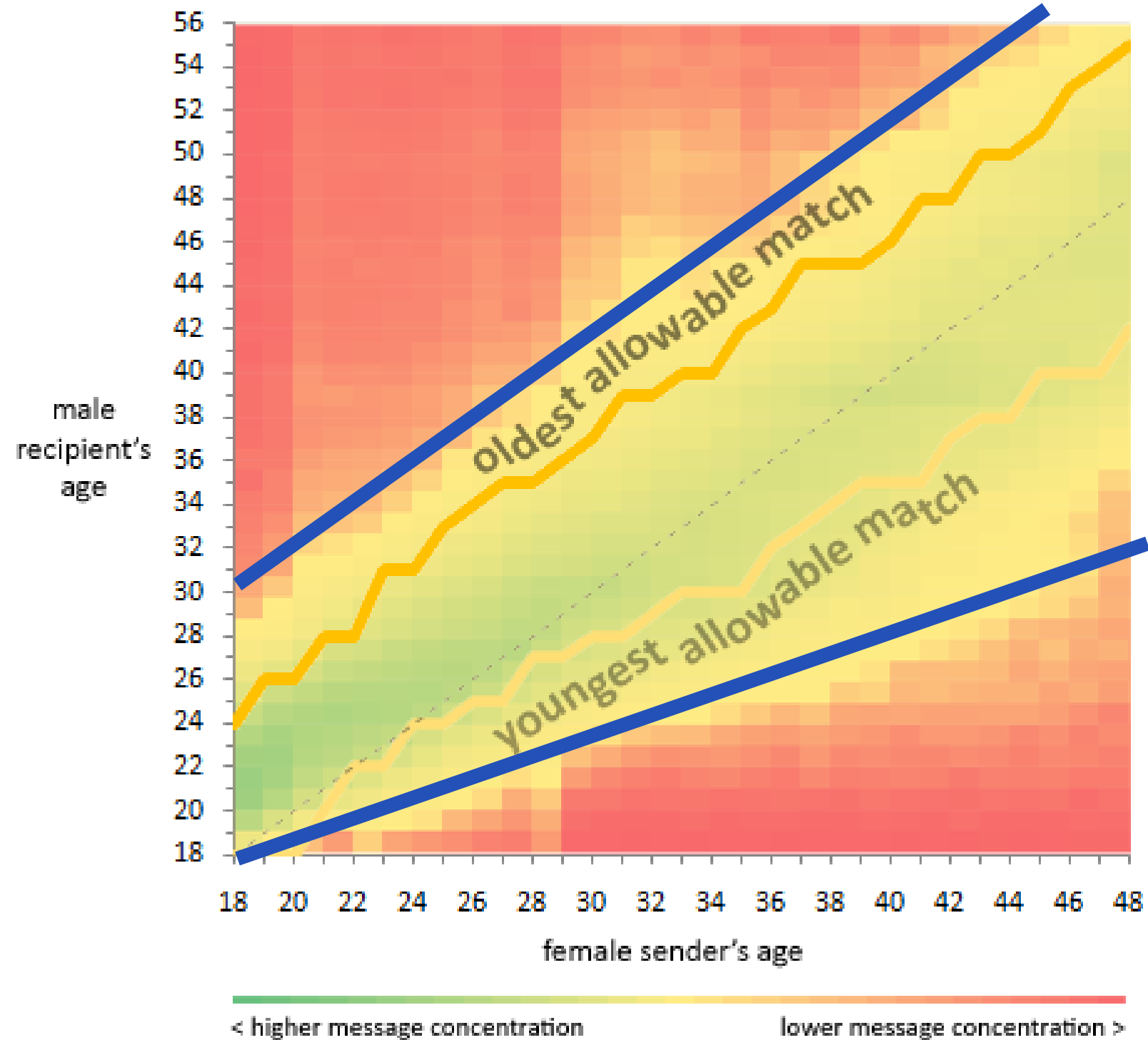
How Female Match Preferences Change With Age

median min. & max. age values in our female users'
"allowable match" settings



Where Women Are Sending Their Messages

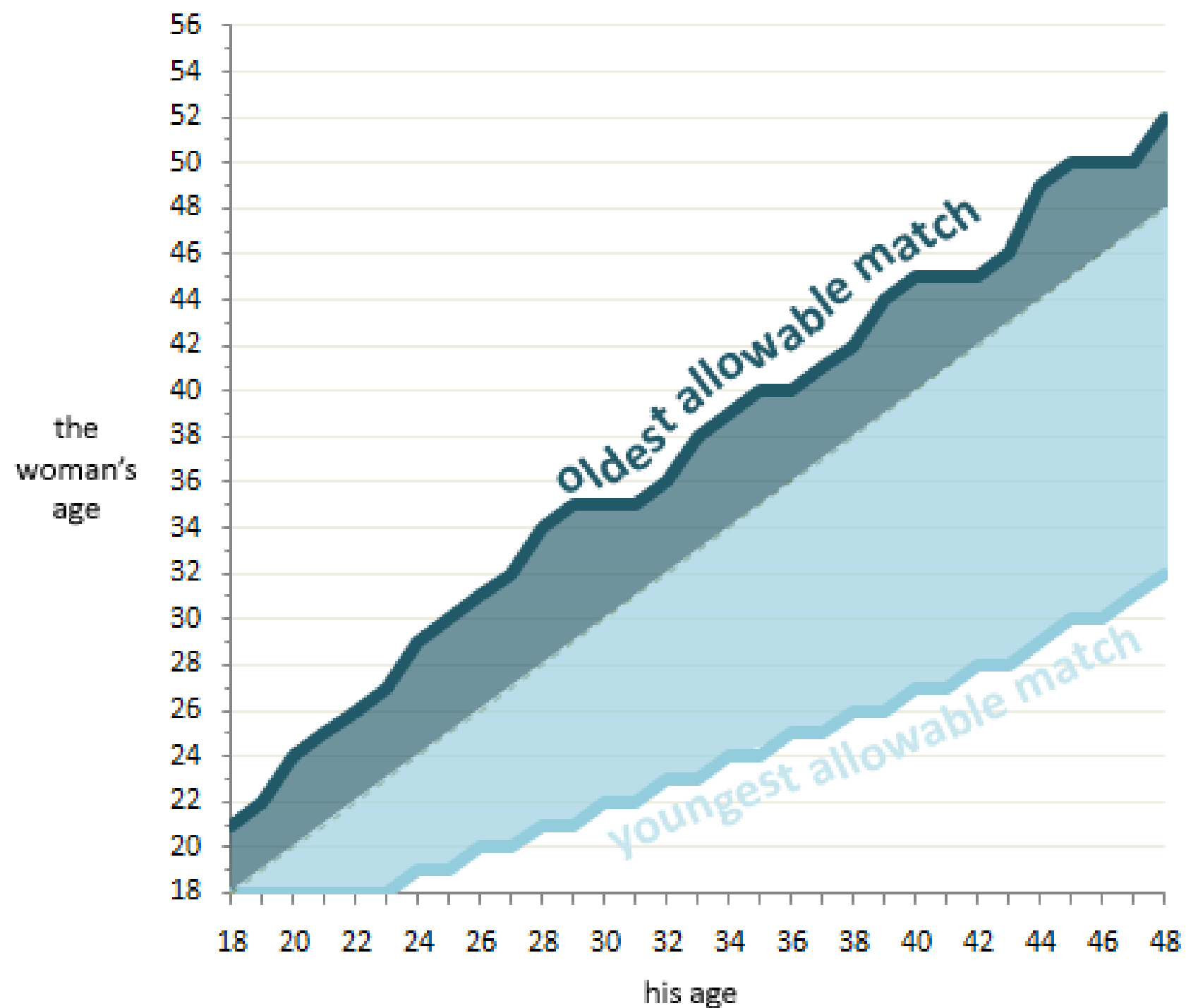
*distribution of female to male first contacts
grouped by age*



How Male Match Preferences Change With Age

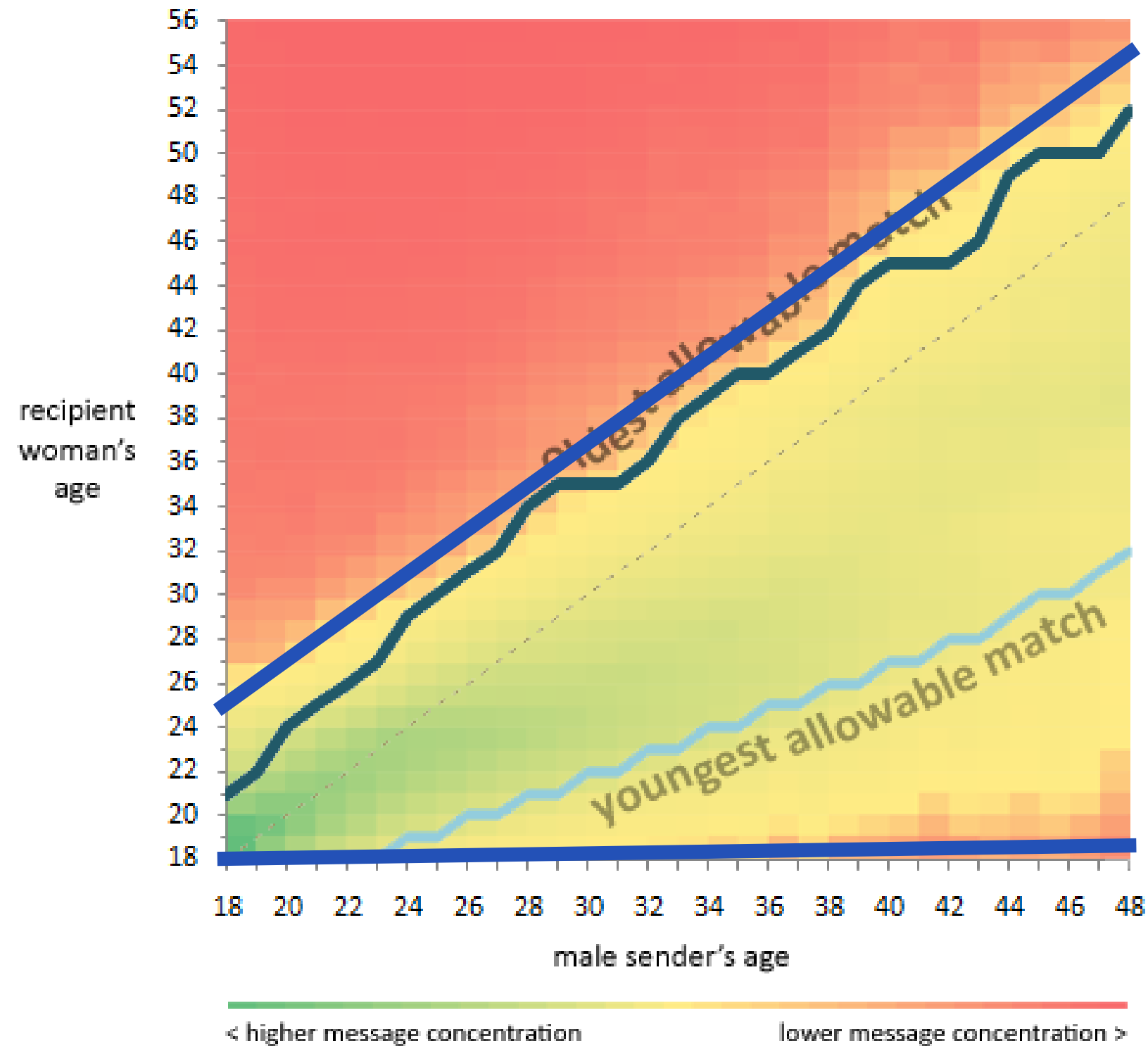
median min. & max. age values in our male users'

"allowable match" settings



Where Men Are Actually Sending Their Messages

*distribution of male to females first contacts
grouped by age*

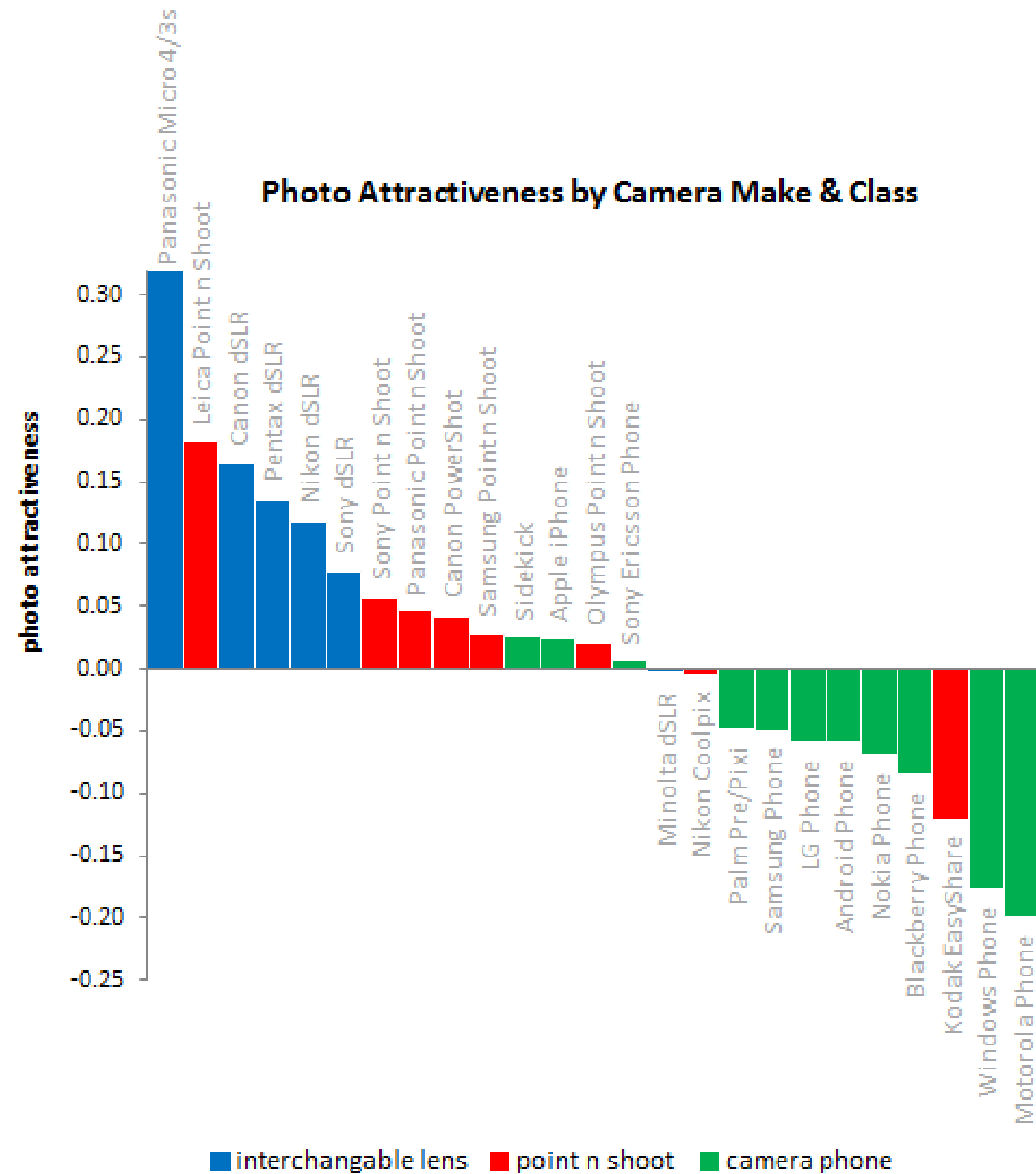


“Don’t be Ugly By Accident!”

OkCupid (blog.okcupid.com)

- Collected 552,000 examples of users pictures
- Asked people to rate their attractiveness
- Compared with the data from the camera





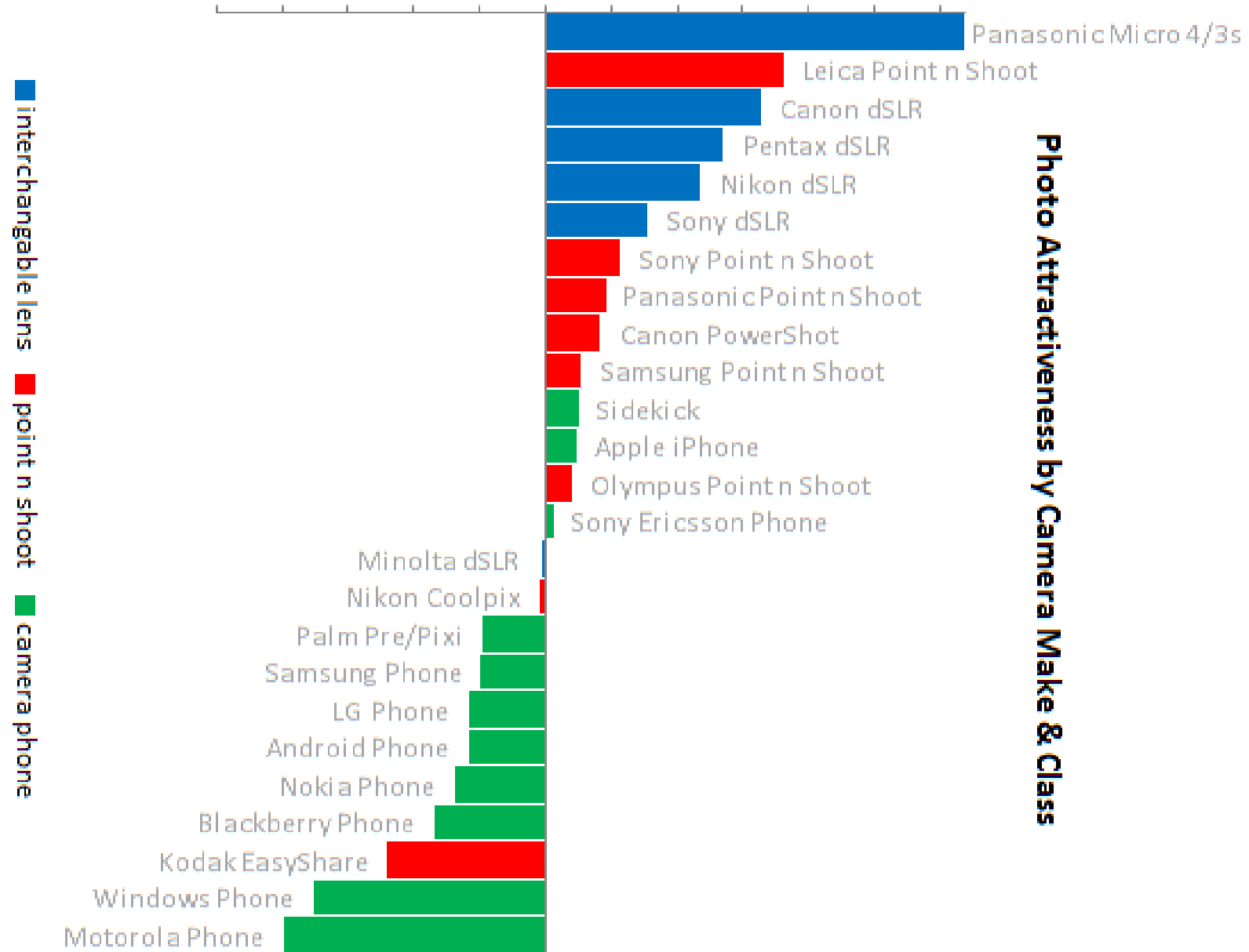
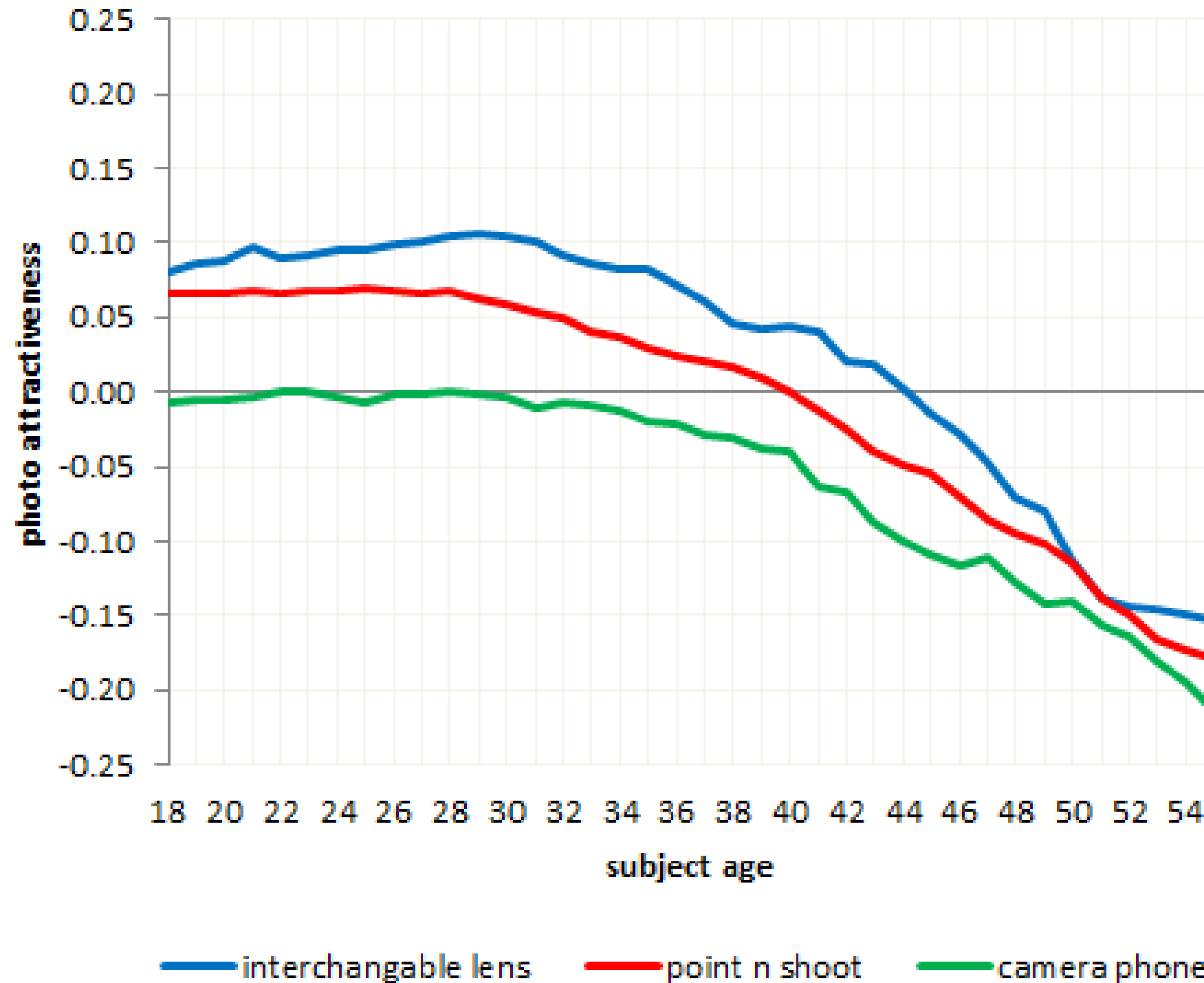


Photo Attractiveness by Camera Class



The Flash Adds 7 Years

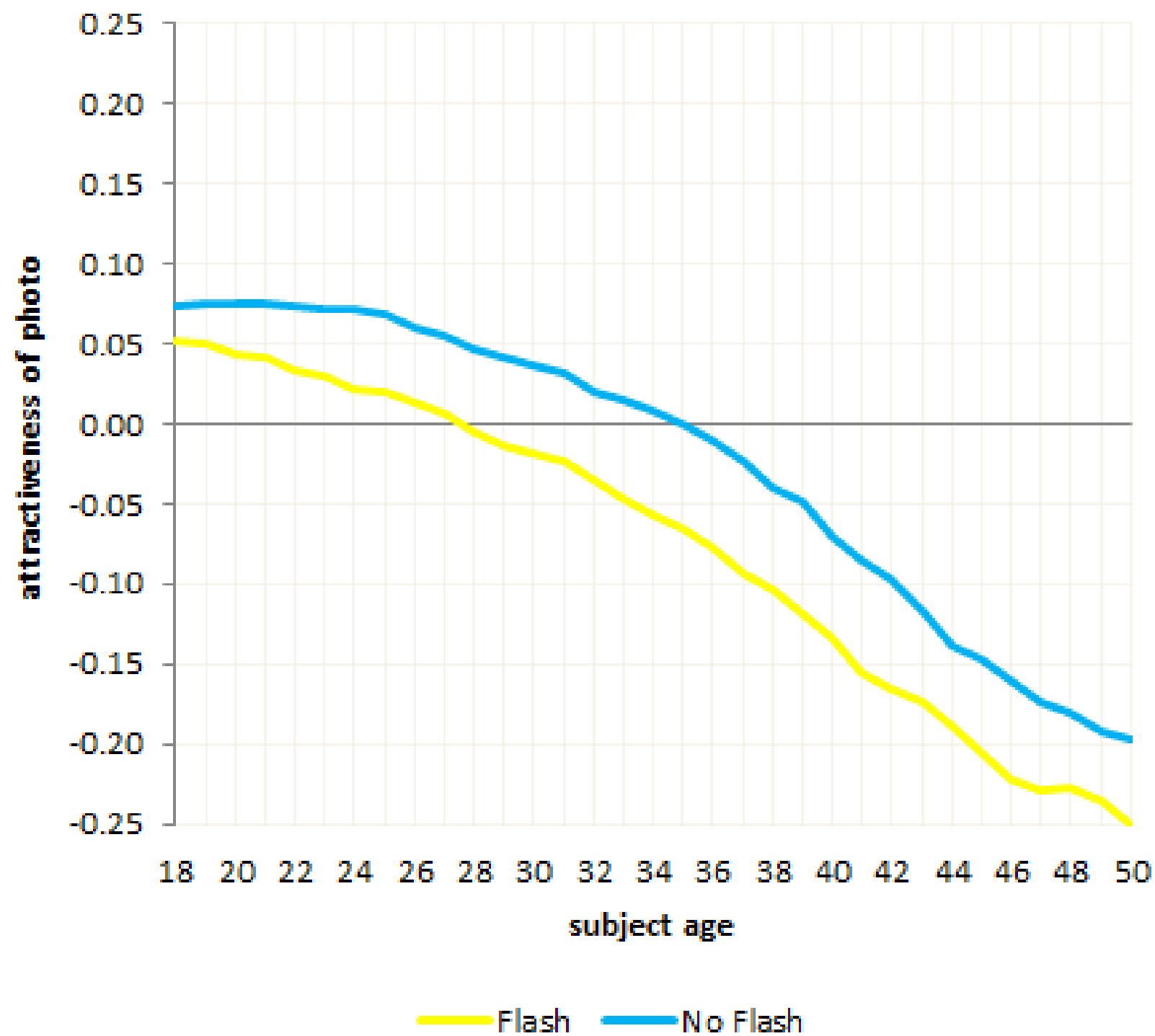
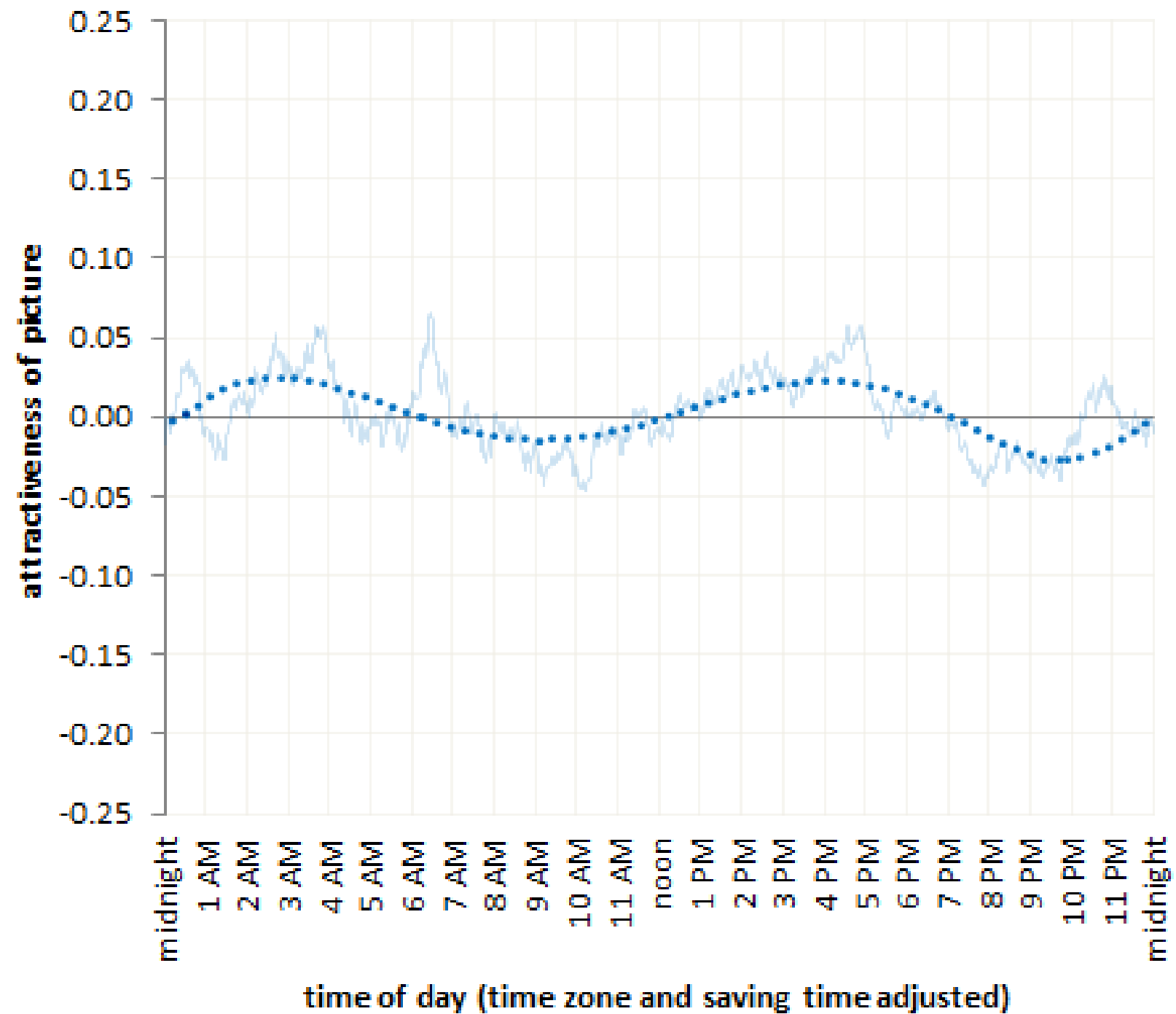
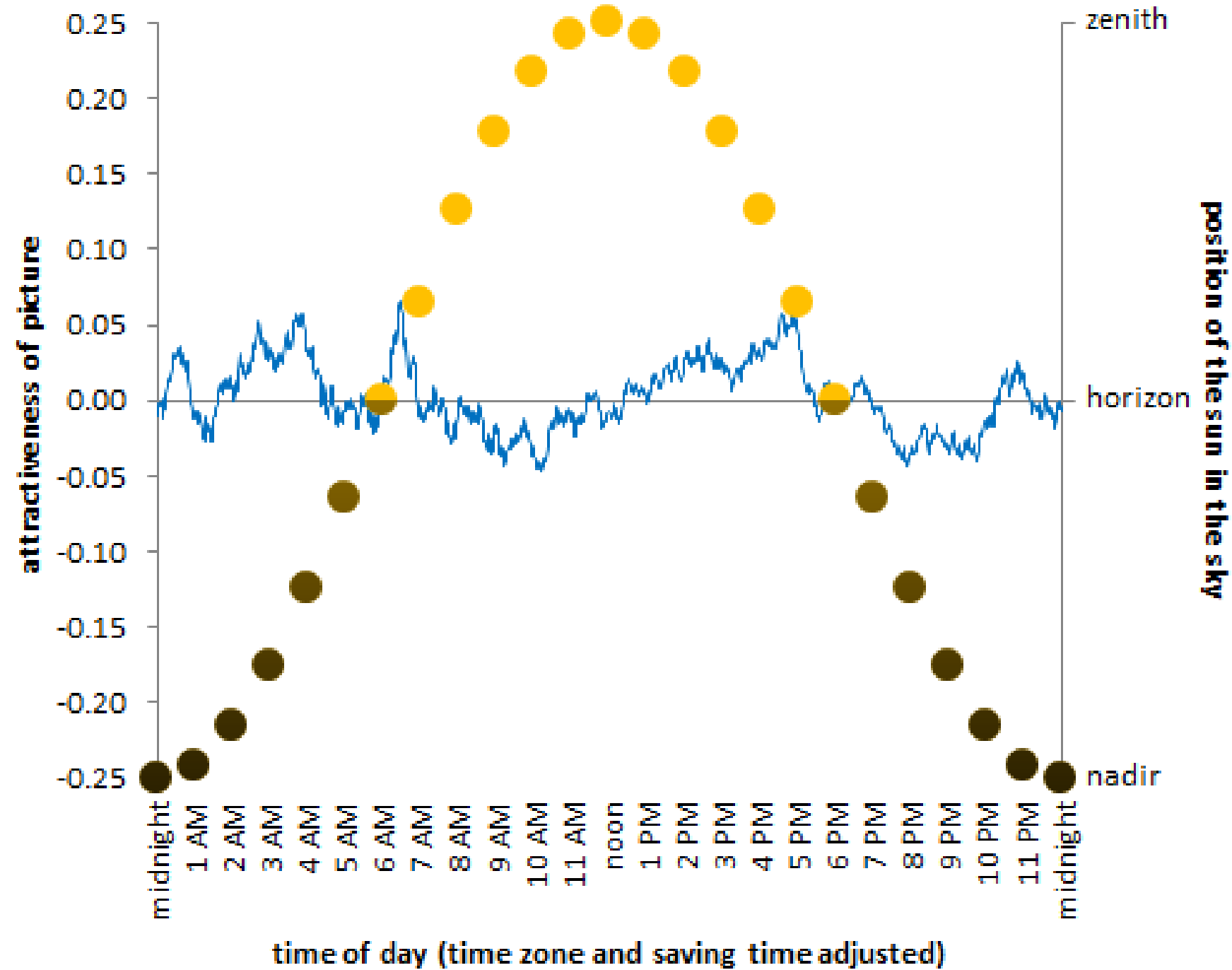


Photo Attractiveness by Time Taken



Sunrise, Sunset, and Photo Attractiveness



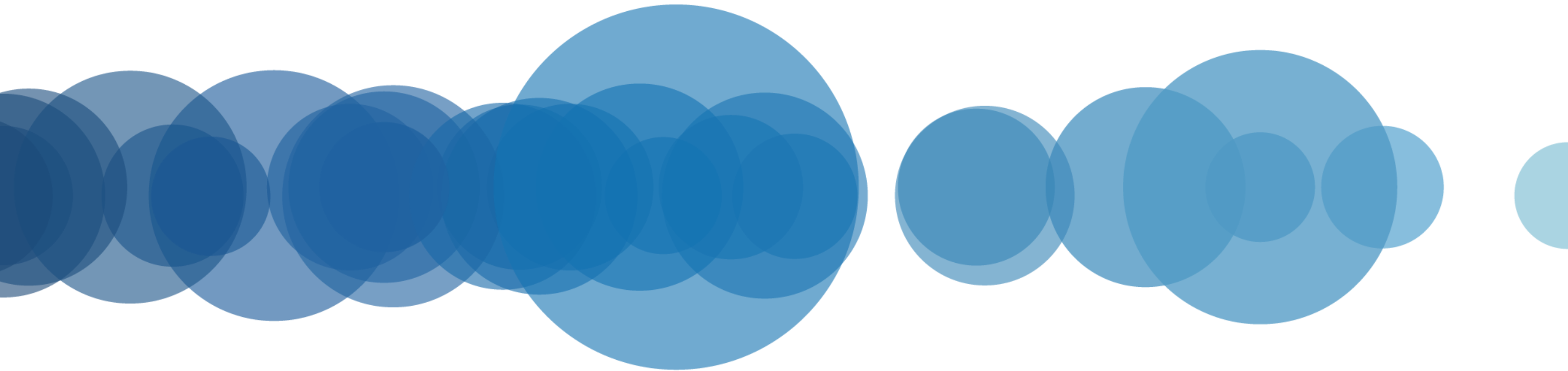
What did we learn from OkCupid?

Take your profile picture with an SLR camera without the flash at 6:30am or 5pm for your best chance of being considered attractive.

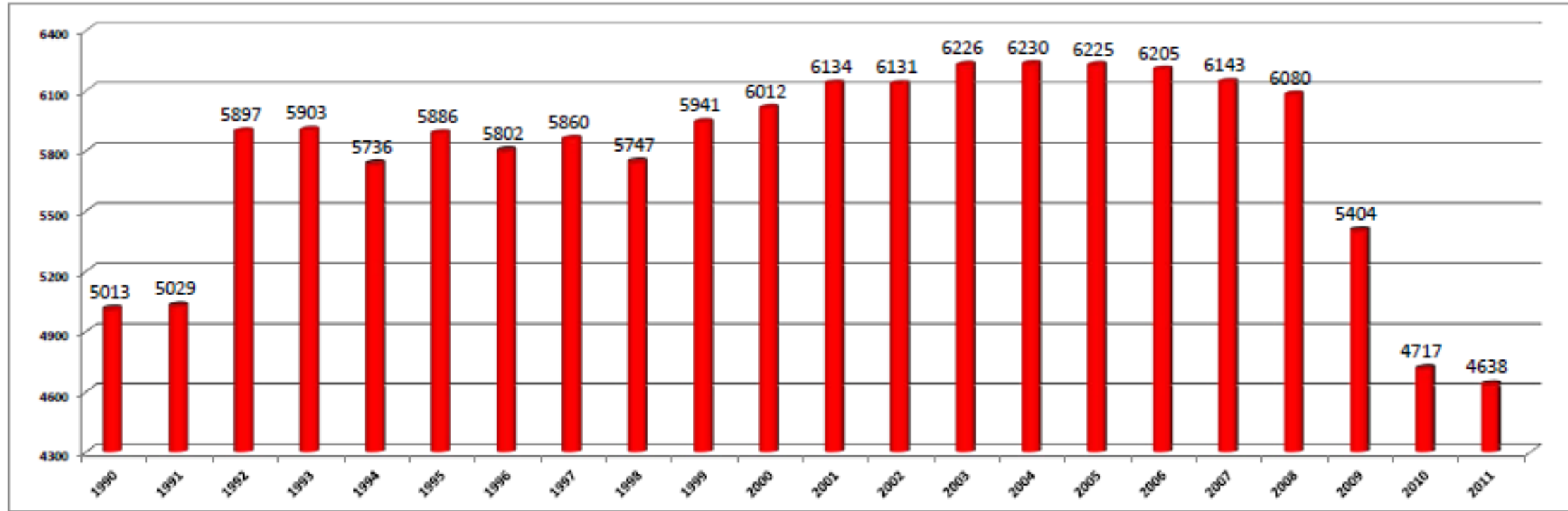


Real-World Example

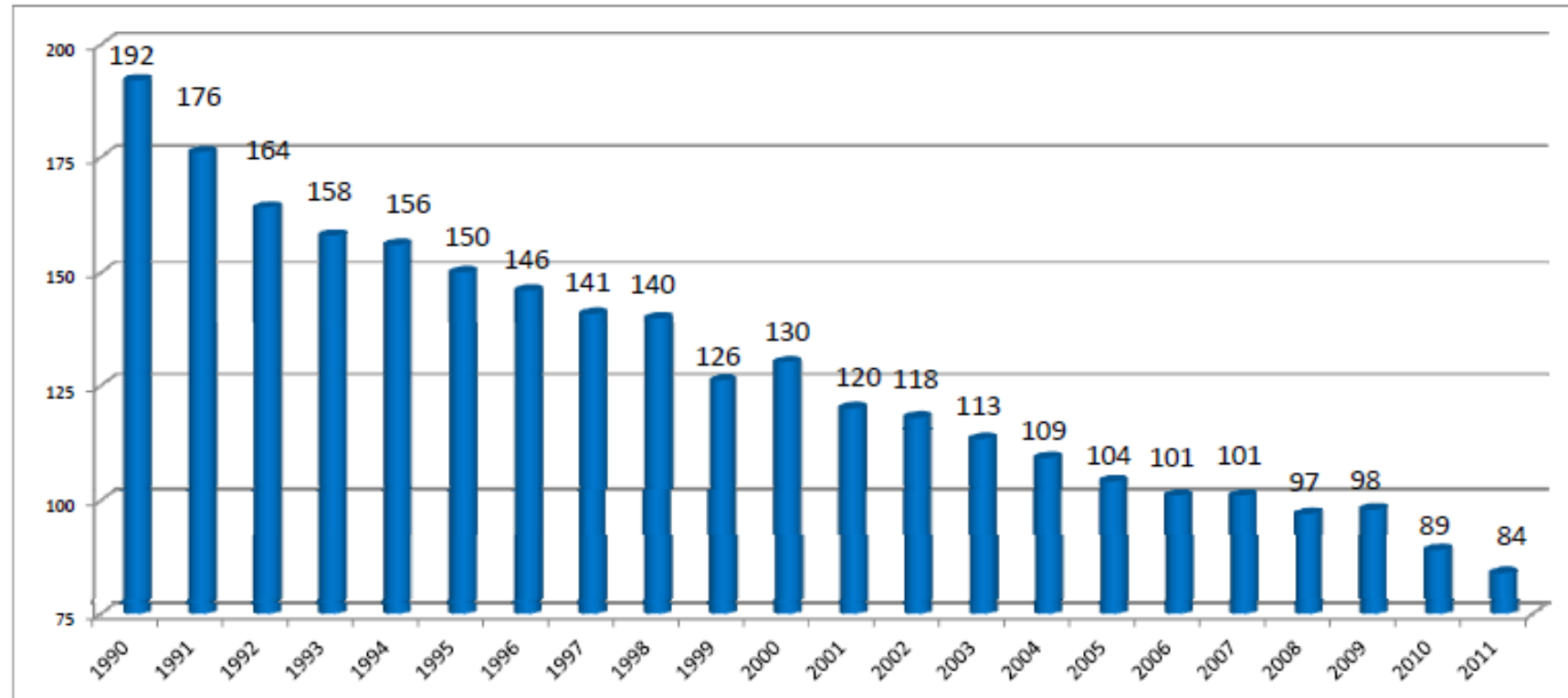
Hamilton County vs. Auditor's Office

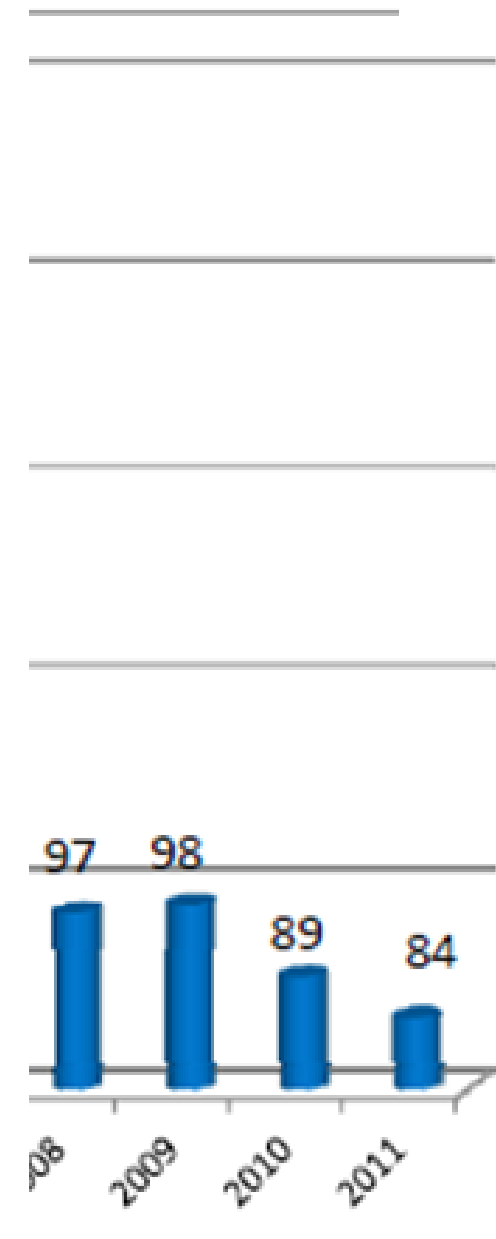
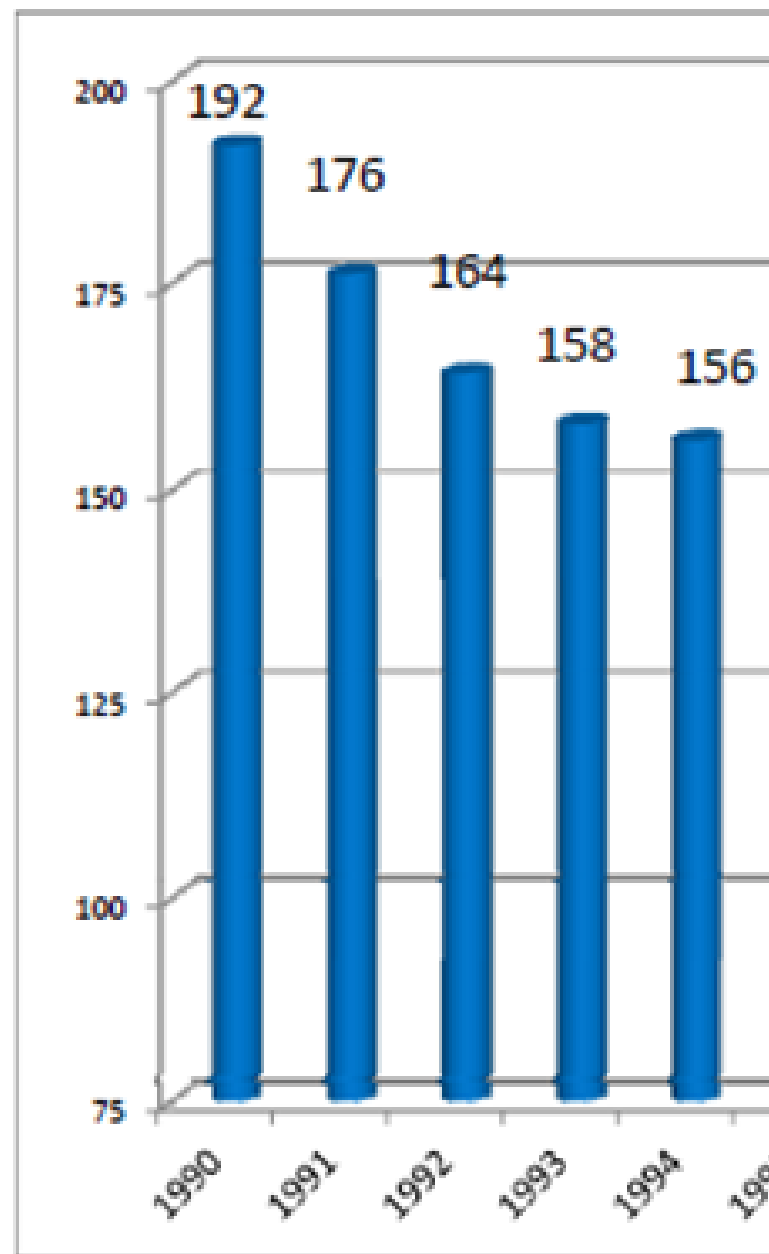
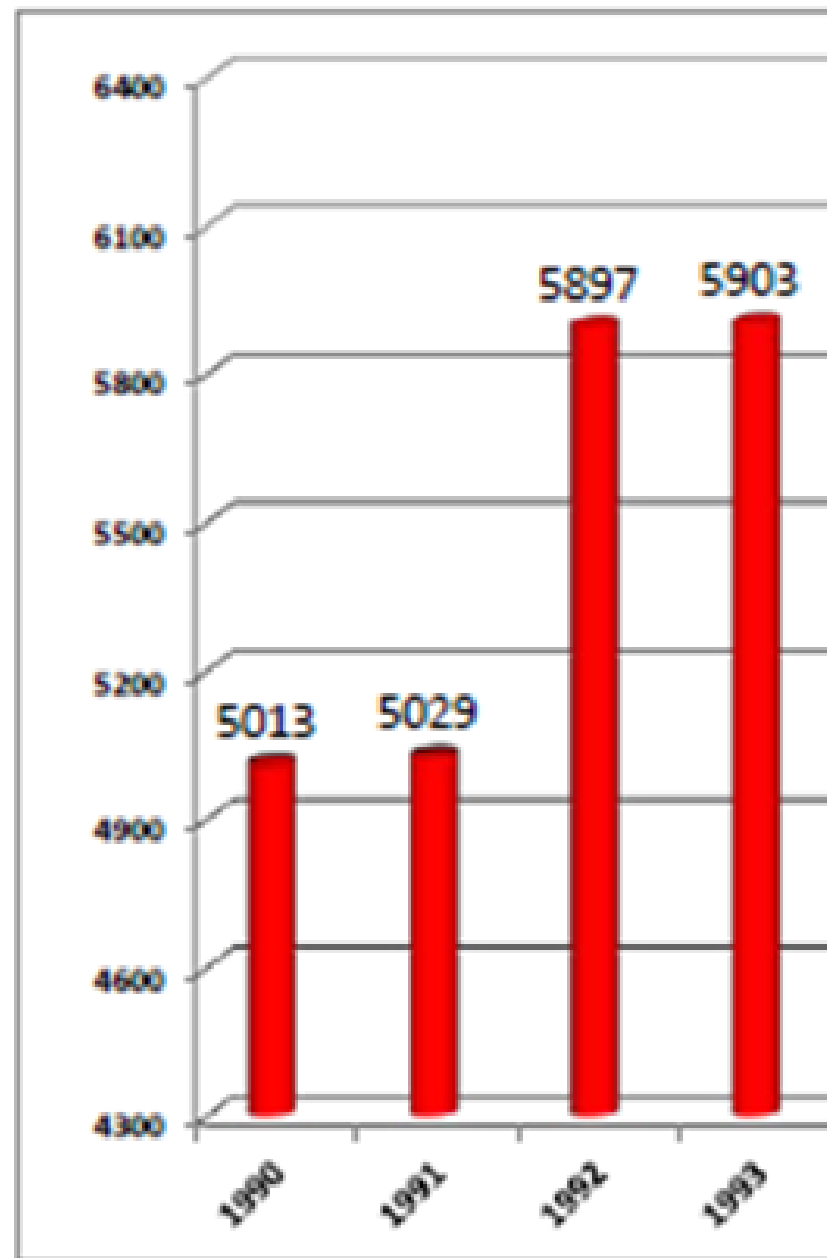


HAMILTON COUNTY POSITIONS MINUS AUDITOR'S



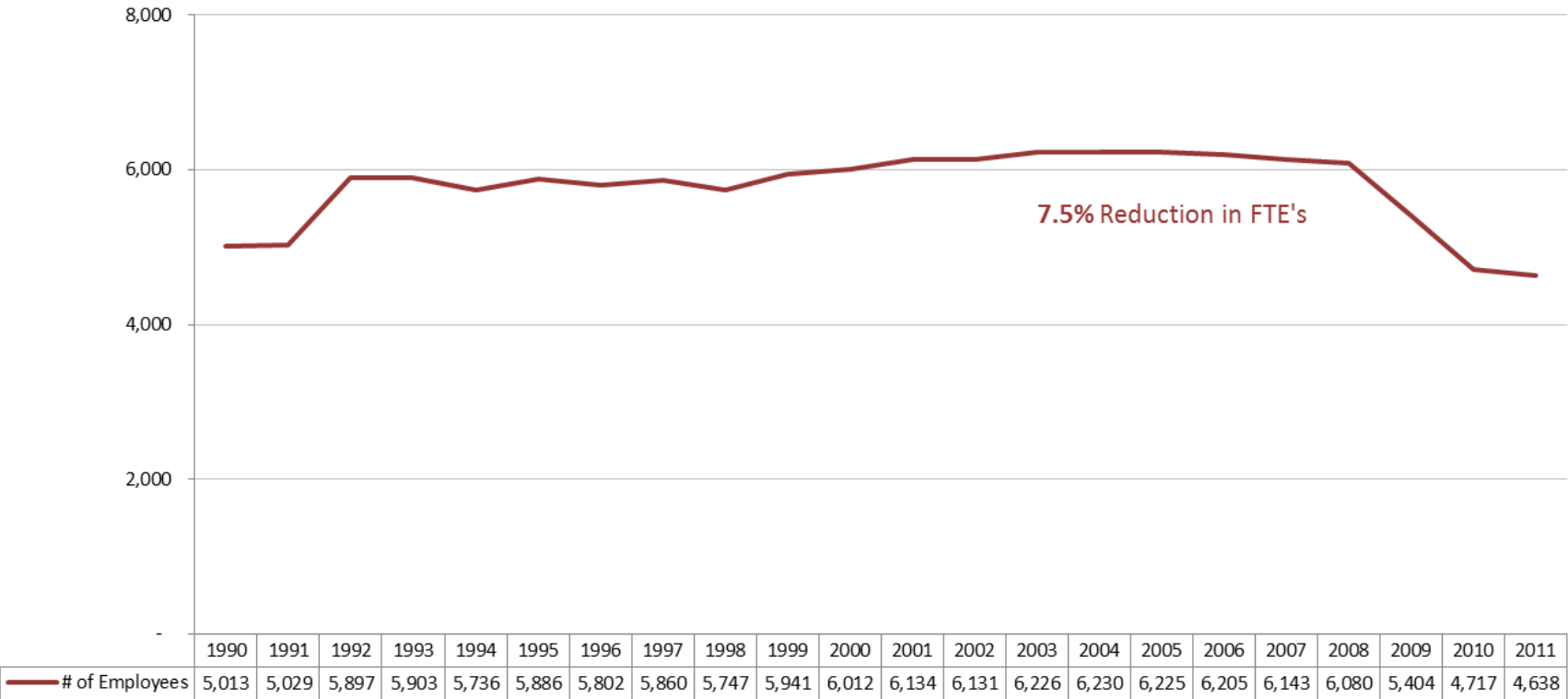
NUMBER OF EMPLOYEES IN THE AUDITOR'S OFFICE



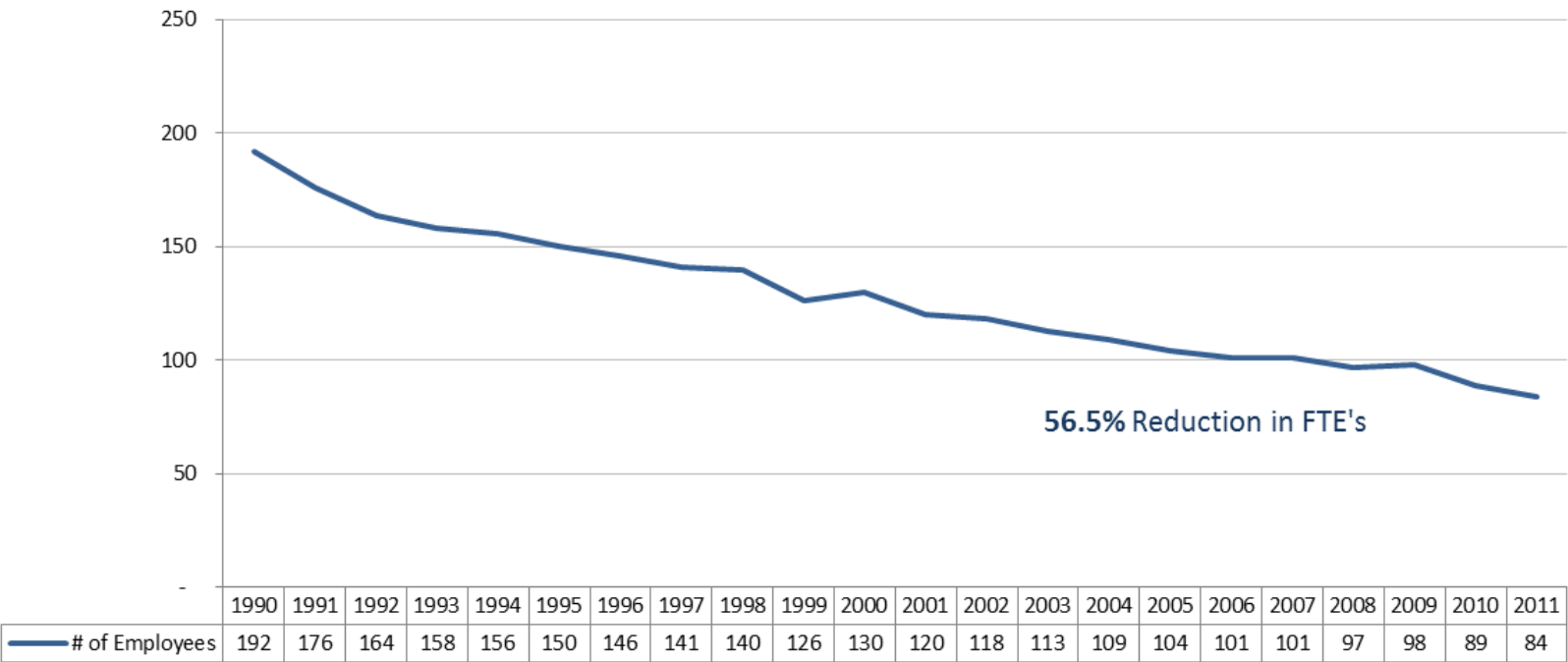


Hamilton County Employees

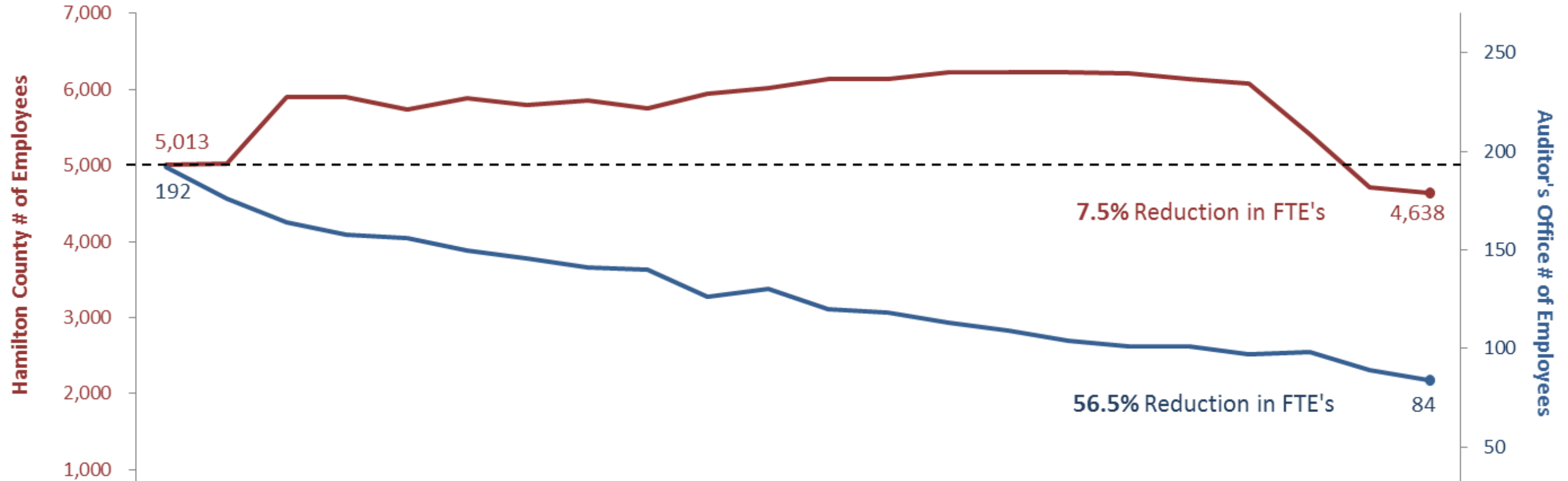
(not including Auditor's office)



Auditor's Office Employees

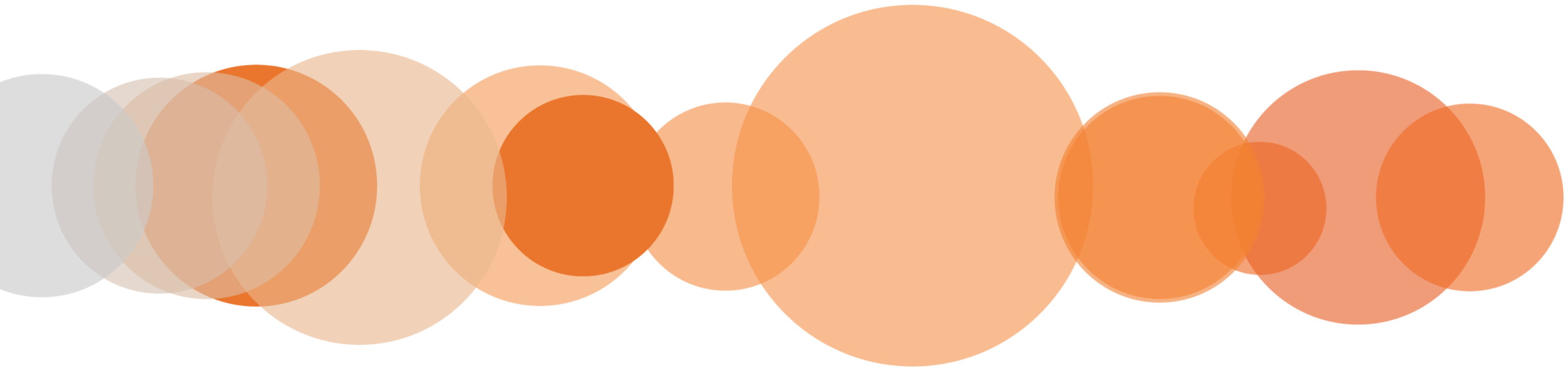


Hamilton County vs. Auditor's Office (# of Employees)



	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
Hamilton County	5,013	5,029	5,897	5,903	5,736	5,886	5,802	5,860	5,747	5,941	6,012	6,134	6,131	6,226	6,230	6,225	6,205	6,143	6,080	5,404	4,717	4,638
Auditor's Office	192	176	164	158	156	150	146	141	140	126	130	120	118	113	109	104	101	101	97	98	89	84

Exercise:

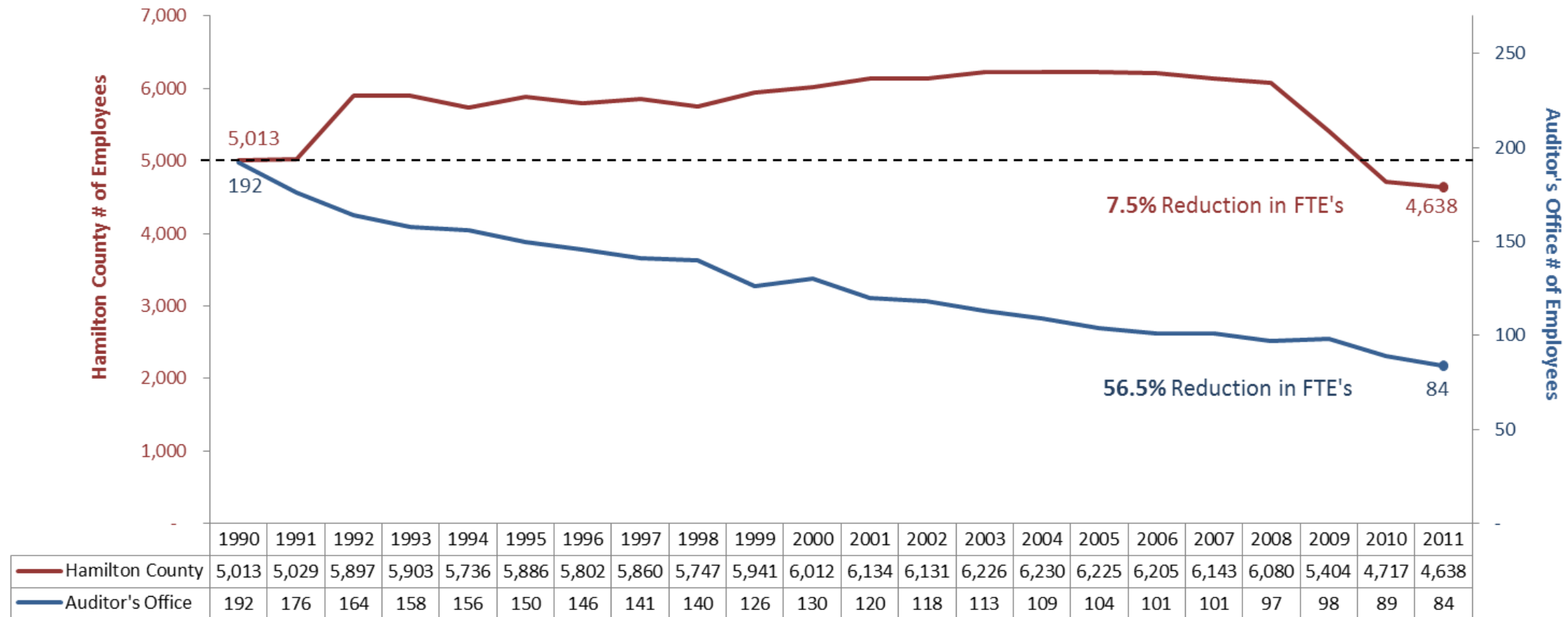


Hamilton County has hired you to create an alternative graph to argue the Auditor's Position.

- Can you use this same data to tell a completely different story?
- Using the same data, design an alternative version for Hamilton County that tells a better looking story for them.



Hamilton County vs. Auditor's Office (# of Employees)



Alternative #1:

Use absolute # of employees instead of %

375 FTE vs. only 108 (nearly 3.5x more)

Graph total number reduced as comparison

Alternative #2:

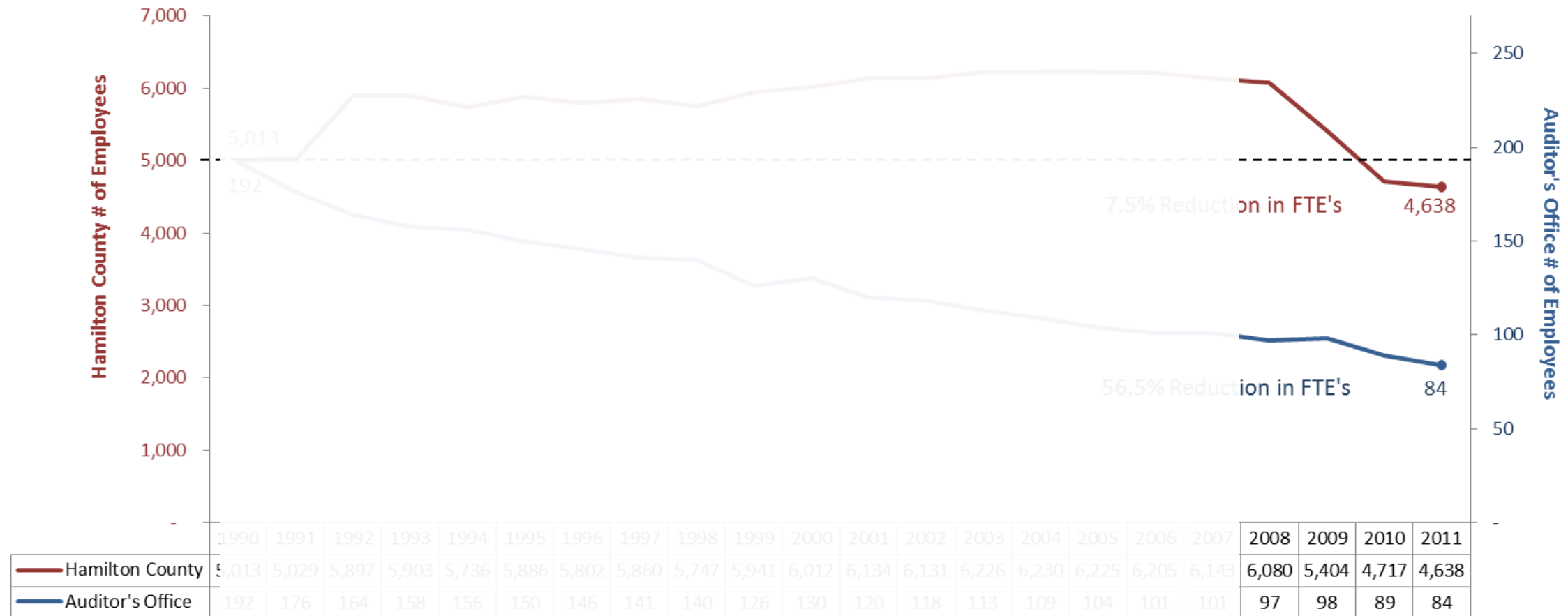
Adjust time period

Maximizes reduction for Hamilton County

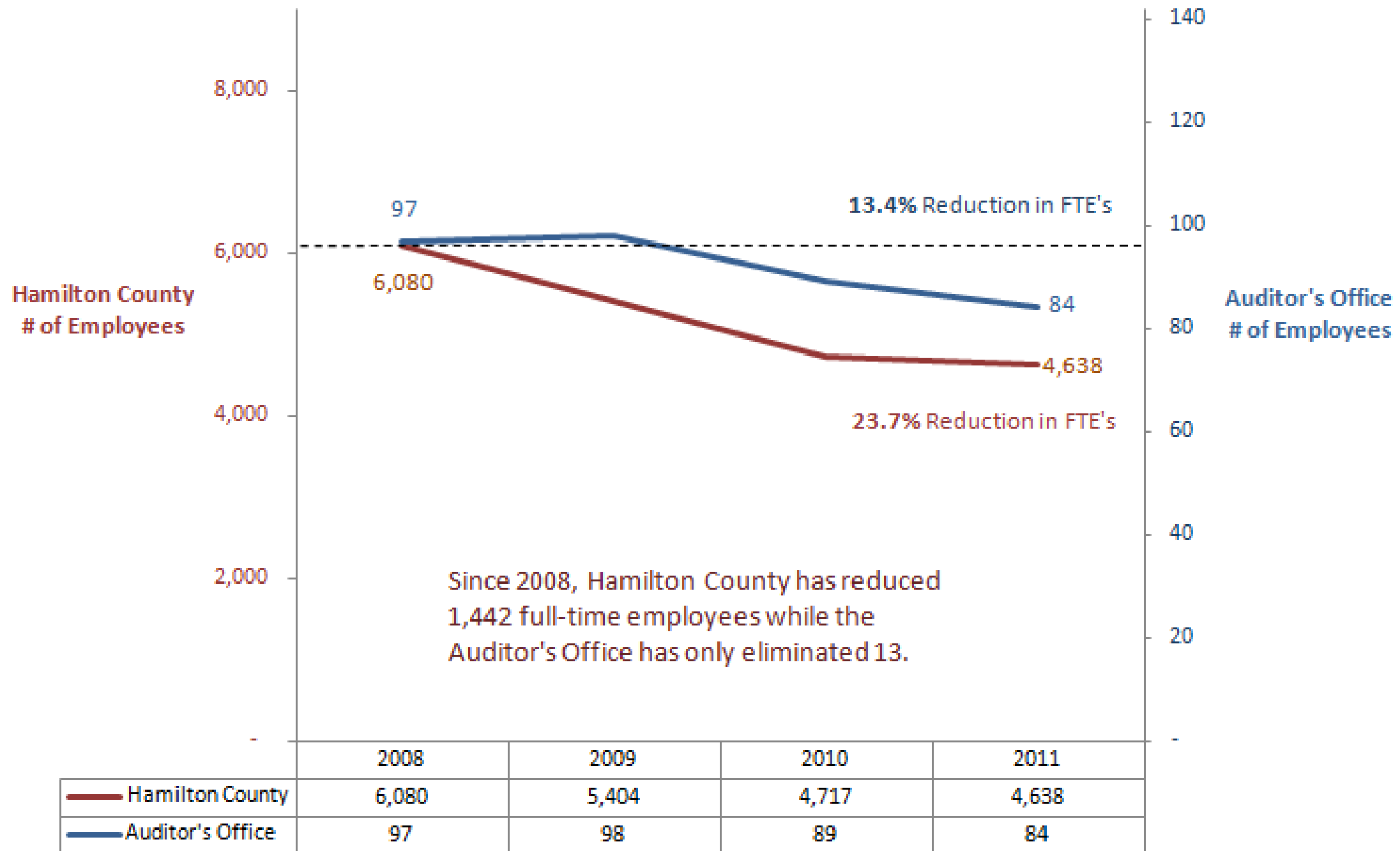
Since 2008 - 1,442 FTE reduced vs. only 13



Hamilton County vs. Auditor's Office (# of Employees)

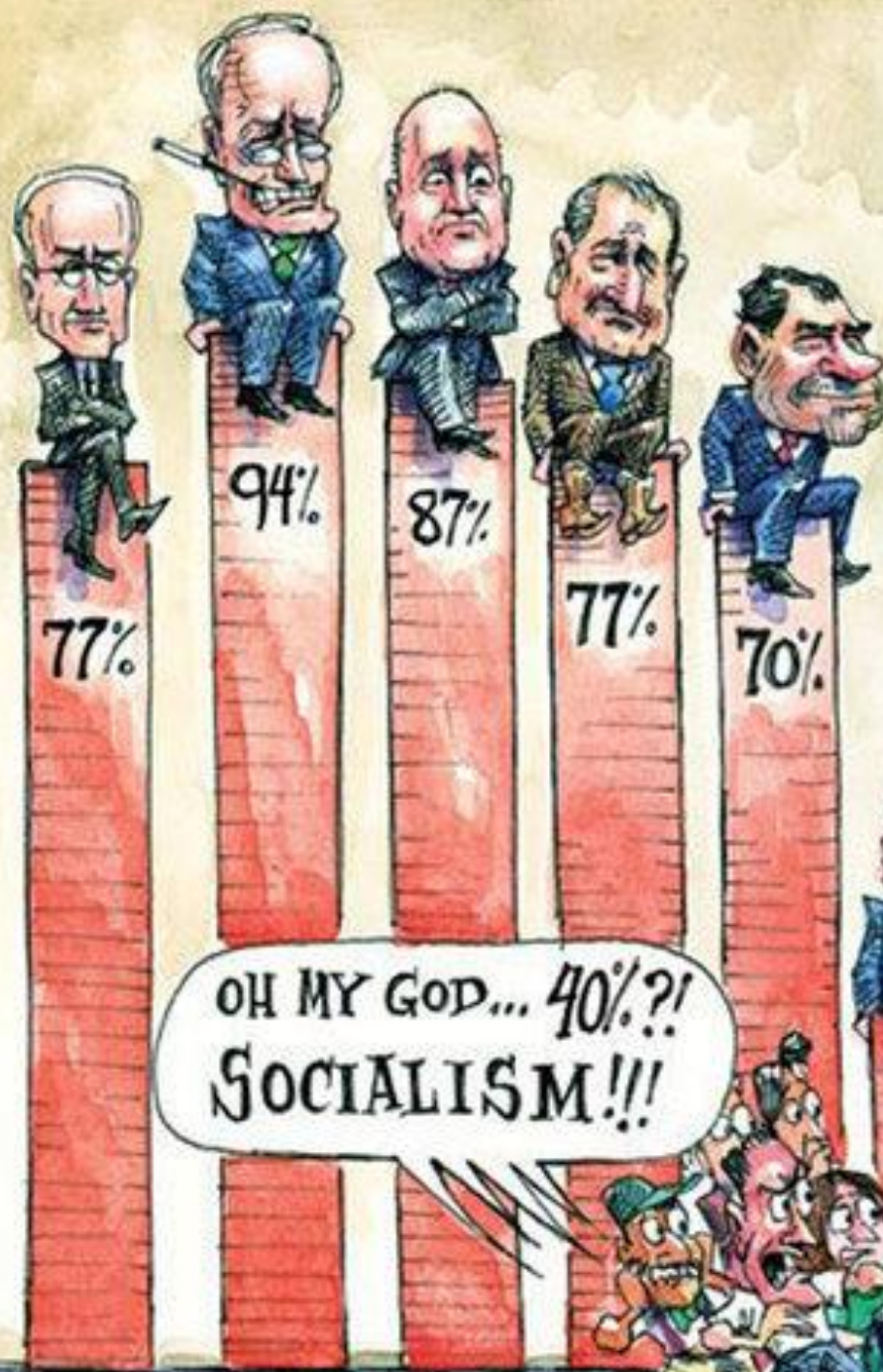


Hamilton County vs. Auditor's Office (# of Employees)

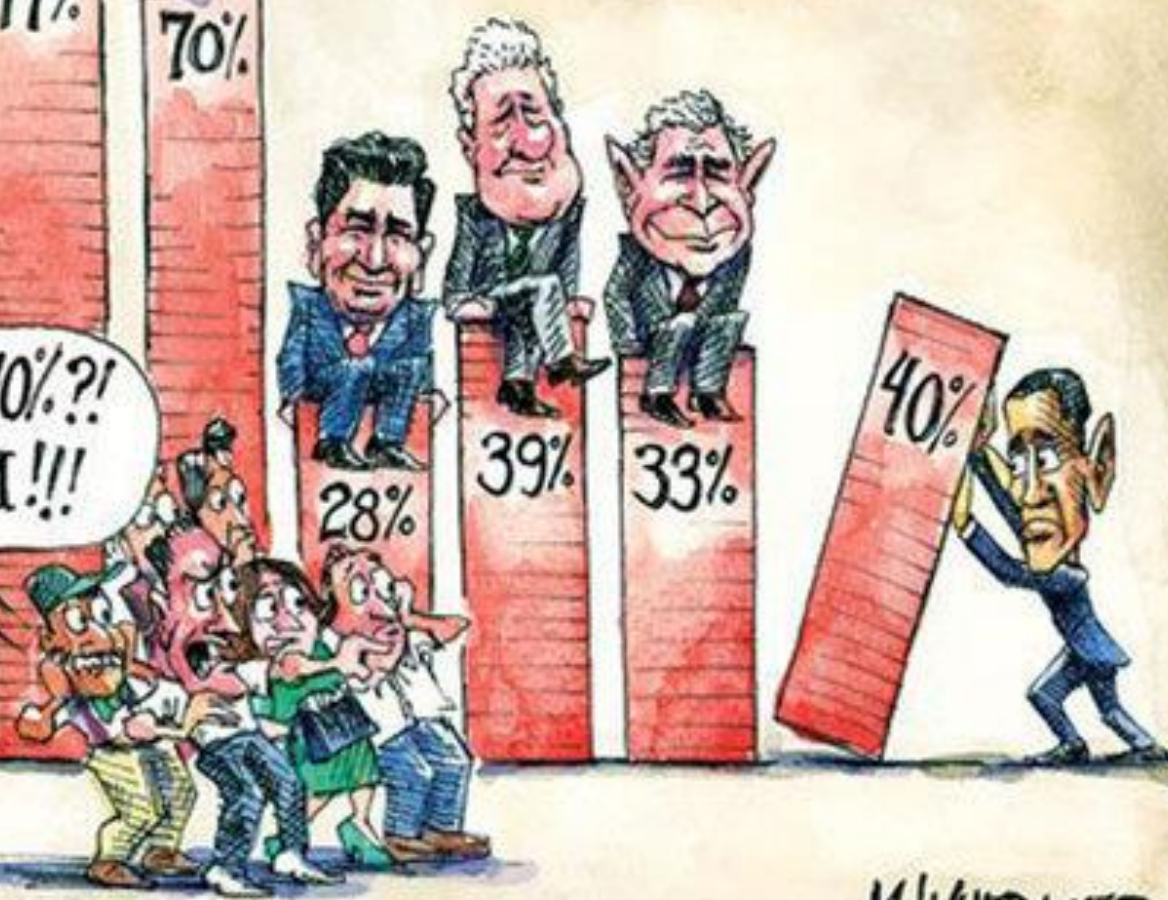


THE TOP TAX RATE

A LITTLE
HISTORICAL PERSPECTIVE



OH MY GOD... 40%?!
SOCIALISM!!!

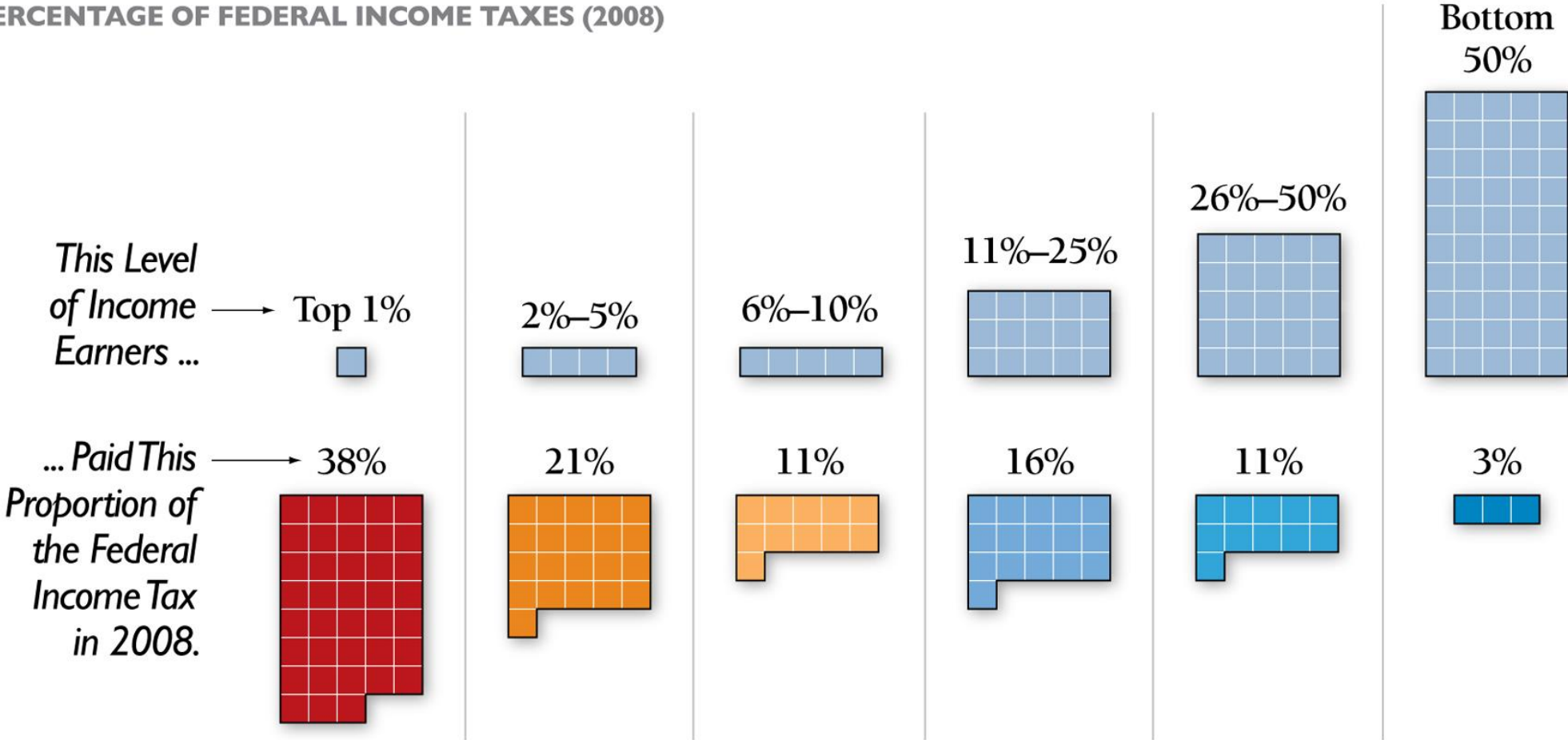


M. WUERKER POLITICO

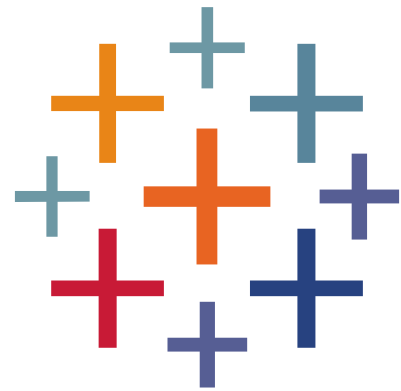
The Top 10 Percent of Earners Paid 70 Percent of Federal Income Taxes

Top earners are the target for new tax increases, but the U.S. tax system is already highly progressive. The top 1 percent of income earners paid 38 percent of all federal income taxes in 2008, while the bottom 50 percent paid only 3 percent. Forty-nine percent of U.S. households paid no federal income tax at all.

PERCENTAGE OF FEDERAL INCOME TAXES (2008)



Source: Tax Foundation and Internal Revenue Service.



+ able au[®]